Week 3

ID: 0b332f00

The function g is defined by $g(x) = 6x$. For what value of x is $g(x) = 54$?

ID: 7e3f8363

In the *xy*-plane, the graph of the linear function f contains the points $(0, 3)$ and $(7, 31)$. Which equation defines f, where $y = f(x)$?

A. $f(x) = 28x + 34$

B. $f(x) = 3x + 38$

C. $f(x) = 4x + 3$

D. $f(x) = 7x + 3$

ID: 0eae6be1

The number y is 84 less than the number x. Which equation represents the relationship between x and y?

A. $y = x + 84$

B. $y = \frac{1}{84}x$

C. $y = 84x$

D. $y = x - 84$

ID: 447fa970

The function f is defined by the equation $f(x) = 7x + 2$. What is the value of $f(x)$ when $x = 4$?

ID: c1bd5301

A model predicts that a certain animal weighed 241 pounds when it was born and that the animal gained 3 pounds per day in irst year of life. This model is defined by an equation in the form $f(x) = a + bx$, where $f(x)$ is the predicted weight, in pounds, of the animal x days after it was born, and a and b are constants. What is the value of a?

ID: bf36c815

The function g is defined by $g(x) = -x + 8$.

What is the value of $g(0)$?

A. -8

B. 0

C. 4

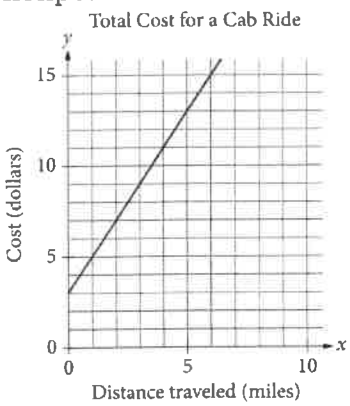
D. 8

ID: 0d391910

The function f is defined by $f(x) = 4x$. For what value of x does $f(x) = 8$?

ID: 3f5375d9

The line graphed in the xy-plane below models the total cost, in dollars, for a cab ride, y, in a certain city during nonpeak hours based on the number of miles traveled, x.



According to the graph, what is the cost for each additional mile traveled, in dollars, of a cab ride?

A. $2.00

B. $2.60

C. $3.00

D. $5.00

ID: 620fe971

A team of workers has been moving cargo off of a ship. The equation below models the approximate number of tons of cargo, y, that remains to be moved x hours after the team started working.

$$y = 120 - 25x$$

The graph of this equation in the xy-plane is a line. What is the best interpretation of the x-intercept in this context?

A. The team will have moved all the cargo in about 4.8 hours.

B. The team has been moving about 4.8 tons of cargo per hour.

C. The team has been moving about 25 tons of cargo per hour.

D. The team started with 120 tons of cargo to move.

ID: b3abf40f

$$F(x) = \frac{9}{5}(x - 273.15) + 32$$

The function F gives the temperature, in degrees Fahrenheit, that corresponds to a temperature of x kelvins. If a temperature increased by 9.10 kelvins, by how much did the temperature increase, in degrees Fahrenheit?

A. 16.38

B. 48.38

C. 475.29

D. 507.29

ID: 997bec28

The perimeter of an isosceles triangle is 83 inches. Each of the two congruent sides of the triangle has a length of 24 inches. What is the length, in inches, of the third side?

ID: 12ee1edc

$(b-2)x=8$

In the given equation, b is a constant. If the equation has no solution, what is the value of b ?

A. 2

B. 4

C. 6

D. 10

ID: 25e1cfed

How many solutions does the equation $10(15x-9)=-15(6-10x)$ have?

A. Exactly one

B. Exactly two

C. Infinitely many

D. Zero

ID: 6ac23de7

$$\frac{4x}{5}=20$$

In the equation above, what is the value of x ?

A. 25

B. 24

C. 16

D. 15

ID: e6cb2402

$$3(kx + 13) = \frac{48}{17}x + 36$$

In the given equation, k is a constant. The equation has no solution. What is the value of k?

ID: 7392dfc1

Which of the following is equivalent to $4x+6=12$?

A. $2x+4=6$

B. $x+3=3$

C. $3x+2=4$

D. $2x+3=6$

ID: 93954cfa

One pound of grapes costs $2. At this rate, how many dollars will c pounds of grapes cost?

A. $2c$

B. $2+c$

C. $\frac{2}{c}$

D. $\frac{c}{2}$

ID: 3d04de9c

A principal used a total of 25 flags that were either blue or yellow for field day. The principal used 20 blue flags. How many yellow flags were used?

A. 5

B. 20

C. 25

D. 30

ID: 60f71697

$$8x = 88$$

What value of x is the solution to the given equation?

A. 11

B. 80

C. 96

D. 704

ID: ae2287e2

A certain product costs a company \$65 to make. The product is sold by a salesperson who earns a commission that is equal to 20% of the sales price of the product. The profit the company makes for each unit is equal to the sales price minus the combined cost of making the product and the commission. If the sales price of the product is \$100, which of the following equations gives the number of units, u, of the product the company sold to make a profit of \$6,840 ?

A. $(100(1-0.2)-65)u=6{,}840$

B. $(100-65)(1-0.8)u=6{,}840$

C. $0.8(100)-65u=6{,}840$

D. $(0.2(100)+65)u=6{,}840$

ID: b23bba4c

$$3a+4b=25$$

A shipping company charged a customer $25 to ship some small boxes and some large boxes. The equation above represents the relationship between a, the number of small boxes, and b, the number of large boxes, the customer had shipped. If the customer had 3 small boxes shipped, how many large boxes were shipped?

A. 3

B. 4

C. 5

D. 6

ID: 24854644

What is the equation of the line that passes through the point $(0, 5)$ and is parallel to the graph of $y = 7x + 4$ in the *xy*-plane?

A. $y = 5x$

B. $y = 7x + 5$

C. $y = 7x$

D. $y = 5x + 7$

ID: 87322577

$$x + y = 75$$

The equation above relates the number of minutes, x, Maria spends running each day and the number of minutes, y, she spends biking each day. In the equation, what does the number 75 represent?

A. The number of minutes spent running each day

B. The number of minutes spent biking each day

C. The total number of minutes spent running and biking each day

D. The number of minutes spent biking for each minute spent running

ID: 9f3cb472

Line t in the *xy*-plane has a slope of $-\frac{1}{3}$ and passes through the point $(9, 10)$. Which equation defines line t?

A. $y = 13x - \frac{1}{3}$

B. $y = 9x + 10$

C. $y = -\frac{x}{3} + 10$

D. $y = -\frac{x}{3} + 13$

ID: ebf8d2b7

A machine makes large boxes or small boxes, one at a time, for a total of 700 minutes each day. It takes the machine 10 minutes to make a large box or 5 minutes to make a small box. Which equation represents the possible number of large boxes, x, and small boxes, y, the machine can make each day?

A. $5x + 10y = 700$

B. $10x + 5y = 700$

C. $(x + y)(10 + 5) = 700$

D. $(10 + x)(5 + y) = 700$

ID: c6b151d4

A total of 364 paper straws of equal length were used to construct two types of polygons: triangles and rectangles. The triangles and rectangles were constructed so that no two polygons had a common side. The equation $3x + 4y = 364$ represents this situation, where x is the number of triangles constructed and y is the number of rectangles constructed. What is the best interpretation of $(x, y) = (24, 73)$ in this context?

A. If 24 triangles were constructed, then 73 rectangles were constructed.

B. If 24 triangles were constructed, then 73 paper straws were used.

C. If 73 triangles were constructed, then 24 rectangles were constructed.

D. If 73 triangles were constructed, then 24 paper straws were used.

ID: 8c98c834

The equation $y = 0.1x$ models the relationship between the number of different pieces of music a certain pianist practices, y, during an x-minute practice session. How many pieces did the pianist practice if the session lasted 30 minutes?

A. 1

B. 3

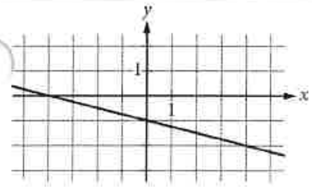
C. 10

D. 30

ID: fdee0fbf

In the xy-plane, line k intersects the y-axis at the point $(0, -6)$ and passes through the point $(2, 2)$. If the point $(20, w)$ lies on line k, what is the value of w ?

ID: b2845d88



Which of the following is an equation of the graph shown in the *xy*-plane above?

A. $y = -\frac{1}{4}x - 1$

B. $y = -x - 4$

C. $y = -x - \frac{1}{4}$

D. $y = -4x - 1$

ID: b450ab03

An employee at a restaurant prepares sandwiches and salads. It takes the employee 1.5 minutes to prepare a sandwich and 1.9 minutes to prepare a salad. The employee spends a total of 46.1 minutes preparing x sandwiches and y salads. Which equation represents this situation?

A. $1.9x + 1.5y = 46.1$

B. $1.5x + 1.9y = 46.1$

C. $x + y = 46.1$

D. $30.7x + 24.3y = 46.1$

ID: ee2f611f

A local transit company sells a monthly pass for $95 that allows an unlimited number of trips of any length. Tickets for individual trips cost $1.50, $2.50, or $3.50, depending on the length of the trip. What is the minimum number of trips per month for which a monthly pass could cost less than purchasing individual tickets for trips?

ID: 563407e5

A bakery sells trays of cookies. Each tray contains at least 50 cookies but no more than 60. Which of the following could be the total number of cookies on 4 trays of cookies?

A. 165

B. 205

C. 245

D. 285

ID: 541bef2f

$$y \leq x + 7$$
$$y \geq -2x - 1$$

Which point (x, y) is a solution to the given system of inequalities in the *xy*-plane?

A. $(-14, 0)$

B. $(0, -14)$

C. $(0, 14)$

D. $(14, 0)$

ID: 6c71f3ec

A salesperson's total earnings consist of a base salary of x dollars per year, plus commission earnings of 11% of the total sales the salesperson makes during the year. This year, the salesperson has a goal for the total earnings to be at least 3 times and at most 4 times the base salary. Which of the following inequalities represents all possible values of total sales s, in dollars, the salesperson can make this year in order to meet that goal?

A. $2x \leq s \leq 3x$

B. $\frac{2}{0.11}x \leq s \leq \frac{3}{0.11}x$

C. $3x \leq s \leq 4x$

D. $\frac{3}{0.11}x \leq s \leq \frac{4}{0.11}x$

ID: 68f2cbaf

set a goal to walk at least 24 kilometers every day to prepare for a multiday hike. On a certain day, Ty plans to walk at an average speed of 4 kilometers per hour. What is the minimum number of hours Ty must walk on that day to fulfill the daily goal?

A. 4

B. 6

C. 20

D. 24

ID: 1a621af4

A number x is at most 2 less than 3 times the value of y. If the value of y is -4, what is the greatest possible value of x?

ID: df32b09c

Tom scored 85, 78, and 98 on his first three exams in history class. Solving which inequality gives the score, G, on Tom's fourth exam that will result in a mean score on all four exams of at least 90 ?

A. $90-(85+78+98) \leq 4G$

B. $4G+85+78+98 \geq 360$

C. $\dfrac{(G+85+78+98)}{4} \geq 90$

D. $\dfrac{(85+78+98)}{4} \geq 90-4G$

ID: bf5f80c6

$$y < -4x + 4$$

Which point (x, y) is a solution to the given inequality in the xy-plane?

A. $(-4, 0)$

B. $(0, 5)$

C. $(2, 1)$

D. $(2, -1)$

ID: 80da233d

A certain elephant weighs 200 pounds at birth and gains more than 2 but less than 3 pounds per day during its first year. Which of the following inequalities represents all possible weights w, in pounds, for the elephant 365 days after birth?

A. $400 < w < 600$

B. $565 < w < 930$

C. $730 < w < 1{,}095$

D. $930 < w < 1{,}295$

ID: b31c3117

$$H = 120p + 60$$

The Karvonen formula above shows the relationship between Alice's target heart rate H, in beats per minute (bpm), and the intensity level p of different activities. When $p = 0$, Alice has a resting heart rate. When $p = 1$, Alice has her maximum heart rate. It is recommended that p be between 0.5 and 0.85 for Alice when she trains. Which of the following inequalities describes Alice's target training heart rate?

A. $120 \leq H \leq 162$

B. $102 \leq H \leq 120$

C. $60 \leq H \leq 162$

D. $60 \leq H \leq 102$

ID: dba8d38a

A petting zoo sells two types of tickets. The standard ticket, for admission only, costs \$5. The premium ticket, which includes admission and food to give to the animals, costs \$12. One Saturday, the petting zoo sold a total of 250 tickets and collected a total of \$2,300 from ticket sales. Which of the following systems of equations can be used to find the number of standard tickets, s, and premium tickets, p, sold on that Saturday?

A. $\begin{aligned} s+p&=250 \\ 5s+12p&=2{,}300 \end{aligned}$

B. $\begin{aligned} s+p&=250 \\ 12s+5p&=2{,}300 \end{aligned}$

C. $\begin{aligned} 5s+12p&=250 \\ s+p&=2{,}300 \end{aligned}$

D. $\begin{aligned} 12s+5p&=250 \\ s+p&=2{,}300 \end{aligned}$

ID: 317e80f9

$$x + y = 18$$
$$5y = x$$

What is the solution (x, y) to the given system of equations?

A. $(15, 3)$

B. $(16, 2)$

C. $(17, 1)$

D. $(18, 0)$

ID: 6a87902f

$$y = 2x + 10$$
$$y = 2x - 1$$

At how many points do the graphs of the given equations intersect in the xy-plane?

A. Zero

B. Exactly one

C. Exactly two

D. Infinitely many

ID: f75bd744

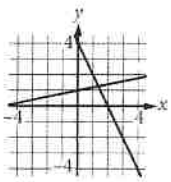
$$4x - 6y = 10y + 2$$
$$ty = \frac{1}{2} + 2x$$

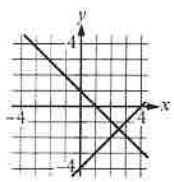
In the given system of equations, t is a constant. If the system has no solution, what is the value of t?

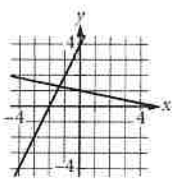
ID: 6e6a3241

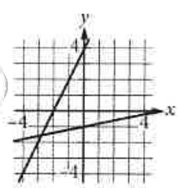
$x + 5y = 5$
$2x - y = -4$

Which of the following graphs in the *xy*-plane could be used to solve the system of equations above?

A. 

B. 

C. 



ID: aff28230

$$x = 10$$
$$y = x + 21$$

The solution to the given system of equations is (x, y). What is the value of y?

A. 2.1

B. 10

C. 21

D. 31

ID: f5929f7a

$$y = -\frac{1}{9}x$$
$$y = \frac{1}{2}x$$

The solution to the given system of equations is (x, y). What is the value of x?

A. -9

B. -7

C. 0

D. 2

ID: 74c03c21

A bus traveled on the highway and on local roads to complete a trip of 160 miles. The trip took 4 hours. The bus traveled at an average speed of 55 miles per hour (mph) on the highway and an average speed of 25 mph on local roads. If x is the time, in hours, the bus traveled on the highway and y is the time, in hours, it traveled on local roads, which system of equations represents this situation?

A. $55x + 25y = 4$
$x + y = 160$

B. $55x + 25y = 160$
$x + y = 4$

C. $25x + 55y = 4$
$x + y = 160$

D. $25x + 55y = 160$
$x + y = 4$

ID: 8abed0fb

$$y = 2x + 3$$
$$x = 1$$

What is the solution (x, y) to the given system of equations?

A. $(1,2)$

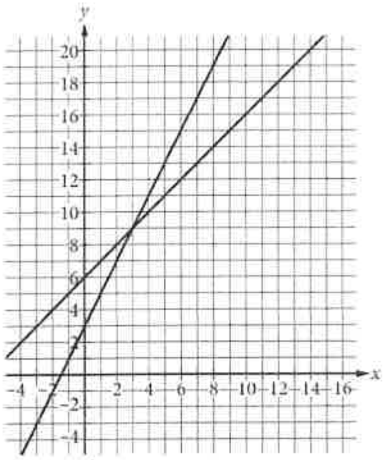
B. $(1,5)$

C. $(2,3)$

D. $(2,7)$

ID: e1259a5a

A system of two linear equations is graphed in the xy-plane below.



Which of the following points is the solution to the system of equations?

A. (3,9)

B. (6,15)

C. (8,10)

D. (12,18)

Mathematics—Module 1

1. An apple falls from a high branch. The height of the apple is modeled by the function $f(x) = -x^2 - 2x + 7$, where $f(x)$ represents the height of the apple and x represents the number of seconds since the fall. After how many seconds will the apple reach the ground?

A) 1.8
B) 2.1
C) 2.3
D) 2.6

2. If $2x + 6 = -2$, what is the value of $10x + 11$?

A) -4
B) -8
C) -29
D) -51

3. Expand the following expression: $(x + 2)(x - 3)$

A) $x^2 - 1$
B) $x^2 - 6$
C) $x^2 - x - 6$
D) $x^2 - 5x - 1$

Refer to the following for question 4:

The table below represents the test scores on a chemistry final, as well as the lab grades for the semester.

Final Exam	78	85	91	77	94	83	87
Lab Grade	84	86	98	90	97	85	90

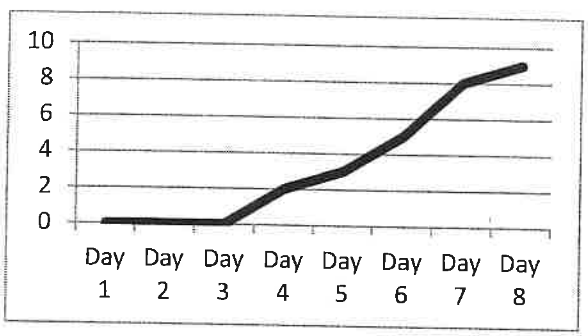
4. What is the difference of the mean and median of the lab grades?

A) 0
B) 1.5
C) 3
D) 4.5

5. The graph of $y = 2x^2 + 3x$ intersects the graph of $y = -x$ at $(0, 0)$ and (a, b). What is the value of b?

6. A population of bacteria tripled every day. After 5 days, there were 2,430 bacteria. How many were there originally?

Refer to the following for question 7:



Beth plants a number of sunflower seeds and checks daily to see if any have sprouted. The graph above shows how many seedlings are growing each day when she checks.

7. Which day did the first seedling appear?

A) Day 1
B) Day 2
C) Day 3
D) Day 4

8. Kasie is filling glasses with water. Each glass is cylindrical with a radius of 1.25 inches and a height of 6 inches. If she fills them to a height of 5 inches, how many glasses can she fill with 3 quarts of water (1 gallon = 231 cubic inches)? Use 3.14 for π.

A) 3
B) 5
C) 7
D) 9

9. **Mandy can buy 4 containers of yogurt and 3 boxes of crackers for \$9.55. She can buy 2 containers of yogurt and 2 boxes of crackers for \$5.90. How much does one box of crackers cost?**

A) \$1.75
B) \$2.00
C) \$2.25
D) \$2.50

10. **A researcher surveyed 350 people at a grocery store, offering them a sample of a well-known juice along with a new brand of juice. 85% of those who tasted the two juices preferred the new brand. Which of the following inferences is a logical conclusion?**

A) 85% of shoppers will prefer the new brand of juice.
B) 15% of shoppers will prefer the new brand of juice.
C) Most shoppers will prefer the new brand of juice.
D) Most shoppers will prefer the well-known juice.

11. **If the solution for the system of equations below is (x, y), what is the value of $x^2 - 3y$?**

$$5x - 6y = 12$$
$$-2x + 3y = -5$$

A) -3
B) -1
C) 3
D) 5

12. **A colony of *Escherichia coli* is inoculated from a Petri dish into a test tube containing 50 mL of nutrient broth. The test tube is placed in a 37 °C incubator and agitator. After one hour, the number of bacteria in the test tube is determined to be 8×10^6. Given that the doubling time of *Escherichia coli* is 20 minutes with agitation at 37 °C, approximately how many bacteria should the test tube contain after eight hours of growth?**

A) 2.56×10^8
B) 2.05×10^9
C) 1.7×10^{14}
D) 1.7×10^{13}

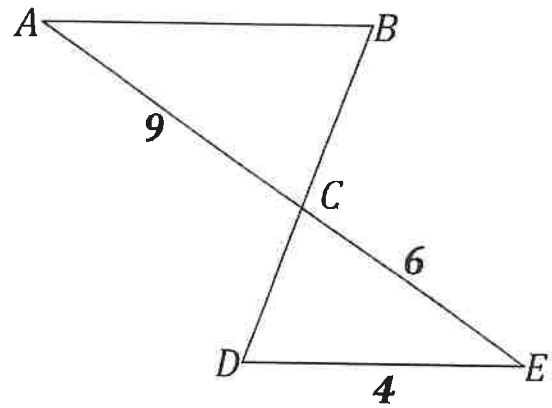
13. **If $3x - 4 = -1$, what is the value of $12x + 3$?**

A) 18
B) 15
C) 6
D) 1

14. **Evaluate the expression $4\sqrt{6} + 8\sqrt{6}$. Simplify your answer as much as possible.**

A) $12\sqrt{12}$
B) 72
C) $12\sqrt{6}$
D) $24\sqrt{3}$

15. **In the figure below (not drawn to scale), ΔABC is similar to ΔEDC. What is the length of AB?**



16. **The equation of a circle is shown below. What is the circle's diameter?**

$$x^2 + y^2 - 4x + 2y = 4$$

A) 2
B) 3
C) 5
D) 6

17. **If $4a + 3 \leq 1$, what is the greatest possible value of $6a - 2$?**

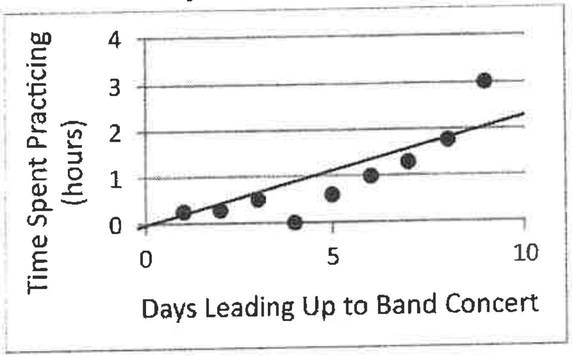
A) -5
B) -3
C) $-\frac{1}{2}$
D) 1

18. **In the equations below, i refers to the amount of money (in dollars) invested in production of x products and p refers to the profits. What is the "break even" number of products, where investments are equal to profits?**

$$i = 15{,}200 + 0.75x$$
$$p = 5.5x$$

A) 2,432
B) 3,075
C) 3,200
D) 17,600

19. The scatterplot below shows Julio's time (in hours) on the y-axis spent practicing his trumpet on the days (x-axis) leading up to a band concert. What is the difference in the amount of time he spent on the final day versus the amount predicted by the line of best fit?



A) 0.5 hours
B) 1 hour
C) 2 hours
D) 3 hours

20. The formula for finding the volume of a cone is $V = \frac{1}{3}\pi r^2 h$. Which of the following equations is correctly solved for r?

A) $r = \frac{1}{3}\pi h$
B) $r = \sqrt{\frac{3V}{\pi h}}$
C) $r = \frac{3V}{\pi h}$
D) $r = V - \frac{1}{3}\pi h$

21. Jan has six pairs of socks for every pair of shoes she owns. If she has 12 pairs of shoes and s pairs of socks, which of the following equations is true?

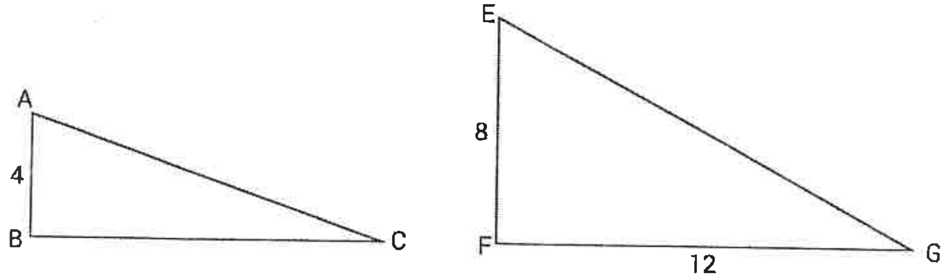
A) $12s = 6$
B) $6s = 12$
C) $\frac{s}{6} = 12$
D) $s + 6 = 12$

22. If $x^{-\frac{1}{2}} = m$, what is x?

A) m^2
B) $1 - m^2$
C) $\frac{1}{m^2}$
D) $\frac{1}{m}$

Mathematics—Module 2

1. In the figures below (not drawn to scale), right triangle ΔABC is similar to ΔEFG. What is the ratio of the area of ΔABC to the area of ΔEFG?



2. The function f is defined by a polynomial. The table below gives values for x and $f(x)$. Which of the following must be a factor of $f(x)$?

x	$f(x)$
0	4
1	2
2	0

A) x
B) $x - 1$
C) $x + 2$
D) $x - 2$

3. In the system of equations below, a is a constant and x and y are variables. For which of the following values of a will the system have no solution?

$$x + 3y = -7$$
$$ax - 2y = 6$$

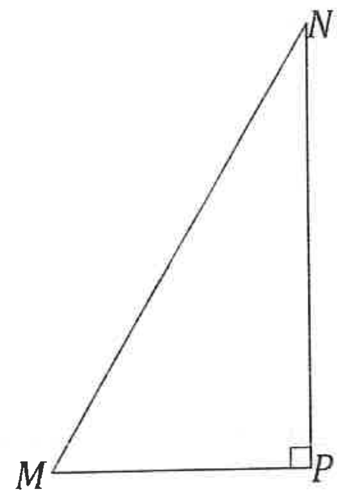
A) $-\frac{2}{3}$
B) $-\frac{1}{3}$
C) 0
D) $\frac{7}{3}$

4. What is the sum of all values of x that satisfy the following equation?

$$5x^2 - 30x - 32 = 3$$

A) -1
B) 0
C) 6
D) 7

5. In right triangle MNP below, angle M measures $a°$ and $\cos a° = \frac{3}{5}$. What is $\cos(90 - a°)$?



6. If 52% of 25 people polled in a coffee survey preferred Brand A and 48% preferred Brand B, which of the following inferences is a logical conclusion?

A) At least 52% of shoppers will prefer Brand A over Brand B.
B) The majority of shoppers will prefer Brand A over Brand B.
C) The majority of shoppers will prefer Brand B over Brand A.
D) The number of shoppers who prefer Brand A may be similar to the number of shoppers who prefer Brand B.

7. Ben bought a bag of assorted rubber balls and measured each one. Each had a diameter between 1.2 and 1.5 inches. What is a possible volume, rounded to the nearest cubic inch, of one of the balls? Use 3.14 for π.

8. How many gallon jugs of water should Julia buy to completely fill a container in the shape of a rectangular prism if the length is 8 feet, the width is 4 feet, and the height is 3 feet (1 gallon = 231 cubic inches)?

A) 5
B) 718
C) 719
D) 165,888

9. Subtract the polynomials: $(2x^2 + 3x + 2) - (x^2 + 2x - 3)$

A) $x^2 + x + 5$
B) $x^2 + x - 1$
C) $x^2 + 5x + 5$
D) $x^2 + 5x - 1$

10. The table below shows the breakdown of orders at a frozen yogurt shop. There were two flavors of yogurt (strawberry and lemon) and two sizes (small and large). If a person among those who ordered frozen yogurt is chosen at random, what is the probability that this person ordered a small serving of lemon yogurt?

	Strawberry	Lemon	Total
Small	12	9	21
Large	20	16	36
Total	32	25	57

A) $\frac{9}{25}$
B) $\frac{3}{7}$
C) $\frac{7}{19}$
D) $\frac{3}{19}$

11. If (x, y) is the solution to the following system of equations, what is the value of $x + y$?

$$3x + y = -2$$
$$y = -x$$

A) -1
B) 0
C) 1
D) 2

Refer to the following for questions 12 - 13:

Elise, Max, and Rylie are each offering lawn care services. Their prices for each service are listed in the table below. Prices are per square foot.

	Elise	Max	Rylie
Mowing	\$0.0025	\$0.002	\$0.003
Edging	\$0.0015	\$0.0015	\$0.002
Weeding	\$0.055	\$0.065	\$0.05

12. Rylie was asked to mow and edge a lawn as well as weed a 1,000 square foot garden. If she will spend \$12 on gas and other overhead costs, how large of a lawn does she need to be able to make a \$75 profit?

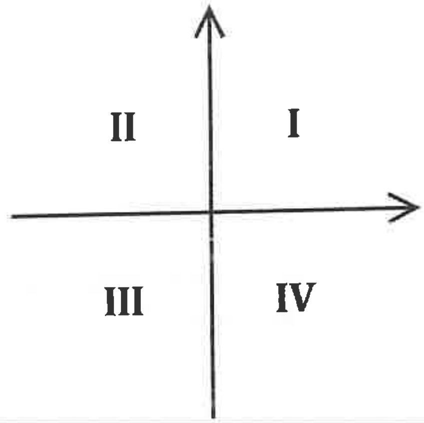
A) 2,600 ft^2
B) 5,000 ft^2
C) 6,875 ft^2
D) 7,400 ft^2

13. Ms. Lin wants to have her 7,500 square foot lawn mowed and edged. She also has a flower bed of x square feet to be weeded. Which of the following inequalities represents x if Elise's services are a better deal than Max's?

A) $x < 250$
B) $x < 375$
C) $x > 250$
D) $x > 375$

14. A number of cash prizes were awarded for the top salespeople at ABC Communications, each in the amount of $100, $200, or $400. If the prizes totaled $1,600 and there were 7 total prizes awarded, what is the greatest possible number of $100 prizes awarded?

15. The graph of the system of inequalities $y \le \frac{2}{3}x - \frac{3}{2}$ and $y \ge -2x - \frac{5}{2}$ has solutions in which quadrants on the xy-plane below?



A) Quadrants I and IV
B) Quadrants II and III
C) Quadrants I, II, and IV
D) Quadrants I, III, and IV

16. The equation below is used to solve for the distance, d (meters), that an object travels in a certain amount of time, t (seconds), with an initial velocity of v meters per second. Which of the following equations gives a in terms of d, v, and t?

$$d = vt + \frac{1}{2}at^2$$

A) $\frac{2d-2vt}{t}$
B) $\frac{2dvt}{t^2}$
C) $\frac{2d}{t^2} - \frac{2v}{t}$
D) $\frac{2d}{t^2} - 2vt$

17. The function f is defined by a polynomial. The table below gives values for x and $f(x)$. Which of the following must be a factor of $f(x)$?

x	$f(x)$
-5	14
-3	0
-1	-6

A) $x + 3$
B) $x - 3$
C) $x + 5$
D) $x - 5$

18. Simplify the following expression.

$$(3x+5)(4x-6)$$

A) $12x^2 - 38x - 30$
B) $12x^2 + 2x - 30$
C) $12x^2 - 2x - 1$
D) $12x^2 + 2x + 30$

19. If (x, y) is the solution to the following system of equations, what is the value of $x - y$?

$$4x - y = 11$$
$$x = 3y$$

A) 0
B) 1
C) 2
D) 3

20. The equation below is used to calculate the period of a pendulum (i.e., the time it takes a pendulum to make one full oscillation). If T refers to the time in seconds, L is the length of the pendulum in meters, and g is the acceleration due to gravity in meters per second squared, which of the following equations gives L in terms of T, g, and π?

$$T = 2\pi\sqrt{\frac{L}{g}}$$

A) $(T - 2\pi)^2 g$
B) $g\sqrt{\frac{T}{2\pi}}$
C) $\frac{T^2 g}{4\pi^2}$
D) $\frac{4\pi^2}{T^2 g}$

21. If $x > 0$ and $x^2 - 4x = 12$, what is the value of x?

22. In the function below, b is a constant. If $f(2) = 0$, what is the value of $f(-1)$?

$$f(x) = \frac{3}{2}x + b$$

A) $-\frac{9}{2}$
B) -3
C) $-\frac{3}{2}$
D) $\frac{3}{2}$

ID: b8a225ff

Circle A in the *xy*-plane has the equation $(x+5)^2 + (y-5)^2 = 4$. Circle B has the same center as circle A. The radius of circle B is two times the radius of circle A. The equation defining circle B in the *xy*-plane is $(x+5)^2 + (y-5)^2 = k$, where k is a constant. What is the value of k?

ID: b0a72bdc

What is the diameter of the circle in the *xy*-plane with equation $(x-5)^2+(y-3)^2=16$?

A. 4

B. 8

C. 16

D. 32

ID: 249d3f80

Point O is the center of a circle. The measure of arc RS on this circle is 100°. What is the measure, in degrees, of its associated angle ROS?

ID: ab176ad6

The equation $(x+6)^2+(y+3)^2=121$ defines a circle in the *xy*-plane. What is the radius of the circle?

ID: 3e577e4a

A circle in the *xy*-plane has its center at $(-4, -6)$. Line k is tangent to this circle at the point $(-7, -7)$. What is the slope of line k?

A. -3

B. $-\frac{1}{3}$

C. $\frac{1}{3}$

D. 3

ID: a0369739

In triangle ABC, the measure of angle B is 90° and $\overline{BD}$ is an altitude of the triangle. The length of $\overline{AB}$ is 15 and the length of $\overline{AC}$ is 23 greater than the length of $\overline{AB}$. What is the value of $\frac{BC}{BD}$?

A. $\frac{15}{38}$

B. $\frac{15}{23}$

C. $\frac{23}{15}$

D. $\frac{38}{15}$

ID: cbe8ca31

In $\triangle XYZ$, the measure of $\angle X$ is 24° and the measure of $\angle Y$ is 98°. What is the measure of $\angle Z$?

A. 58°

B. 74°

C. 122°

D. 212°

ID: 94364a79

Two nearby trees are perpendicular to the ground, which is flat. One of these trees is 10 feet tall and has a shadow that is 5 feet long. At the same time, the shadow of the other tree is 2 feet long. How tall, in feet, is the other tree?

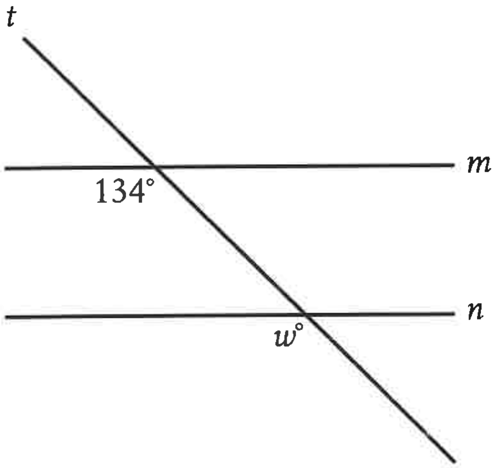
A. 3

B. 4

C. 8

D. 27

ID: c24e1bda



Note: Figure not drawn to scale.

In the figure, line m is parallel to line n. What is the value of w?

A. 13

B. 34

C. 66

D. 134

ID: 901c3215

In triangles ABC and DEF, angles B and E each have measure 27° and angles C and F each have measure 41°. Which additional piece of information is sufficient to determine whether triangle ABC is congruent to triangle DEF?

A. The measure of angle A

B. The length of side AB

C. The lengths of sides BC and EF

D. No additional information is necessary.

ID: f7dbde16

In triangles LMN and RST, angles L and R each have measure 60°, $LN = 10$, and $RT = 30$. Which additional piece of information is sufficient to prove that triangle LMN is similar to triangle RST?

A. $MN = 7$ and $ST = 7$

B. $MN = 7$ and $ST = 21$

C. The measures of angles M and S are 70° and 60°, respectively.

D. The measures of angles M and T are 70° and 50°, respectively.

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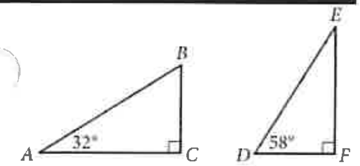

Triangles *ABC* and *DEF* are shown above. Which of the following is equal to the ratio $\frac{BC}{AB}$?

A. $\frac{DE}{DF}$

B. $\frac{DF}{DE}$

C. $\frac{DF}{EF}$

D. $\frac{EF}{DE}$

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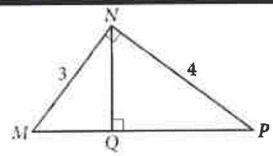


In the figure above, what is the length of $\overline{NQ}$?

A. 2.2

B. 2.3

C. 2.4

D. 2.5

ID: 3b225698

Triangle XYZ is similar to triangle RST such that X, Y, and Z correspond to R, S, and T, respectively. The measure of $\angle Z$ is 20° and $2XY = RS$. What is the measure of $\angle T$?

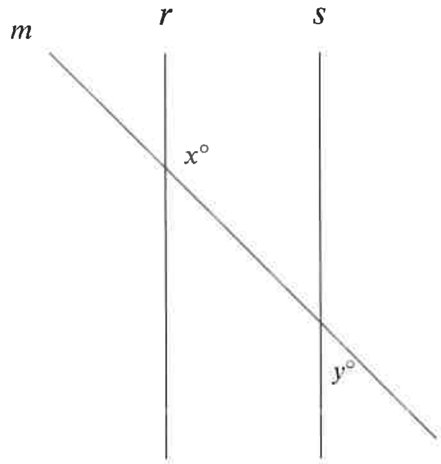
A. 2°

B. 10°

C. 20°

D. 40°

ID: a4c05a1b



Note: Figure not drawn to scale.

In the figure shown, lines r and s are parallel, and line m intersects both lines. If $y < 65$, which of the following must be true?

A. $x < 115$

B. $x > 115$

C. $x + y < 180$

D. $x + y > 180$

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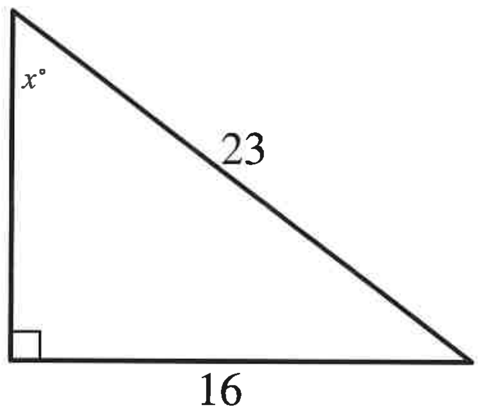

Note: Figure not drawn to scale.

In the triangle shown, what is the value of $\sin x^{\circ}$?

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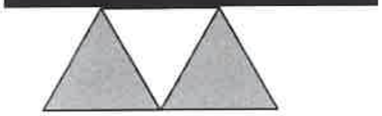

A graphic designer is creating a logo for a company. The logo is shown in the figure above. The logo is in the shape of a trapezoid and consists of three congruent equilateral triangles. If the perimeter of the logo is 20 centimeters, what is the combined area of the shaded regions, in square centimeters, of the logo?

A. $2\sqrt{3}$

B. $4\sqrt{3}$

C. $8\sqrt{3}$

D. 16

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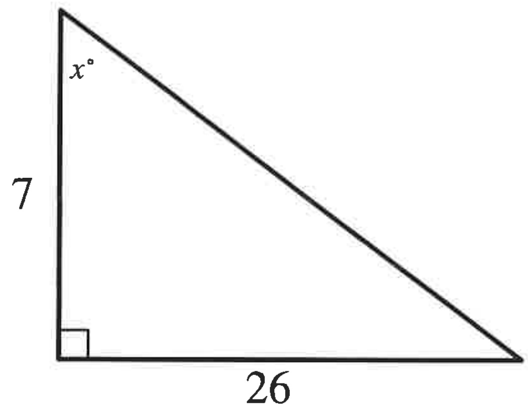


Note: Figure not drawn to scale.

In the triangle shown, what is the value of $\tan x°$?

A. $\frac{1}{26}$

B. $\frac{19}{26}$

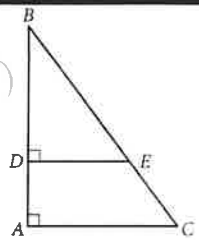
C. $\frac{26}{7}$

D. $\frac{33}{7}$

ID: a4bd60a3

The perimeter of an equilateral triangle is 624 centimeters. The height of this triangle is $k\sqrt{3}$ centimeters, where k is a constant. What is the value of k?

ID: 55bb437a



In the figure above, $\tan B = \frac{3}{4}$. If $BC = 15$ and $DA = 4$, what is the length of $\overline{DE}$?

Mathematics—Module 1

1. The ratio of new car sales to used car sales at the car lot is 3 : 5. If the total car sales were $287,400 last month, what was the total of the used car sales?

2. A company has been asked to design a building for an athletic event. The building is in the shape of a square pyramid. The pyramid has a height of 481 feet, and the length of each side of the base is 756 feet. What is the approximate volume of the pyramid?

A) 1.21×10^5 ft^3
B) 4.85×10^5 ft^3
C) 9.16×10^7 ft^3
D) 2.75×10^8 ft^3

3. Simplify the following expression: $(2x^2 + 3x + 2) - (x^2 + 2x - 3)$

A) $x^2 + x + 5$
B) $x^2 + x - 1$
C) $x^2 + 5x + 5$
D) $x^2 + 5x - 1$

4. Simplify the following: $\frac{x^2}{y^2} + \frac{x}{y^3}$

A) $\frac{x^3+x}{y^3}$
B) $\frac{x^2+xy}{y^3}$
C) $\frac{x^2y+xy}{y^3}$
D) $\frac{x^2y+x}{y^3}$

5. The graph of $y = -x^2 + 5x$ intersects the graph of $y = 2x$ at $(0, 0)$ and (a, b). What is the value of b?

6. Max reads three books averaging 360 pages. Lucy reads five books averaging 200 pages. What is the average length of all the books that Max and Lucy read?

A) 212 pages
B) 232 pages
C) 260 pages
D) 295 pages

7. Solve: $7x^2 + 6x = -2$.

A) $x = \frac{-3\pm\sqrt{23}}{7}$
B) $x = \pm i\sqrt{5}$
C) $x = \pm\frac{2i\sqrt{2}}{7}$
D) $x = \frac{-3\pm i\sqrt{5}}{7}$

8. Ride Service A charges a flat rate of $10 for the first 10 miles, plus 25 cents per mile for anything over 10 miles. Ride Service B charges 40 cents per mile. Both services charge the same for a trip that is how long?

A) 40 miles
B) 45 miles
C) 50 miles
D) 55 miles

9. What is the perimeter of a 45-45-90 triangle if the hypotenuse is 4 inches?

A) 4 inches
B) 8 inches
C) $4 + 4\sqrt{2}$ inches
D) $4 + 2\sqrt{2}$ inches

10. Which of the following is equivalent to $x^2 + 3 > 2x + 2$?

A) $x < -1$
B) $x \neq 1$
C) $x > 1$
D) $x < -1$ or $x > 1$

11. In the system of equations below, n is a constant and x and y are variables. For which of the following values of n will the system have no solution?

$$3x - y = 2$$
$$nx + 3y = -5$$

A) $-\frac{1}{3}$
B) 3
C) $-\frac{5}{3}$
D) -9

12. Gillian is deciding between two data plans for her cellphone. Plan A provides 2.5 GB of data for a flat rate of $20/month and charges $15 per GB for any extra use. Plan B provides unlimited data for $50/month. What amount of data would Gillian have to use in a month for both plans to cost the same amount?

A) 2 GB
B) 3.5 GB
C) 3.75 GB
D) 4.5 GB

13. Every person attending a meeting hands out a business card to every other person at the meeting. If a total of 30 cards are handed out, how many people are at the meeting?

A) 5 people
B) 6 people
C) 10 people
D) 15 people

14. Which of the following is a solution to the inequality $4x - 12 < 4$?

A) 7
B) 6
C) 4
D) 3

15. The table below shows the breakdown of soup orders at a restaurant. There were two kinds of soup (chicken and veggie) and two sizes (cup and bowl). If a person among those who ordered soup is chosen at random, what is the probability that this person ordered a cup of veggie soup?

	Chicken	Veggie	Total
Cup	7	8	15
Bowl	15	12	27
Total	22	20	42

A) $\frac{8}{15}$
B) $\frac{1}{6}$
C) $\frac{2}{5}$
D) $\frac{4}{21}$

16. Jamal plants a white petunia for every three red petunias in the flowerbed. If he plants 8 white petunias and r red petunias, which of the following equations is true?

A) $3r = 8$
B) $8r = 3$
C) $\frac{r}{3} = 8$
D) $r + 3 = 8$

17. Solve $\sqrt{2x} - 3 = \sqrt{2x - 15}$.

A) $x = 0$
B) $x = 4$
C) $x = 8$
D) $x = 10$

18. What is the solution to the equation: $4\sqrt{x} + 8 = 24$?

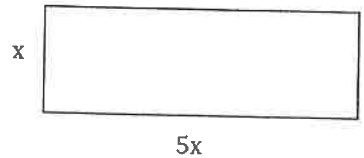
A) $x = 2$
B) $x = 4$
C) $x = 12$
D) $x = 16$

19. What is the sum of all values of x that satisfy the following equation?

$$3x^2 - 3x - 34 = 2$$

A) -3
B) 0
C) 1
D) 4

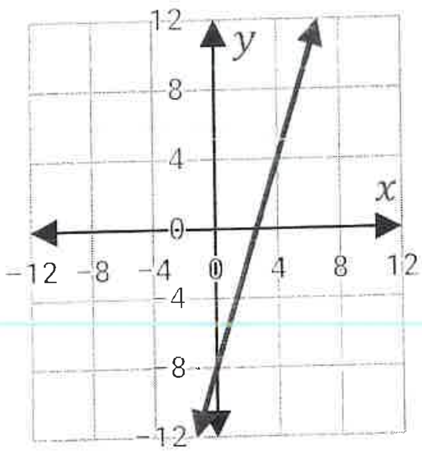
20. The rectangle below has an area of 245 inches^2. What is the length of the longest side of the rectangle?



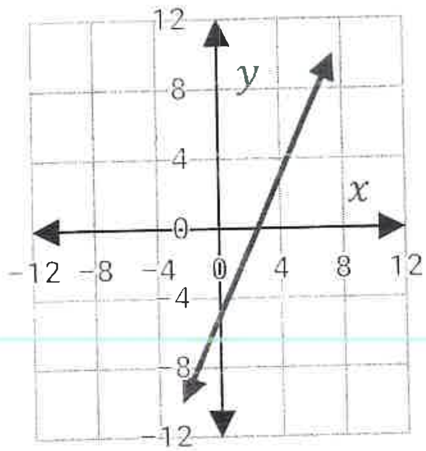
A) 25 inches
B) 30 inches
C) 35 inches
D) 40 inches

21. The variables x and y have a linear relationship. The table below shows a few sample values. Which of the following graphs correctly represents the linear equation relating x and y?

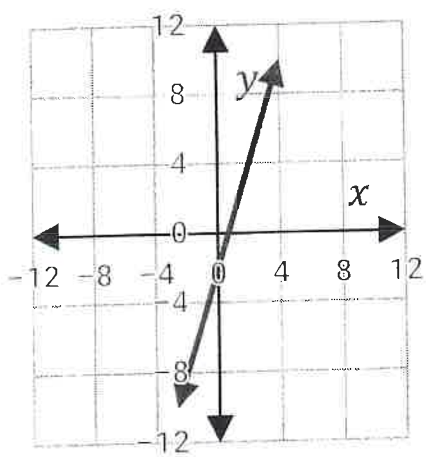
x	y
−2	−11
−1	−8
0	−5
1	−2
2	1



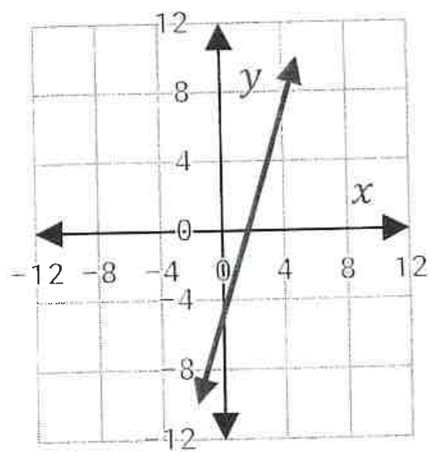
A.



B.



C.



D.

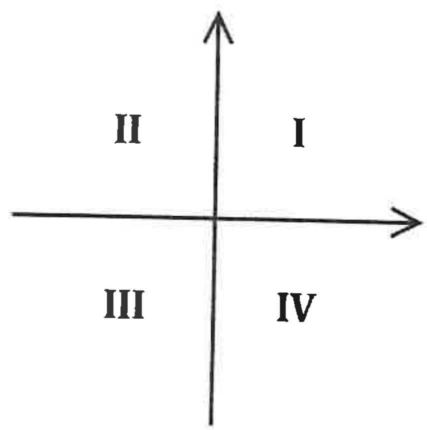
22. Roxana walks x meters west and $x + 20$ meters south to get to her friend's house. On a neighborhood map which has a scale of 1 cm: 10 m, the direct distance between Roxana's house and her friend's house is 10 cm. How far did Roxana walk to her friend's house?

Mathematics—Module 2

1. If $\frac{1}{\sqrt[3]{x}} = a$, what is x?

A) a^3
B) $1 - a^3$
C) $\frac{1}{a^2}$
D) $\frac{1}{a^3}$

2. The graph of the system of inequalities $y \le \frac{1}{4}x - 2$ and $y > 3x - \frac{7}{5}$ has solutions in which quadrants on the xy-plane below?



A) Quadrant III only
B) Quadrants II and III
C) Quadrants III and IV
D) Quadrants II, III, and IV

Refer to the following for questions 3 - 4:

Nurseries A, B, and C offer various plants for sale, as well as landscaping services to plant the trees and bushes. Prices are listed in the table below.

	A	B	C
Trees	\$25	\$30	\$20
Bushes	\$15	\$20	\$15
Landscaping (per hour)	\$45	\$55	\$50

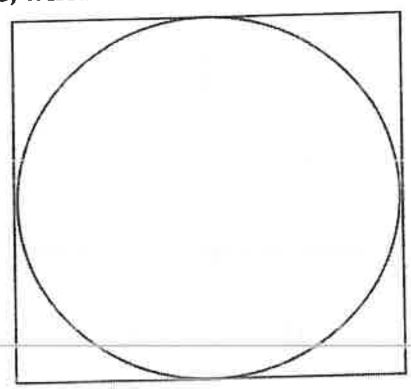
3. Nursery B was hired to plant 100 trees and 100 bushes. If their overhead cost of the plants, equipment, and personnel is \$5,500, how many hours will they have to work to make a profit of \$820?

A) 15 hours
B) 21.5 hours
C) 24 hours
D) 32.5 hours

4. Isabella plans to hire one of the companies to plant 10 trees and 8 bushes. This project will take x hours. Which of the following inequalities represents x if Nursery A offers a better deal than Nursery C?

A) $x < 5$
B) $x < 10$
C) $x > 5$
D) $x > 10$

5. A circle is inscribed within a square, as shown. What is the difference between the area of the square and that of the circle, where r is the radius of the circle?



A) 2π
B) $\frac{4}{3}\pi r^3$
C) $r^2(4-\pi)$
D) $2\pi r$

6. The equation below shows Emma's savings plan. She set aside an initial lump sum and adds to it on a monthly basis. If i is the total investment in cents and m is the number of months since she began, how much does she save each month?

$$i = 50,000 + 4,500m$$

A) $45
B) $500
C) $4,500
D) $50,000

7. Solve for n in the equation: $4n - p = 3r$

A) $\frac{3r}{4} - p$
B) $p + 3r$
C) $p - 3r$
D) $\frac{3r}{4} + \frac{p}{4}$

8. If the solution for the system of equations below is (x, y), what is the value of $2x^2 - y$?

$$3x - 2y = 0$$
$$-2x + 4y = -8$$

A) -3
B) -2
C) 5
D) 11

9. What is the y-coordinate of the center of the circle defined in the equation below?

$$x^2 + y^2 - y - 6x = -\frac{21}{4}$$

10. Riley has several \$5 bills, \$10 bills, and \$20 bills. If she has a total of 9 bills that add up to \$80, what is the greatest number of \$5 bills she could have?

11. Which of the following represents the factored form of the expression $x^2 - 3x - 40$?

A) $(x - 8)(x + 5)$
B) $(x - 7)(x + 4)$
C) $(x + 10)(x - 4)$
D) $(x + 6)(x - 9)$

12. A new business is calculating the amount of products that must sell to break even (so that investments equal profits). If d dollars are invested to produce a products that can be sold for p profit, how much investment is required, according to the equations below?

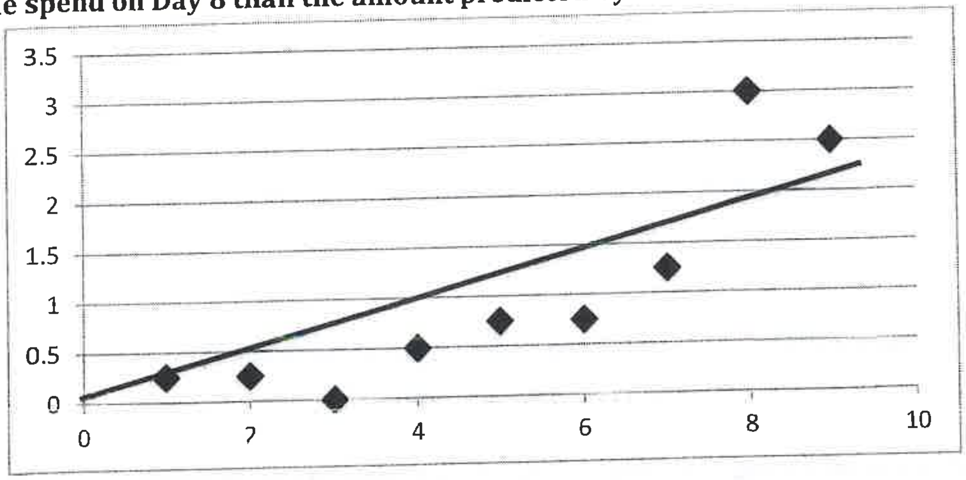
$$d = 9{,}660 + 1.2a$$
$$p = 4a$$

A) \$3,450
B) \$4,140
C) \$11,260
D) \$13,800

13. If 3 times the square of a positive number is 48, what is the result when twice the number is subtracted from 15?

A) -7
B) 4
C) 7
D) 11

14. The scatterplot below shows Zac's time (in hours) on the y-axis spent working on his science project on the days (x-axis) leading up to the due date. Approximately how much longer did he spend on Day 8 than the amount predicted by the line of best fit?



A) 0.5 hours
B) 1 hour
C) 2 hours
D) 3 hours

15. If $x > 0$ and $x^2 - 7 = 9$, what is the value of x?

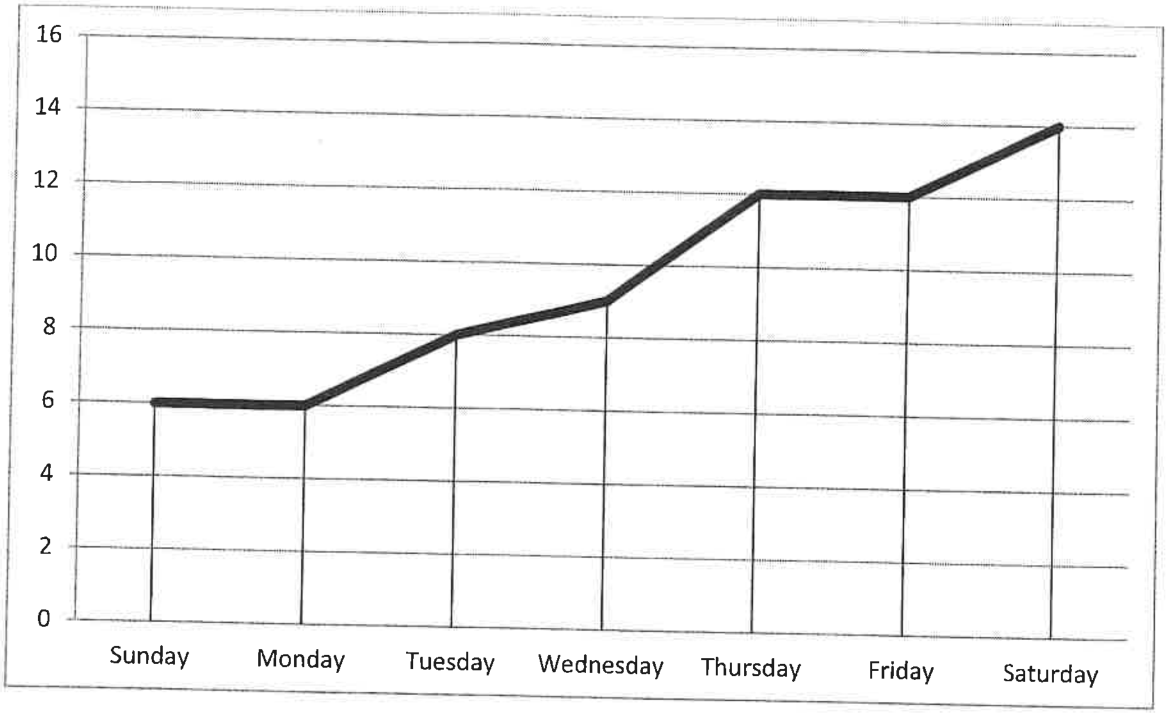
16. If $\frac{3}{7}q = -6$, what is the value of q?

A) -14
B) $-\frac{18}{7}$
C) $-\frac{45}{7}$
D) -42

17. What is the expanded form of $(x + 6)(x - 6)$?

A) $x^2 - 12x - 36$
B) $x^2 + 12x - 36$
C) $x^2 + 12x + 36$
D) $x^2 - 36$

Refer to the following for questions 18 - 19:



Every time Miguel gets quarters in change, he puts them in a jar. The chart above shows how many quarters he has in the jar at the end of each day during a particular week.

18. How many quarters did Miguel add on Monday?

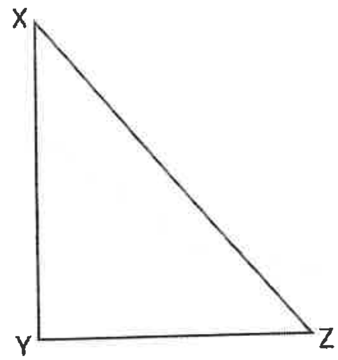
A) 0
B) 1
C) 3
D) 6

19. Which day did Miguel add the most quarters to the jar?

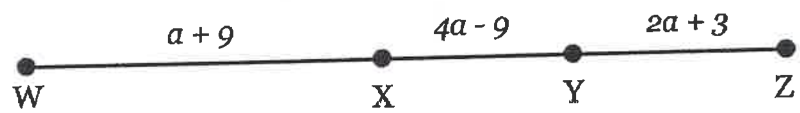
A) Monday
B) Tuesday
C) Thursday
D) Saturday

20. The half-life of caffeine in the human body is 5 hours. If Rafe drinks a cup of coffee at 7:00 a.m., what percentage of the caffeine is still in his system at 2:00 p.m.? Round your answer to the nearest whole percent.

21. In right triangle ΔXYZ below, angle Z measures $z°$ and $\sin z° = \frac{4}{5}$. What is $\cos(90 - z)°$?



22. On line WZ below, $WX = XZ$. What is the length of WY? (Note: figure not drawn to scale)



ID: c178d4da

Value	Data set A frequency	Data set B frequency
30	2	9
34	4	7
38	5	5
42	7	4
46	9	2

Data set A and data set B each consist of 27 values. The table shows the frequencies of the values for each data set. Which of the following statements best compares the means of the two data sets?

A. The mean of data set A is greater than the mean of data set B.

B. The mean of data set A is less than the mean of data set B.

C. The mean of data set A is equal to the mean of data set B.

D. There is not enough information to compare the means of the data sets.

ID: 457d2f2c

A data set of 27 different numbers has a mean of 33 and a median of 33. A new data set is created by adding 7 to each number in the original data set that is greater than the median and subtracting 7 from each number in the original data set that is less than the median. Which of the following measures does NOT have the same value in both the original and new data sets?

A. Median

B. Mean

C. Sum of the numbers

D. Standard deviation

ID: 35bec412

73, 74, 75, 77, 79, 82, 84, 85, 91

What is the median of the data shown?

ID: 07f2829b

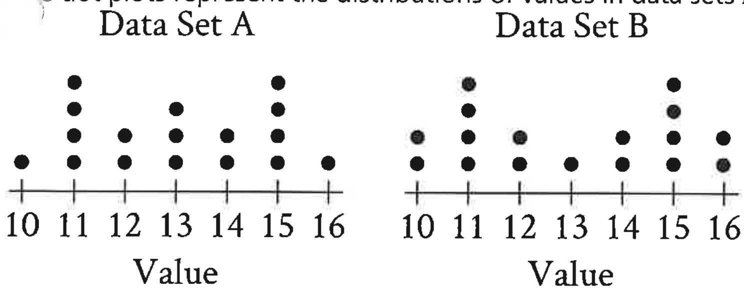
International Tourist Arrivals, in millions

Country	2012	2013
France	83.0	84.7
United States	66.7	69.8
Spain	57.5	60.7
China	57.7	55.7
Italy	46.4	47.7
Turkey	35.7	37.8
Germany	30.4	31.5
United Kingdom	26.3	32.2
Russia	24.7	28.4

The table above shows the number of international tourist arrivals, rounded to the nearest tenth of a million, to the top nine tourist destinations in both 2012 and 2013. Based on the information given in the table, how much greater, in millions, was the median number of international tourist arrivals to the top nine tourist destinations in 2013 than the median number in 2012, to the nearest tenth of a million?

ID: d65b9a87

The dot plots represent the distributions of values in data sets A and B.



Which of the following statements must be true?

I. The median of data set A is equal to the median of data set B.

II. The standard deviation of data set A is equal to the standard deviation of data set B.

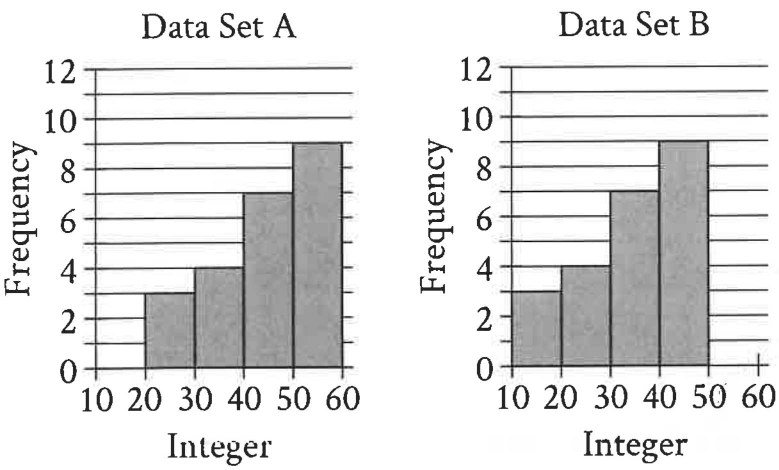
A. I only

B. II only

C. I and II

D. Neither I nor II

ID: f8a322d9



Two data sets of 23 integers each are summarized in the histograms shown. For each of the histograms, the first interval represents the frequency of integers greater than or equal to 10, but less than 20. The second interval represents the frequency of integers greater than or equal to 20, but less than 30, and so on. What is the smallest possible difference between the mean of data set A and the mean of data set B?

A. 0

B. 1

C. 10

D. 23

ID: 12dae628

$$2, 9, 14, 23, 32$$

What is the mean of the data shown?

A. 14

B. 16

C. 17

D. 32

ID: 1142af44

Value	Frequency
1	a
2	$2a$
3	$3a$
4	$2a$
5	a

The frequency distribution above summarizes a set of data, where a is a positive integer. How much greater is the mean of the set of data than the median?

A. 0

B. 1

C. 2

D. 3

ID: 651d83bb

Two different teams consisting of 10 members each ran in a race. Each member's completion time of the race was recorded. The mean of the completion times for each team was calculated and is shown below.

Team A: 3.41 minutes

Team B: 3.79 minutes

Which of the following MUST be true?

1. Every member of team A completed the race in less time than any member of team B.
2. The median time it took the members of team B to complete the race is greater than the median time it took the members of team A to complete the race.
3. There is at least one member of team B who took more time to complete the race than some member of team A.

A. III only

B. I and III only

C. II and III only

D. I, II, and III

ID: 1e8ccffd

The mean score of 8 players in a basketball game was 14.5 points. If the highest individual score is removed, the mean score of the remaining 7 players becomes 12 points. What was the highest score?

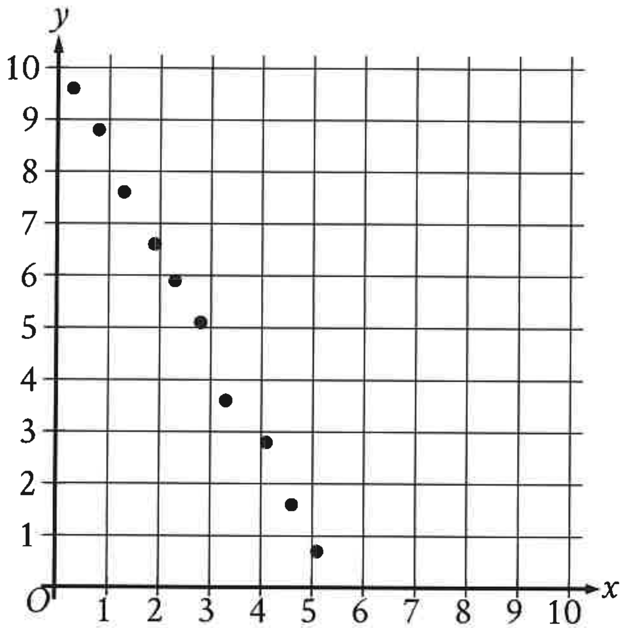
A. 20

B. 24

C. 32

D. 36

ID: 5f3ee607



Which of the following equations is the most appropriate linear model for the data shown in the scatterplot?

A. $y = -1.9x - 10.1$

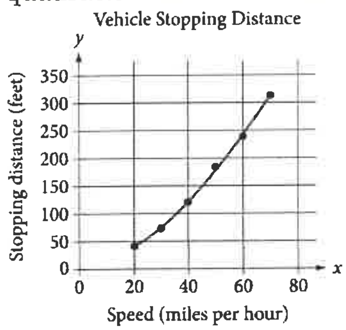
B. $y = -1.9x + 10.1$

C. $y = 1.9x - 10.1$

D. $y = 1.9x + 10.1$

ID: 5c24c861

A study was done to determine a new car's stopping distance when it was traveling at different speeds. The study was done on a dry road with good surface conditions. The results are shown below, along with the graph of a quadratic function that models the data.



According to the model, which of the following is the best estimate for the stopping distance, in feet, if the vehicle was traveling 55 miles per hour?

A. 25

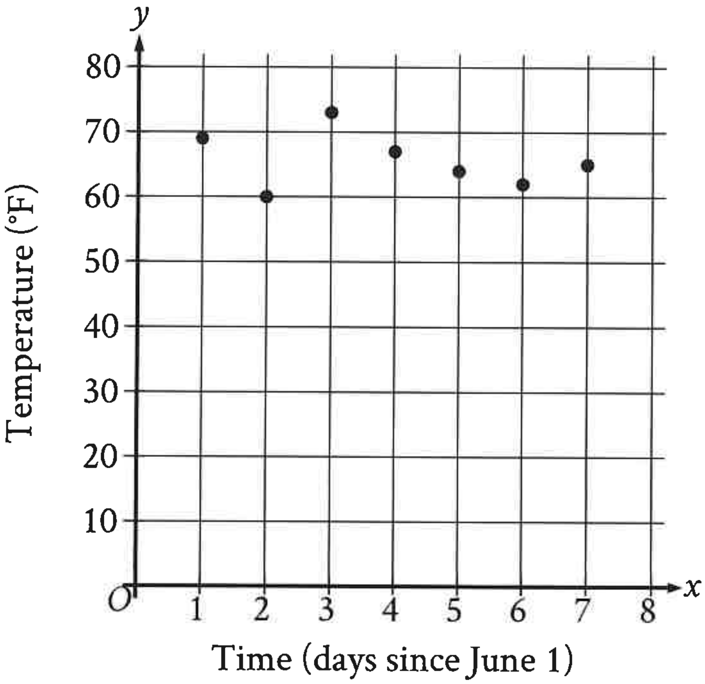
B. 30

C. 210

D. 250

ID: d112bc9d

The scatterplot shows the temperature y, in $^\circ\mathrm{F}$, recorded by a meteorologist at various times x, in days since June 1.



During which of the following time periods did the greatest increase in recorded temperature take place?

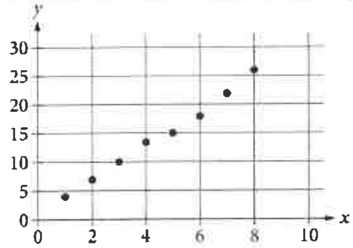
A. From $x = 6$ to $x = 7$

B. From $x = 5$ to $x = 6$

C. From $x = 2$ to $x = 3$

D. From $x = 1$ to $x = 2$

ID: 9eb896c5



Which of the following could be the equation for a line of best fit for the data shown in the scatterplot above?

A. $y = 3x + 0.8$

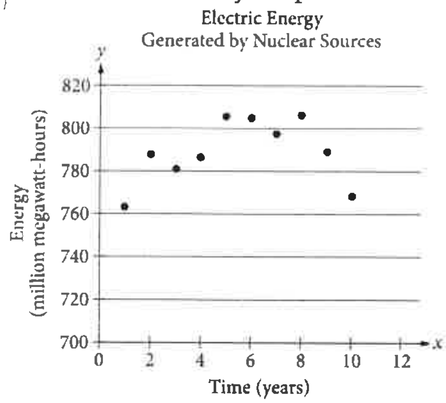
B. $y = 0.8x + 3$

C. $y = -0.8x + 3$

D. $y = -3x + 0.8$

ID: e821a26d

The scatterplot below shows the amount of electric energy generated, in millions of megawatt-hours, by nuclear sources over a 10-year period.



Of the following equations, which best models the data in the scatterplot?

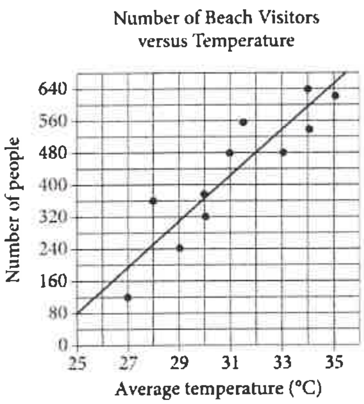
A. $y = 1.674x^2 + 19.76x - 745.73$

B. $y = -1.674x^2 - 19.76x - 745.73$

C. $y = 1.674x^2 + 19.76x + 745.73$

D. $y = -1.674x^2 + 19.76x + 745.73$

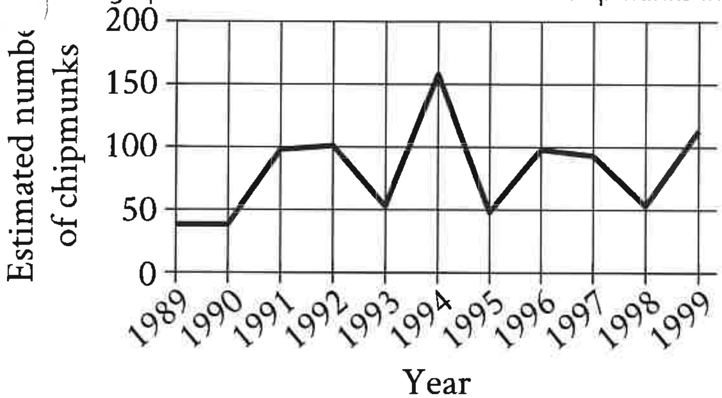
ID: 8156d446



Each dot in the scatterplot above represents the temperature and the number of people who visited a beach in Lagos, Nigeria, on one of eleven different days. The line of best fit for the data is also shown. According to the line of best fit, what is the number of people, rounded to the nearest 10, predicted to visit this beach on a day with an average temperature of 32°C?

ID: 2e511919

The line graph shows the estimated number of chipmunks in a state park on April 1 of each year from 1989 to 1999.



Based on the line graph, in which year was the estimated number of chipmunks in the state park the greatest?

A. 1989

B. 1994

C. 1995

D. 1998

ID: ab7740a8

In which of the following tables is the relationship between the values of x and their corresponding y-values nonlinear?

A.

x	1	2	3	4
y	8	11	14	17

B.

x	1	2	3	4
y	4	8	12	16

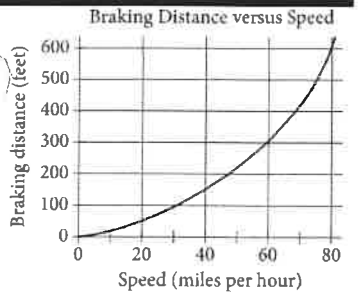
C.

x	1	2	3	4
y	8	13	18	23

D.

x	1	2	3	4
y	6	12	24	48

ID: d6121490



The graph above shows the relationship between the speed of a particular car, in miles per hour, and its corresponding braking distance, in feet. Approximately how many feet greater will the car's braking distance be when the car is traveling at 50 miles per hour than when the car is traveling at 30 miles per hour?

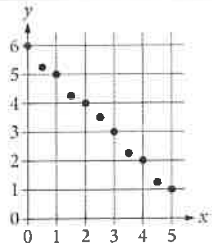
A. 75

B. 125

C. 175

D. 250

ID: 9296553d



Which of the following could be an equation for a line of best fit for the data in the scatterplot?

A. $y = -x + 6$

B. $y = -x - 6$

C. $y = 6x + 1$

D. $y = 6x - 1$

ID: be35c117

A wind turbine completes 900 revolutions in 50 minutes. At this rate, how many revolutions per minute does this turbine complete?

A. 18

B. 850

C. 950

D. $1{,}400$

ID: 3f236a64

x	y
1	4
3	12
5	20
40	k

In the table above, the ratio of y to x for each ordered pair is constant. What is the value of k ?

A. 28

B. 36

C. 80

D. 160

ID: c9fb15ad

Species of tree	Growth factor
Red maple	4.5
River birch	3.5
Cottonwood	2.0
Black walnut	4.5
White birch	5.0
American elm	4.0
Pin oak	3.0
Shagbark hickory	7.5

One method of calculating the approximate age, in years, of a tree of a particular species is to multiply the diameter of the tree, in inches, by a constant called the growth factor for that species. The table above gives the growth factors for eight species of trees. If a white birch tree and a pin oak tree each now have a diameter of 1 foot, which of the following will be closest to the difference, in inches, of their diameters 10 years from now? (1 foot = 12 inches)

A. 1.0

B. 1.2

C. 1.3

D. 1.4

ID: 6310adbc

The ratio of t to u is 1 to 2, and $t=10$.

What is the value of u ?

A. 2

B. 5

C. 10

D. 20

ID: 2d16d62c

A special camera is used for underwater ocean research. When the camera is at a depth of 58 fathoms, what is the camera's depth in feet? ($1 \text{ fathom} = 6 \text{ feet}$)

ID: 69f6717f

A sample of oak has a density of 807 kilograms per cubic meter. The sample is in the shape of a cube, where each edge has a length of 0.90 meters. To the nearest whole number, what is the mass, in kilograms, of this sample?

A. 588

B. 726

C. 897

D. $1,107$

ID: fea831fc

On April 18, 1775, Paul Revere set off on his midnight ride from Charlestown to Lexington. If he had ridden straight to Lexington without stopping, he would have traveled 11 miles in 26 minutes. In such a ride, what would the average speed of his horse have been, to the nearest tenth of a mile per hour?

ID: aeeaec96

How many yards are equivalent to 612 inches? $(1 \text{ yard} = 36 \text{ inches})$

A. 0.059

B. 17

C. 576

D. $22{,}032$

ID: 181cc4d6

Rectangle A has length 15 and width w. Rectangle B has length 20 and the same length-to-width ratio as rectangle A. What is the width of rectangle B in terms of w ?

A. $\frac{4}{3}w$

B. $w+5$

C. $\frac{3}{4}w$

D. $w-5$

ID: e9841407

Shaquan has 7 red cards and 28 blue cards. What is the ratio of red cards to blue cards that Shaquan has?

A. 1 to 4

B. 4 to 1

C. 1 to 7

D. 7 to 1

ID: d693f563

Last year, Cedric had 35 plants in his garden. This year, the number of plants in Cedric's garden is 60% greater than the number of plants in his garden last year. How many plants does Cedric have in his garden this year?

ID: bb7c8186

What is 23% of 100?

A. 23

B. 46

C. 77

D. 123

ID: 7cab9fe1

Which expression represents the result of increasing a positive quantity w by 43%?

A. $1.43w$

B. $0.57w$

C. $43w$

D. $0.43w$

ID: 121dc44f

The population of City A increased by 7% from 2015 to 2016. If the 2016 population is k times the 2015 population, what is the value of k?

A. 0.07

B. 0.7

C. 1.07

D. 1.7

ID: 29c177e6

What is 10% of 470?

A. 37

B. 47

C. 423

D. 460

ID: 8a714fa1

Which of the following represents the result of increasing the quantity x by 9%, where $x > 0$?

A. $1.09x$

B. $0.09x$

C. $x + 9$

D. $x + 0.09$

ID: 8e2e424e

The number k is 36% greater than 50. If k is the product of 50 and r, what is the value of r ?

A. 36

B. 3.6

C. 1.36

D. 0.36

ID: 67c0200a

The number a is 70% less than the positive number b. The number c is 80% greater than a. The number c is how many tim
b?

ID: 40e7a1a9

210 is $p\%$ greater than 30. What is the value of p?

ID: 709e04de

The value of z is 1.13 times 100. The value of z is what percent greater than 100?

A. 11.3

B. 13

C. 130

D. 213

ID: 90eed2e5

A city has 50 city council members. A reporter polled a random sample of 20 city council members and found that 6 of those polled supported a specific bill. Based on the sample, which of the following is the best estimate of the number of city council members in the city who support the bill?

A. 6

B. 9

C. 15

D. 30

ID: 53d97af5

A study was done on the weights of different types of fish in a pond. A random sample of fish were caught and marked in order to ensure that none were weighed more than once. The sample contained 150 largemouth bass, of which 30% weighed more than 2 pounds. Which of the following conclusions is best supported by the sample data?

A. The majority of all fish in the pond weigh less than 2 pounds.

B. The average weight of all fish in the pond is approximately 2 pounds.

C. Approximately 30% of all fish in the pond weigh more than 2 pounds.

D. Approximately 30% of all largemouth bass in the pond weigh more than 2 pounds.

ID: f8f79e11

A park ranger asked a random sample of visitors how far they hiked during their visit. Based on the responses, the estimated mean was found to be 4.5 miles, with an associated margin of error of 0.5 miles. Which of the following is the best conclusion from these data?

A. It is likely that all visitors hiked between 4 and 5 miles.

B. It is likely that most visitors hiked exactly 4.5 miles.

C. It is not possible that any visitor hiked less than 3 miles.

D. It is plausible that the mean distance hiked for all visitors is between 4 and 5 miles.

ID: e1ad3d41

Coat color	Eye color		
	Deep blue	Light brown	Total
Cream-tortoiseshell	16	16	32
Chocolate	12	4	16
Total	28	20	48

The data on the coat color and eye color for 48 Himalayan kittens available for adoption were collected and summarized in the table above. What fraction of the chocolate-colored kittens has deep blue eyes?

A. $\frac{12}{48}$

B. $\frac{12}{28}$

C. $\frac{16}{32}$

D. $\frac{12}{16}$

ID: 46545dd6

Number of High School Students Who Completed Summer Internships

High school	Year				
	2008	**2009**	**2010**	**2011**	**2012**
Foothill	87	80	75	76	70
Valley	44	54	65	76	82
Total	131	134	140	152	152

The table above shows the number of students from two different high schools who completed summer internships in each of five years. No student attended both schools. Of the students who completed a summer internship in 2010, which of the following represents the fraction of students who were from Valley High School?

A. $\frac{10}{140}$

B. $\frac{65}{140}$

C. $\frac{75}{140}$

D. $\frac{65}{75}$

ID: 16cea46c

Voice type	Number of singers
Countertenor	4
Tenor	6
Baritone	10
Bass	5

A total of 25 men registered for singing lessons. The frequency table shows how many of these singers have certain voice types. If one of these singers is selected at random, what is the probability he is a baritone?

A. 0.10

B. 0.40

C. 0.60

D. 0.67

ID: b680e76d

A survey taken by 1,000 students at a school asked whether they played school sports. The table below summarizes all 1,000 responses from the students surveyed.

	Males	Females
Play a school sport	312	220
Do not play a school sport	?	216

How many of the males surveyed responded that they do not play a school sport?

A. 109

B. 252

C. 468

D. 688

ID: d4413871

	Blood type			
Rhesus factor	A	B	AB	O
$+$	33	9	3	37
$-$	7	2	1	x

Human blood can be classified into four common blood types—A, B, AB, and O. It is also characterized by the presence $(+)$ or absence $(-)$ of the rhesus factor. The table above shows the distribution of blood type and rhesus factor for a group of people. If one of these people who is rhesus negative $(-)$ is chosen at random, the probability that the person has blood type B is $\frac{1}{9}$. What is the value of x ?

ID: b4f5a7ca

A survey was conducted using a sample of history professors selected at random from the California State Universities. The professors surveyed were asked to name the publishers of their current texts. What is the largest population to which the results of the survey can be generalized?

A. All professors in the United States

B. All history professors in the United States

C. All history professors at all California State Universities

D. All professors at all California State Universities

ID: 9bf4c545

The members of a city council wanted to assess the opinions of all city residents about converting an open field into a dog park. The council surveyed a sample of 500 city residents who own dogs. The survey showed that the majority of those sampled were in favor of the dog park. Which of the following is true about the city council's survey?

A. It shows that the majority of city residents are in favor of the dog park.

B. The survey sample should have included more residents who are dog owners.

C. The survey sample should have consisted entirely of residents who do not own dogs.

D. The survey sample is biased because it is not representative of all city residents.

Math

35 MINUTES, 22 QUESTIONS

DIRECTIONS

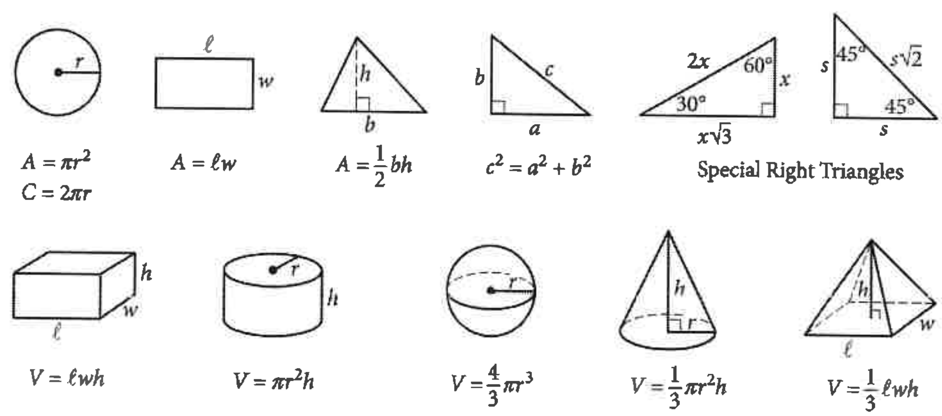
The questions in this section address a number of important math skills. Use of a calculator is permitted for all questions.

NOTES

Unless otherwise indicated:

- All variables and expressions represent real numbers.
- Figures provided are drawn to scale.
- All figures lie in a plane.
- The domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

REFERENCE



The number of degrees of arc in a circle is 360.
The number of radians of arc in a circle is 2π.
The sum of the measures in degrees of the angles of a triangle is 180.

2 Module 1 2

For multiple-choice questions, solve each problem, choose the correct answer from the choices provided, and then circle your answer in this book. Circle only one answer for each question. If you change your mind, completely erase the circle. You will not get credit for questions with more than one answer circled, or for questions with no answers circled.

For student-produced response questions, solve each problem and write your answer next to or under the question in the test book as described below.

- Once you've written your answer, circle it clearly. You will not receive credit for anything written outside the circle, or for any questions with more than one circled answer.
- If you find more than one correct answer, write and circle only one answer.
- Your answer can be up to 5 characters for a positive answer and up to 6 characters (including the negative sign) for a negative answer, but no more.
- If your answer is a fraction that is too long (over 5 characters for positive, 6 characters for negative), write the decimal equivalent.
- If your answer is a decimal that is too long (over 5 characters for positive, 6 characters for negative), truncate it or round at the fourth digit.
- If your answer is a mixed number (such as $3\frac{1}{2}$), write it as an improper fraction (7/2) or its decimal equivalent (3.5).
- Don't include symbols such as a percent sign, comma, or dollar sign in your circled answer.

2 Module 1 2

1

$$3(y - \frac{5}{3}) = 10$$

What value of y is the solution to the given equation?

A) 5

B) 3

C) $\frac{5}{3}$

D) 10

2

$$|x - 5| \leq 7$$

Which of the following inequalities is equivalent to the inequality above?

A) $-7 \leq x \leq 7$

B) $-2 \leq x \leq 12$

C) $x \leq 2$ or $x \geq 12$

D) $x \leq -12$ or $x \geq 2$

3

If $f(x) = -8x + 9$ and $f(k) = -15$, what is the value of k?

4

The length of a swimming pool is 4 times its width. If the width is 25 meters, what is the area of the swimming pool, in square meters?

A) 625

B) 1875

C) 2500

D) 1250

5

Let the function f be defined by $f(x) = (x^6 - 8x^9 + x^2) + (x^2 + 8x^9 - x^6)$. What is the value of $f(8)$?

A) 128

B) 4096

C) 64

D) 256

6

In the xy-plane, line l passes through the point (5,1) and is parallel to the line with the equation $y = -\frac{2}{5}x + \frac{7}{2}$. What is the equation of line l?

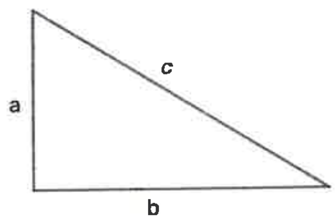
A) $y = \frac{2}{5}x + 3$

B) $y = -\frac{2}{5}x - 3$

C) $y = -\frac{2}{5}x + 3$

D) $y = \frac{5}{2}x + 3$

7



The length of three sides of the triangle are a, b, and c. Which of the following is/are true?

I. $a + b > c$

II. $|a - c| < b$

III. $a^2 + b^2 = c^2$

A) I only

B) II only

C) I and II and III

D) I and III

8

In the $x\,y$-plane, what is the y-intercept of the function $y = (x + 3)^2 - 6$?

A) $(-3, -6)$

B) $(0, -6)$

C) $(0,3)$

D) $(\sqrt{6},0)$

9

In the $x\,y$-plane, a circle has center $(0,8)$ and radius 9. Point $(0,k)$ is on the circle and k is a positive constant. What is the value of k?

A) 17

B) -1

C) 9

D) 1

10

$$y = 80(1 + 12\%)^t$$

The Royal Yacht Club was established in 1985 with 80 members. The equation above models the number of the members y, where t is the number of years since 1985. Which of the following equations models the number of members in the yacht club m months since 1985?

A) $y = 80(1 + 12\%)^{\frac{m}{6}}$

B) $y = 80(1 + 12\%)^{12m}$

C) $y = 80(1 + 12\%)^{\frac{m}{12}}$

D) $y = 80(1 + 12\%)^{\frac{m}{4}}$

11

Jay exercises by briskly walking everyday at a constant speed of 5 mph. If Jay walks 3 hours per day, how many miles has Jay walked in the past 20 days?

A) 100

B) 15

C) 300

D) 60

12

23, 25, 32, 27, 40, 37, 35

What is the median of the given data set?

A) 27

B) 32

C) 29.5

D) 33.5

13

A group of 90 people go hiking. A sample of people among the group was selected at random and asked whether they prefer hike a 5 km trail or a 10 km trail. The results show that 70% of the sample respond that they prefer the 10 km trail. Based on the survey, how many people in the group would prefer the 5 km trail?

A) 63

B) 90

C) 27

D) 20

14

$$y - 6 = 5$$
$$(y - 6)^2 = x + 3$$

Which of the following coordinates, (x, y), is a solution to the given system of equations?

A) (22,11)

B) (25,11)

C) (19,11)

D) (11,11)

15

The function is defined by $f(x) = x^3 + x^2 + 3$. What is the value of $f(-2)$?

A) -1

B) 7

C) 15

D) -9

16

76 students went on a field trip together in 5 school buses. Some of the buses can hold 12 students each, while the rest of the buses can hold 20 students each. How many of the buses can hold 20 students? (assuming each bus reached its maximum capacity)

17

A donut machine makes donuts at a rate of 5760 donuts in 8 hours. At this rate, how many donuts does this machine produce per minute?

18

Joanna buys 80 holiday cards. If she gives 55% of the cards to her friends, how many cards does she have left?

A) 44

B) 28

C) 36

D) 30

19

Which expression is equivalent to $8x^3 + 6x^2$?

A) $2x^3(4 + 6x)$

B) $2x^2(4x + 3)$

C) $4x^2(2x + 3)$

D) $2x^2(4 + 3x)$

20

A rectangle has an area of 48 centimeters and length of 6 centimeters. What is the perimeter of this rectangle, in centimeters?

21

Which expression is equivalent to $\frac{5}{2}x^2 - (\frac{3}{2}x^2 - \frac{1}{6}x^2)$?

A) $\frac{5}{3}x^2$

B) $2x^2$

C) $\frac{2}{3}x^2$

D) $3x^2$

22

A company plans on holding an anniversary party. The total cost, y, for the party can be estimated by the linear equation $y = A + Bx$, where x represents the number of participants and A and B are constants. If there are 50 participants, the total cost is \$1525. If there are 80 participants, the total cost is \$2290. What is the value of A?

No Test Material On This Page

Math

35 MINUTES, 22 QUESTIONS

DIRECTIONS

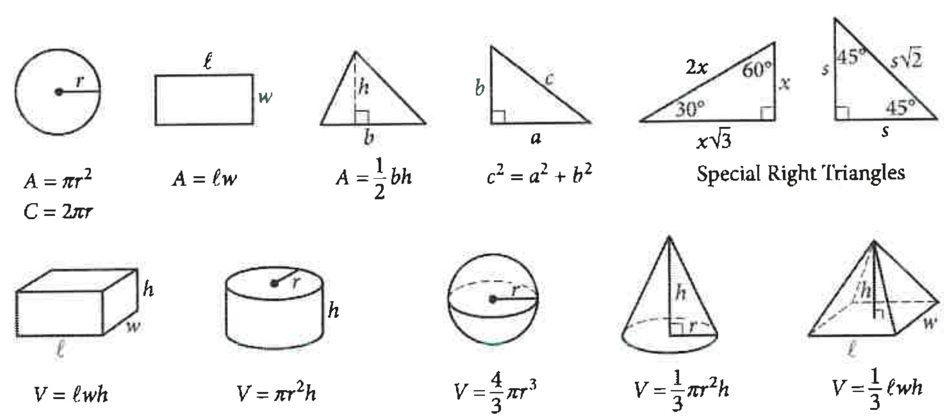
The questions in this section address a number of important math skills.
Use of a calculator is permitted for all questions.

NOTES

Unless otherwise indicated:

- All variables and expressions represent real numbers.
- Figures provided are drawn to scale.
- All figures lie in a plane.
- The domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

REFERENCE



The number of degrees of arc in a circle is 360.
The number of radians of arc in a circle is 2π.
The sum of the measures in degrees of the angles of a triangle is 180.

For multiple-choice questions, solve each problem, choose the correct answer from the choices provided, and then circle your answer in this book. Circle only one answer for each question. If you change your mind, completely erase the circle. You will not get credit for questions with more than one answer circled, or for questions with no answers circled.

For student-produced response questions, solve each problem and write your answer next to or under the question in the test book as described below.

- Once you've written your answer, circle it clearly. You will not receive credit for anything written outside the circle, or for any questions with more than one circled answer.
- If you find more than one correct answer, write and circle only one answer.
- Your answer can be up to 5 characters for a positive answer and up to 6 characters (including the negative sign) for a negative answer, but no more.
- If your answer is a fraction that is too long (over 5 characters for positive, 6 characters for negative), write the decimal equivalent.
- If your answer is a decimal that is too long (over 5 characters for positive, 6 characters for negative), truncate it or round at the fourth digit.
- If your answer is a mixed number (such as $3\frac{1}{2}$), write it as an improper fraction (7/2) or its decimal equivalent (3.5).
- Don't include symbols such as a percent sign, comma, or dollar sign in your circled answer.

2 Module 2 2

1

If $2 + \frac{3}{25}x^2 = 5$, what is the negative value of $x + 1$?

2

If $f(x) = 2x^2 + bx - 2$, where b is a constant and $f(2) = 8$, what is the value of $f(5)$?

3

A store sells apples for x dollars per pound and oranges for y dollars per pound. When Emily buys 3 pounds of apples and 5 pounds of oranges, the total cost is 17 dollars. When Emily buys 5 pounds of apples and 3 pounds of oranges, the total cost is 15 dollars. Which of the following systems of equations represents this situation?

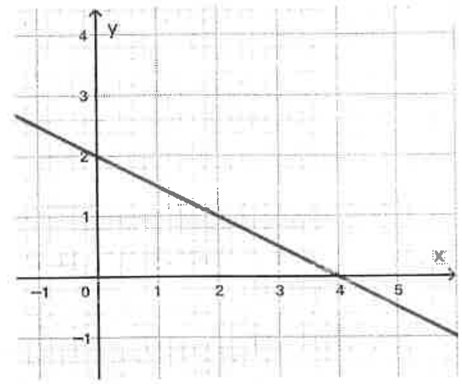
A) $5x + 3y = 17$
$3x + 5y = 15$

B) $5x + 3y = 15$
$5x - 3y = 17$

C) $3x + 5y = 17$
$5x + 3y = 15$

D) $3x + 5y = 15$
$5x + 3y = 17$

4



The equation $ax - by = 12$ is shown in the graph above, where a and b are constants. What is the value of a?

A) 4

B) 2

C) 6

D) 3

5

Data set A contains the heights of the 10 students in Ms. Kelly's class who participated in the basketball team, in which the mean height is 175 cm. Data set B contains the heights of the 40 students in Ms. Kelly's class who are not in the basketball team, in which the mean height is 165 cm. What is the mean height, in cm, of the 50 students in Ms. Kelly's class?

6

$$y = ax^2 + 16x + 5$$

The given equation relates the variables x and y, where a is a constant. The value of y reaches its maximum value when $x = 4$. What is the maximum value of y?

A) 4

B) 37

C) 2

D) 8

7

In the right triangle MQP, the length of the hypotenuse is $13x$ and the length of one leg is $5x$. What is the length of the other leg?

A) $4x$

B) $12x$

C) 12

D) $8x$

8

Alex paid \$614.88 for a watch, of which the 5% HST tax was \$27.45 and the 7% QST tax was \$38.43. Which of the following expressions does not provide the price-before-tax of the watch, x?

A) $x(1 + 7\% + 5\%) = 614.88$

B) $x \cdot 7\% = 38.43$

C) $x \cdot 5\% = 27.45$

D) $x = 614.88(1 + 7\% + 5\%)$

9

A bacteria population doubles every five hours. If the initial bacteria population is 1200 and the function $P(t)$ gives the bacteria population size after t hours, which of the following functions defines $P(t)$?

A) $P(t) = 1200(2)^{5t}$

B) $P(t) = 1200(\frac{1}{2})^{5t}$

C) $P(t) = 1200(\frac{1}{2})^{\frac{t}{5}}$

D) $P(t) = 1200(2)^{\frac{t}{5}}$

10

The function $y = f(x)$ is defined by $f(x) = 8x - k$, where k is a constant. The y-intercept of the graph in the xy-plane is $(0,5)$. What is the value of k?

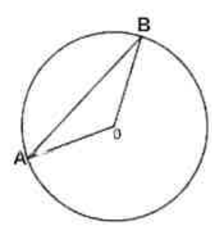
A) 5

B) -5

C) -8

D) 8

11



In the figure shown above, the point O is the center of the circle and the angle of AOB is $\frac{2}{3}\pi$ radians. How many degrees is the angle AOB?

2 Module 2 2

12

$$(2x + 5) - (3 - x)$$

Which of the following is equivalent to the given expression?

A) $x + 8$

B) $3x + 8$

C) $3x + 2$

D) $x + 2$

13

$$d = 45t$$

The equation above can be used to calculate the distance d, in miles, traveled by a car moving at an average speed of 45 miles per hour. For any positive constant n, the distance the car traveled after $6n$ hours is how many times the distance the car traveled after $2n$ hours?

A) 2

B) $3k$

C) 3

D) $2k$

14

When the quadratic function f is graphed in the $x\,y$-plane, where $y = f(x)$, its vertex is (2,4). If one of the x-intercepts of this graph is $(-1,0)$, what is the other x-intercept?

A) (4,0)

B) (7,0)

C) (5,0)

D) (6,0)

15

$$y = 6x + 24$$
$$3y = 14x + 2y$$

The solution to the given system of equations is (m, n), where m and n are constants. What is the value of $m - n$?

A) 39

B) 45

C) -39

D) 24

16

There are three metal spheres with radii of 6, 8, and 10, respectively. If these three metal spheres are melted together to make one large metal sphere, what is the radius of the large metal sphere?

A) 12

B) 24

C) 1720

D) 18

17

Amy wrote 26 Valentine's Day cards to her friends on Saturday and Sunday. On Sunday, she wrote 2 more than 2 times the number of cards that she wrote on Saturday. How many Valentine's Day cards did Amy write on Sunday?

18

Which expression is equivalent to $9x^2 + 3$?

A) $(3x + 5)(3x - 5) - 28$

B) $(3x + 5)(3x - 5) + 28$

C) $3(x + 2)(x - 2) - 15$

D) $3(x + 2)(x - 2) + 9$

19

If $2^{2m} = a^{\frac{2}{5}}$ and $2^{2n} = b^{\frac{2}{5}}$, which of the following is equivalent to $(\frac{a}{b})^2$?

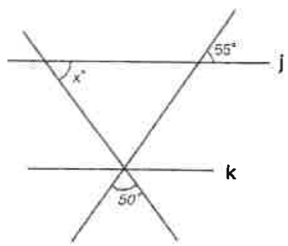
A) 4^{m-n}

B) 32^{m-n}

C) 64^{m-n}

D) 1024^{m-n}

20



In the figure above, lines j and k are parallel. What is the value of x?

21

Data Set A

Value	Frequency
15	2
16	2
17	2
18	2
19	1
20	2

Data Set B

Value	Frequency
15	1
16	1
17	6
18	3
19	2
20	2

Data Set A and B are shown in the tables above. Which of the following statements best compares the standard deviation and the means of the two data sets?

A) The mean of Data Set A is greater than the mean of Data Set B, and the standard deviation of Data Set A is greater than the standard deviation of Data Set B.
B) The mean of Data Set A is greater than the mean of Data Set B, and the standard deviation of Data Set A is less than the standard deviation of Data Set B.
C) The mean of Data Set A is less than the mean of Data Set B, and the standard deviation of Data Set A is greater than the standard deviation of Data Set B.
D) The mean of Data Set A is less than the mean of Data Set B, and the standard deviation of Data Set A is less than the standard deviation of Data Set B.

22

$$y = \frac{2x + p}{x - p}$$

The equation above expresses y in terms of p and x. Which of the following equations expresses x in terms of p and y?

A) $x = \dfrac{p(y-1)}{y-2}$

B) $x = \dfrac{p(y+1)}{y+2}$

C) $x = \dfrac{py}{y+p}$

D) $x = \dfrac{p(y+1)}{y-2}$

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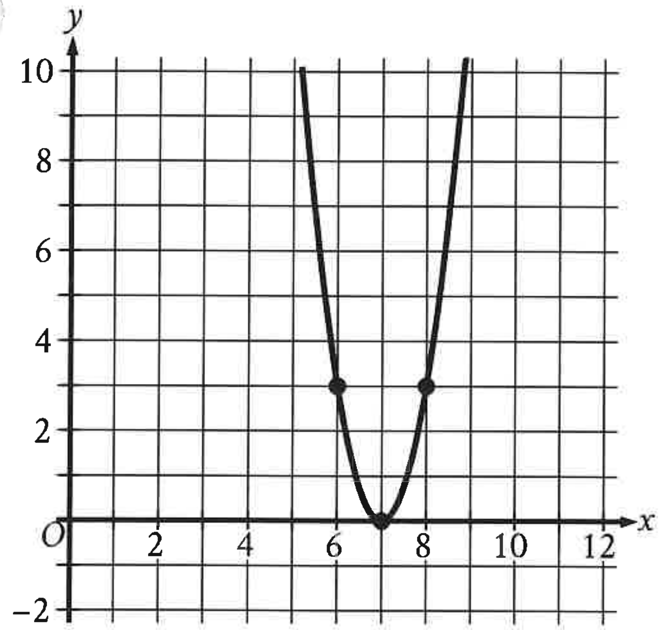

The x-intercept of the graph shown is $(x, 0)$. What is the value of x?

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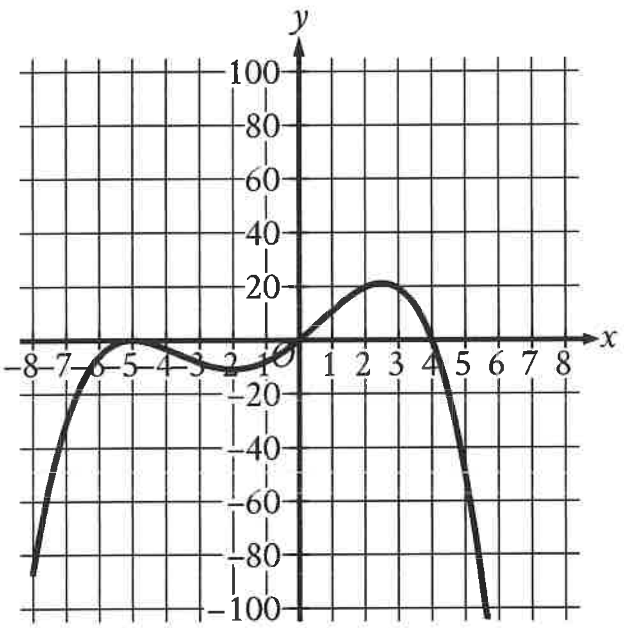


Which of the following could be the equation of the graph shown in the *xy*-plane?

A. $y = -\frac{1}{10}x(x-4)(x+5)$

B. $y = -\frac{1}{10}x(x-4)(x+5)^2$

C. $y = -\frac{1}{10}x(x-5)(x+4)$

D. $y = -\frac{1}{10}x\mathbf{msup}(x+4)$

ID: 841ef26c

$$f(x) = 4x^2 + 64x + 262$$

e function g is defined by $g(x) = f(x+5)$. For what value of x does $g(x)$ reach its minimum?

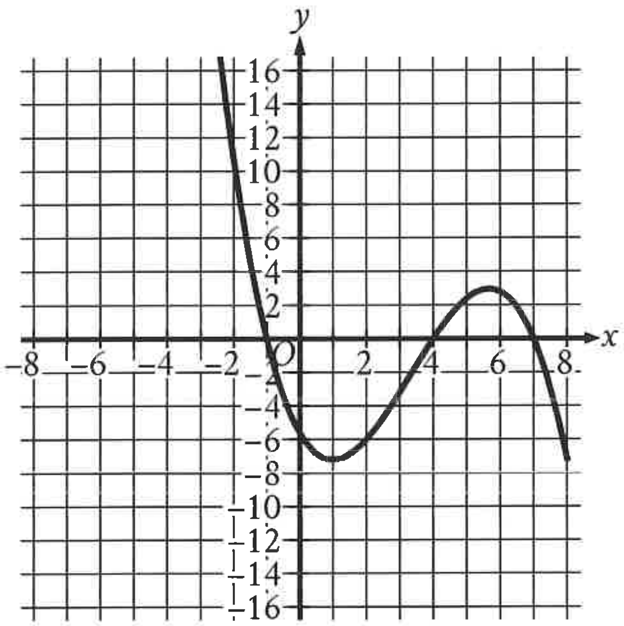
A. -13

B. -8

C. -5

D. -3

ID: cc6ccd71



The graph of $y = f(x)$ is shown, where the function f is defined by $f(x) = ax^3 + bx^2 + cx + d$ and a, b, c, and d are constants. For how many values of x does $f(x) = 0$?

A. One

B. Two

C. Three

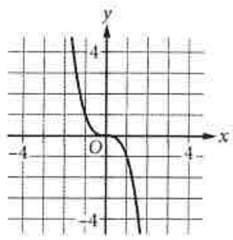
D. Four

ID: b39d74a0

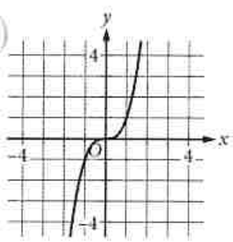
x	y
0	0
1	1
2	8
3	27

The table shown includes some values of x and their corresponding values of y. Which of the following graphs in the xy-plane could represent the relationship between x and y ?

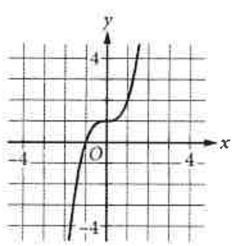
A.



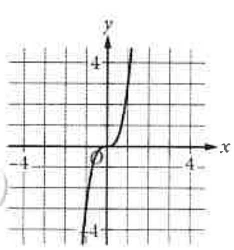
B.



C.



D.



ID: d135f4bf

The function f is defined by $f(x) = (x - 6)(x - 2)(x + 6)$. In the *xy*-plane, the graph of $y = g(x)$ is the result of translating the graph of $y = f(x)$ up 4 units. What is the value of $g(0)$?

ID: d4950429

A rectangle has a length of x units and a width of $(x - 15)$ units. If the rectangle has an area of 76 square units, what is the value of x?

A. 4

B. 19

C. 23

D. 76

ID: 752055d1

A scientist initially measures $12{,}000$ bacteria in a growth medium. 4 hours later, the scientist measures $24{,}000$ bacteria. Assuming exponential growth, the formula $P = C(2)^{rt}$ gives the number of bacteria in the growth medium, where r and C are constants and P is the number of bacteria t hours after the initial measurement. What is the value of r?

A. $\frac{1}{12{,}000}$

B. $\frac{1}{4}$

C. 4

D. $12{,}000$

ID: 271ffad7

A quadratic function models a projectile's height, in meters, above the ground in terms of the time, in seconds, after it was launched. The model estimates that the projectile was launched from an initial height of 7 meters above the ground and reached a maximum height of 51.1 meters above the ground 3 seconds after the launch. How many seconds after the launch does the model estimate that the projectile will return to a height of 7 meters?

A. 3

B. 6

C. 7

D. 9

ID: ee857afb

$$y = x^2 - 14x + 22$$

The given equation relates the variables x and y. For what value of x does the value of y reach its minimum?

ID: 70482e20

Which expression is equivalent to $11x^3 - 5x^3$?

A. $16x^3$

B. $6x^3$

C. $6x^6$

D. $16x^6$

ID: 8452c42b

Which expression is equivalent to $50x^2 + 5x^2$?

A. $250x^2$

B. $10x^2$

C. $45x^2$

D. $55x^2$

ID: ea6d05bb

The expression $(3x - 23)(19x + 6)$ is equivalent to the expression $ax^2 + bx + c$, where a, b, and c are constants. What is the value of b?

ID: 0536ad4f

Which expression is equivalent to $20w - (4w + 3w)$?

A. $10w$

B. $13w$

C. $19w$

D. $21w$

ID: 433184f1

Which expression is equivalent to $\frac{4}{4x-5} - \frac{1}{x+1}$?

A. $\frac{1}{(x+1)(4x-5)}$

B. $\frac{3}{3x-6}$

C. $-\frac{1}{(x+1)(4x-5)}$

D. $\frac{9}{(x+1)(4x-5)}$

ID: 1d3fee25

Which of the following is equivalent to $3(x+5)-6$?

A. $3x-3$

B. $3x-1$

C. $3x+9$

D. $15x-6$

ID: d8789a4c

$$\frac{x^2 - c}{x - b}$$

In the expression above, b and c are positive integers. If the expression is equivalent to $x + b$ and $x \neq b$, which of the following could be the value of c ?

A. 4

B. 6

C. 8

D. 10

ID: a520ba07

$$\sqrt[3]{x^3y^6}$$

Which of the following expressions is equivalent to the expression above?

A. y^2

B. xy^2

C. y^3

D. xy^3

ID: 5b6af6b1

Which expression is equivalent to $(d - 6)(8d^2 - 3)$?

A. $8d^3 - 14d^2 - 3d + 18$

B. $8d^3 - 17d^2 + 48$

C. $8d^3 - 48d^2 - 3d + 18$

D. $8d^3 - 51d^2 + 48$

ID: a255ae72

If $x^2 = a + b$ and $y^2 = a + c$, which of the following is equal to $(x^2 - y^2)^2$?

A. $a^2 - 2ac + c^2$

B. $b^2 - 2bc + c^2$

C. $4a^2 - 4abc + c^2$

D. $4a^2 - 2abc + b^2c^2$

ID: fcdf87b7

$$y = x^2 - 4x + 4$$
$$y = 4 - x$$

If the ordered pair (x, y) satisfies the system of equations above, what is one possible value of x ?

ID: 652054da

An oceanographer uses the equation $s = \frac{3}{2}p$ to model the speed s, in knots, of an ocean wave, where p represents the period of the wave, in seconds. Which of the following represents the period of the wave in terms of the speed of the wave?

A. $p = \frac{2}{3}s$

B. $p = \frac{3}{2}s$

C. $p = \frac{2}{3} + s$

D. $p = \frac{3}{2} + s$

ID: 3de7a7d7

Which of the following is a solution to the equation $2x^2 - 4 = x^2$?

A. 1

B. 2

C. 3

D. 4

ID: 70f98ab4

$$q - 29r = s$$

The given equation relates the positive numbers q, r, and s. Which equation correctly expresses q in terms of r and s?

A. $q = s - 29r$

B. $q = s + 29r$

C. $q = 29rs$

D. $q = -\frac{s}{29r}$

ID: 2c05d312

$$57x^2 + (57b + a)x + ab = 0$$

In the given equation, a and b are positive constants. The product of the solutions to the given equation is kab, where k is a constant. What is the value of k?

A. $\frac{1}{57}$

B. $\frac{1}{19}$

C. 1

D. 57

ID: 1fe32f7d

$$-x^2 + bx - 676 = 0$$

In the given equation, b is a positive integer. The equation has no real solution. What is the greatest possible value of b?

ID: 95ed0b69

$$p = \frac{k}{4j+9}$$

The given equation relates the distinct positive numbers p, k, and j. Which equation correctly expresses $4j + 9$ in terms of p and k?

A. $4j + 9 = \frac{k}{p}$

B. $4j + 9 = kp$

C. $4j + 9 = k - p$

D. $4j + 9 = \frac{p}{k}$

ID: c303ad23

If $3x^2 - 18x - 15 = 0$, what is the value of $x^2 - 6x$?

ID: 6e02cd78

In the xy-plane, what is the y-coordinate of the point of intersection of the graphs of $y=(x-1)^2$ and $y=2x-3$?

ID: 7bd10ef3

$$2x^2 - 4x = t$$

In the equation above, t is a constant. If the equation has no real solutions, which of the following could be the value of t ?

A. -3

B. -1

C. 1

D. 3

NO TEST MATERIAL ON THIS PAGE

The SAT

Practice Test 1

- Make time to take the practice test. (It is one of the best ways to get ready for the SAT)
- After you have taken the practice test, score it using answer keys provided at the end of the test.
- This version of the SAT Practice Test is for students using guideline for the real collegeboard digital SAT.
- You will be given 22 questions in each module but two questions won't be scored in the real collegeboard digital SAT (They are for research purposes) but you count all of the 44 questions on both modules to score this practice test to use the score conversion table at the end of the test.
- The use of a calculator is permitted throughout the Math sections

Math

22 QUESTIONS
(TIME: 35 MIN)

DIRECTIONS

The questions in this section address a number of important math skills.
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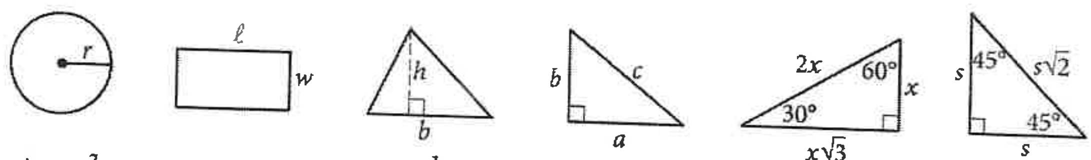
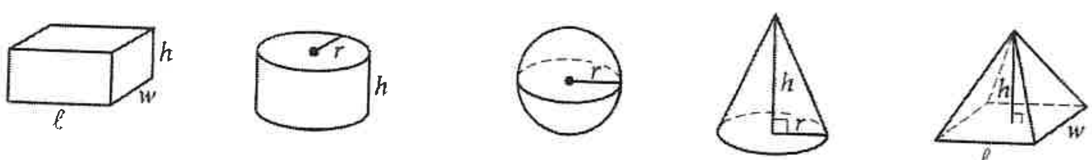
REFERENCE



$A = \pi r^2$, $C = 2\pi r$ | $A = \ell w$ | $A = \frac{1}{2}bh$ | $c^2 = a^2 + b^2$ | Special Right Triangles



$V = \ell wh$ | $V = \pi r^2 h$ | $V = \frac{4}{3}\pi r^3$ | $V = \frac{1}{3}\pi r^2 h$ | $V = \frac{1}{3}\ell wh$

The number of degrees of arc in a circle is 360.
The number of radians of arc in a circle is 2π.
The sum of the measures in degrees of the angles of a triangle is 180.

CONTINUE

For multiple-choice questions, solve each problem, choose the correct answer from the choices provided, and then circle your answer in this book. Circle only one answer for each question. If you change your mind, completely erase the circle. You will not get credit for questions with more than one answer circled, or for questions with no answers circled.

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- Your answer can be up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer, but no more.
- If your answer is a **fraction** that is too long (over 5 characters for positive, 6 characters for negative), write the decimal equivalent.
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- Don't include **symbols** such as a percent sign, comma, or dollar sign in your circled answer.

1

$$\sqrt{4(x-m)} = x - m$$

In the given equation above, where m is a positive constant. The greatest solution is 6. What is the value of m?

2

A computer processes at a constant rate of 82 data per second. At what rate, in data per minutes, does the computer processes the data?

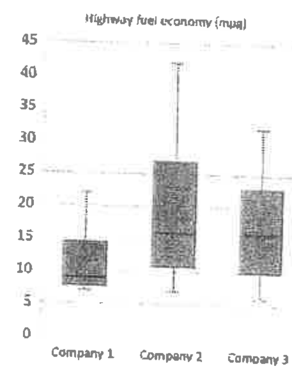
3

$$x - 2y = -1$$
$$-3 + 3x = 4y - 2$$

The solution to the system of equations above is (x, y). What is the value of $2x$?

A) 2
B) 3
C) 4
D) 6

4



Customer Reports released highway fuel economy data (mpg) of three companies for 10 years (2011-2021). Which company shows the largest **range** of highway fuel economy from the boxplot above?

A) Company 1
B) Company 2
C) Company 3
D) All companies have the same range

5

ANNUAL PERCENT CHANGE IN SALES AT TIRE STORES FROM 2020 TO 2022

STORE	Percent change 2020-2021	Percent change 2021-2022
A	20	-20
B	-10	10
C	15	-15

If the sales in dollar at Store B was $12,000 for 2020, what was the dollar amount of sales at the store for 2022?

A) $12,000
B) $11,880
C) $11,730
D) $11,520

6

$$y + mx = 3$$
$$y = x^2 - 2m$$

In the system of equations above, If the equations are graphed in the XY-plane, their graphs intersect at two points assuming m is a constant. Which of the following **cannot** be the value of m?

A) 0
B) -1
C) -3
D) -7

7

PETER'S MATH SCORE REPORT

	Percent of grade	Cumulative Scores
Homework	10	90
Attendance	10	100
Tests / Quizzes	60	87
Final Test	20	

The table above shows the distribution of the cumulative scores of each portion of Peter's Math Grade before his final test. If grade A is required to get the cumulative combined scores greater than or equal to 90 in his math class, what is the **lowest score** he need to get for the final test in order to achieve grade A?

A) 96
B) 95
C) 94
D) 93

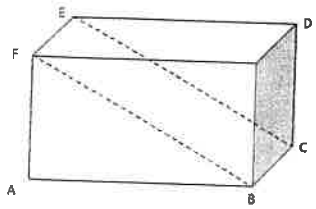
8

Which of the following expressions is equivalent to $8x^3y^4 - 32x^7y^2$?

A) $8x^7y^4(y^3 - x^2)$
B) $-8x^3y^4(-1 + 4x^4y^4)$
C) $8x^3y^2(y - 2x^2)(y + 2x^2)$
D) $8x^3y^2(y^2 - 4x^2)$

CONTINUE

9



In the rectangular solid above, If AB = 4, BC = 5, and CD = 3, What is the area of rectangular region BCEF?

A) 20
B) 25
C) $4\sqrt{34}$
D) $3\sqrt{41}$

10

If the difference of two angles U and N is $\frac{5\pi}{36}$ radians $(U > N)$ and the measure of angle is U is $\frac{\pi}{3}$ radians, what is the measure of angle N, in degrees?

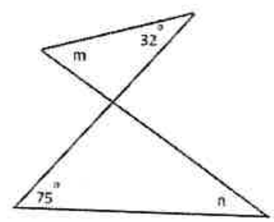
A) 20
B) 25
C) 30
D) 35

11

$$f(x) = -2x^2 + 4x + m$$

In the function above, if m is a unknown constant, for what value of x does the function have the same value of $f(-2)$?

12

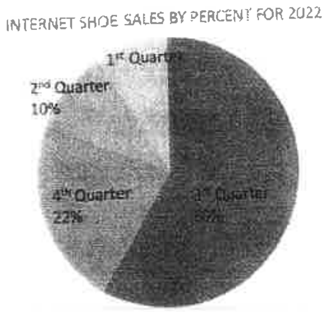


In the figure above, what is the value of $m - n$?

A) -33
B) -43
C) 33
D) 43

CONTINUE

13



The circle graph above shows the distribution of Shoe sales for ABC shop in 2022. What is the nearest degree of the central angle for 1st Quarter sales?

14

$$5f + 6s + 12j = 1{,}230$$

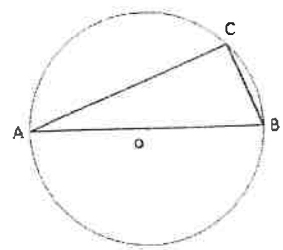
A school consists of 5 freshmen classes, 6 sophomore classes, and 12 junior classes. The total number of students who has a social media account is 1,230. The equation above represents the situation. What is the best interpretation of s in the context?

A) The total number of sophomore students in the school
B) The total number of sophomore students who has social media account in the school
C) The average number of sophomore students who has social media account in the school
D) The average number of students who has social media account in the school

15

List A and list B contain 50 numbers and 60 numbers respectively. If the mean of list A is 25 and the mean of list B is 34, what is the positive difference in their sums?

16



If the diameter ($\overline{AB}$) of the circle O above is 10 and BC = 6, what is the area of triangle ABC?

A) 24
B) 28
C) 30
D) 32

CONTINUE

17

$$\frac{x(xy^2)^4}{x^4y^3} = x^m y^n$$

In the equation above, If $xy \neq 0$, what is the value of $m + n$?

18

The function f has the property for all x that

$$2f(x) = f(2x)$$

If $f(2) = 4$, what is the value of $f(1)$?

19

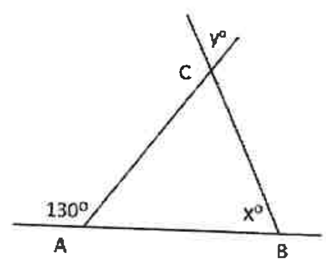
$$\frac{1}{2}k - y = \frac{7}{2}$$

$$-2x + 4y = -14$$

In the system of equations above, for a real number k, which of the following points lies on the graph of each equation?

A) $\left(k, \frac{k}{2} + \frac{7}{2}\right)$
B) $(2k + 7, k)$
C) $(k, 7 + 2k)$
D) $(2k - 7, k)$

20



Note: not drawn to scale

In the figure above, assuming AB = BC, what is the value of $x + y$?

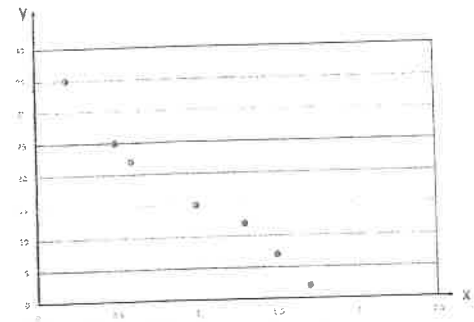
21

$$2x^2 + 4x + 2y^2 - 8y = -4$$

In the circle equation above, what is the area of the circle?

A) π
B) 3π
C) 6π
D) $\sqrt{6}\pi$

22



Which of the following equations is the most accurate linear model for the scatterplot shown above?

A) $y = 21.6x + 35$
B) $y = 21.6x - 35$
C) $y = -21.6x + 35$
D) $y = -21.6x - 35$

STOP

If you finish before time is called, you may check your work on this module only. Do not turn to any other module in the test.

CONTINUE

Math

22 QUESTIONS
(TIME: 35 MIN)

DIRECTIONS

The questions in this section address a number of important math skills.
Use of a calculator is permitted for all questions.

NOTES

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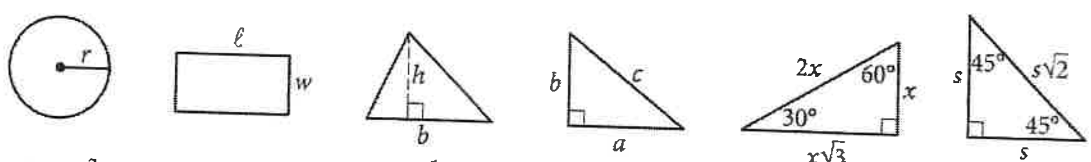
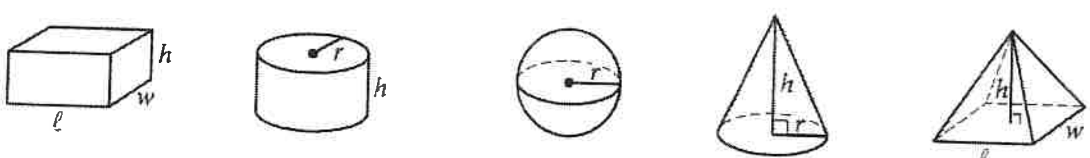
- Figures provided are drawn to scale.
- All figures lie in a plane.
- The domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

REFERENCE



$A = \pi r^2$, $C = 2\pi r$ | $A = \ell w$ | $A = \frac{1}{2}bh$ | $c^2 = a^2 + b^2$ | Special Right Triangles



$V = \ell wh$ | $V = \pi r^2 h$ | $V = \frac{4}{3}\pi r^3$ | $V = \frac{1}{3}\pi r^2 h$ | $V = \frac{1}{3}\ell wh$

The number of degrees of arc in a circle is 360.
The number of radians of arc in a circle is 2π.
The sum of the measures in degrees of the angles of a triangle is 180.

CONTINUE

For multiple-choice questions, solve each problem, choose the correct answer from the choices provided, and then circle your answer in this book. Circle only one answer for each question. If you change your mind, completely erase the circle. You will not get credit for questions with more than one answer circled, or for questions with no answers circled.

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1

$$-2(1 - x) - 3 = ax + 1$$

If the equation above has no solutions, for a constant a, what is the value of a?

2

For an exponential function f, where k is a constant. If the value of $f(-2)$ is k, which of the following forms of the function f shows the value of k as **the coefficient**?

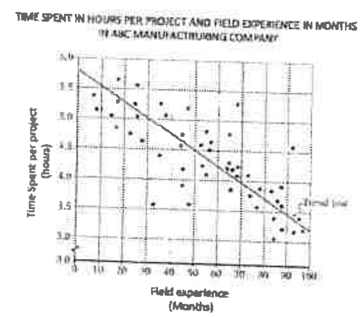
A) $f(x) = 202(2.1)^{x-2}$
B) $f(x) = 12.5(2.1)^{x}$
C) $f(x) = -302(2.1)^{-x}$
D) $f(x) = 4(2.1)^{x+2}$

3

If Sally earns k dollars for t hour in her job, which of the following expression represents the amount of money, in dollars, Sally makes for **$4t + 1$** hours of work if she earns only hourly pay?

A) $4k$
B) $4k + 1$
C) $4k + t/k$
D) $4k + k/t$

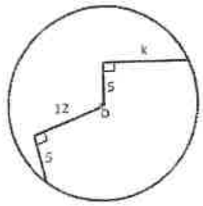
4



In the scatterplot above, the data represent the time spent, in hours, per project for 50 workers in ABC Company. What fraction of workers could finish one project less than the expected time, in hours, if the workers had at least 80 hours of field experience?

CONTINUE

5



Note: Not drawn to scale.
O is the center of the circle

In the circle O above, what is the value of k?

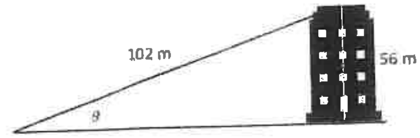
6

$$3x = 4y = 2z = 20$$

In the equation above, what is the value of $15xy^2z$?

A) 8,000
B) 15,000
C) 20,000
D) 25,000

7



The guywire length stretched from the ground to the top of the building is 102 m and the height of the building is 56m. which of the following expressions represents the angle of elevation of the guywire from the ground to the top of the building?

A) $\sin^{-1}\left(\frac{56}{102}\right)$
B) $\cos^{-1}\left(\frac{56}{102}\right)$
C) $\tan\left(\frac{56}{102}\right)$
D) $\sin\left(\frac{56}{102}\right)$

8

$$2x^2 - (m+1)x + 2m = 0$$

In the equation above, m is a constant. If the sum of two roots is 4, then what is the product of the roots?

A) 4
B) 5
C) 7
D) 8

9

$$y < \frac{2}{3}x + 2$$

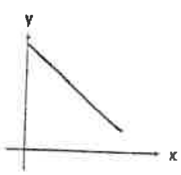
$$y > \frac{2}{3}x - 1$$

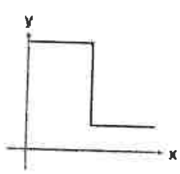
In the inequalities shown above, if $(0, a)$ is a solution of the system of inequalities, assuming a is positive integer, what is the value of a?

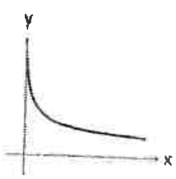
A) 1
B) 2
C) 3
D) 4

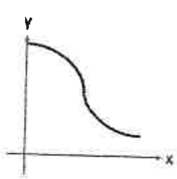
10

A person is injected with a certain drug. The amount of drug left in the person's body decreases in half every hour at a constant rate. Which of the following graphs best represents the amount of drug left after a certain time?

A)


B)


C)


D)


11

Jeremiah builds catapult for his physics project. The restriction of the catapult must measure 20 cm in height, with 1.5 cm margin of error. Which absolute inequalities show the possible height of the catapult he can build?

A) $|20 - x| < 1.5$
B) $|x - 1.5| < 20$
C) $|x - 20| > 1.5$
D) $|x - 1.5| < 20$

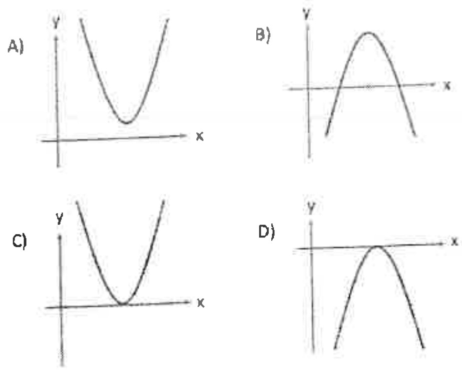
12

An author wants to publish his book. A book publisher informs that the thickness of pages is 0.2 millimeters and both front cover and back cover need to have 3 millimeters thick respectively. Which of the following functions shows the total thickness of the book in millimeter which has x pages?

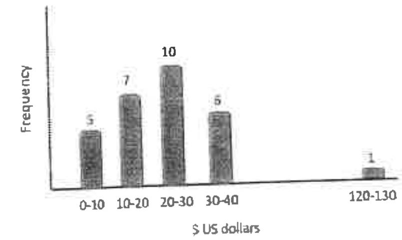
E) $f(x) = 3 + 0.2x$
F) $f(x) = 6 + 0.2x$
G) $f(x) = 3 + 0.4x$
H) $f(x) = 6 + 0.4x$

13

The function f is defined by $f(x) = -(x-1)^2 + 1$. Which of the following graphs below could be the graph of $y = f(x) - 1$?



15



The graph above shows the frequency distribution of 29 students weekly allowance

If the student who gets the largest weekly allowance in the data (outlier) is removed from the research, which value could be **affected the most**?

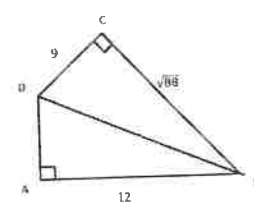
A) Median
B) Range
C) Interquartile Range
D) Mode

14

A carpenter needs to make 4 wooden dog houses. The length of wood stick required for each dog house is 2.9 ft. if the wood stick is purchased as a single length in feet and sells for \$2 per foot. what is the total cost of the wood stick needs to be purchased for the 4 dog houses?

A) \$24.00
B) \$23.20
C) \$23.00
D) \$22.90

16



In the figure above, what is the area of triangle ABD?

E) 28
F) 29
G) 30
H) 31

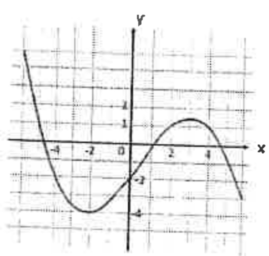
CONTINUE

17

Adrian's monthly earning is t dollars. If he spends $\frac{2}{5}t$ dollars for taxes and expenses and saves the rest every month. If this continues at the same rate, how many **years** will it take him to save $3600?

A) $\frac{6000}{t}$
B) $\frac{1000}{t}$
C) $\frac{500}{t}$
D) $\frac{3600}{t}$

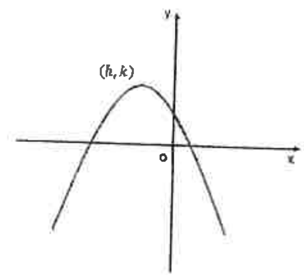
18



The figure above shows the complete graph of the function f in the xy-plane. How many values of x exist if $f(x) = f(f(-2))$?

A) 0
B) 1
C) 2
D) 3

19



The graph of $y = a(x+3)(x-1)$ is shown in the XY-plane above, where a is a constant. If the graph passes through $(0,2)$ and the coordinates of the vertex is (h, k), what is the value of k?

20

During a one-year study, the price of a certain car 10 percent was decreased in the first half year and then increased 10 percent in the second half year. From the beginning of the year to the end of the year, the price of car was

A) did not change
B) decreased by 1%
C) increased by 1%
D) decreased by 10%

21

$$x^2 + 6x + y^2 - 2y = -1$$

In the circle equation above, what is the circumference of the circle?

E) π
F) 3π
G) 6π
H) 9π

22

$$2x - y = 1$$
$$y = x^2 - 9$$

The graphs of the system of equations intersects at the points (x, y) in the $XY - plane$. What is the product of two y values?

A) -8
B) 8
C) -35
D) 35

STOP

If you finish before time is called, you may check your work on this module only. Do not turn to any other module in the test.

Week 4

2 Module 1 2

Math

35 MINUTES, 22 QUESTIONS

DIRECTIONS

The questions in this section address a number of important math skills.
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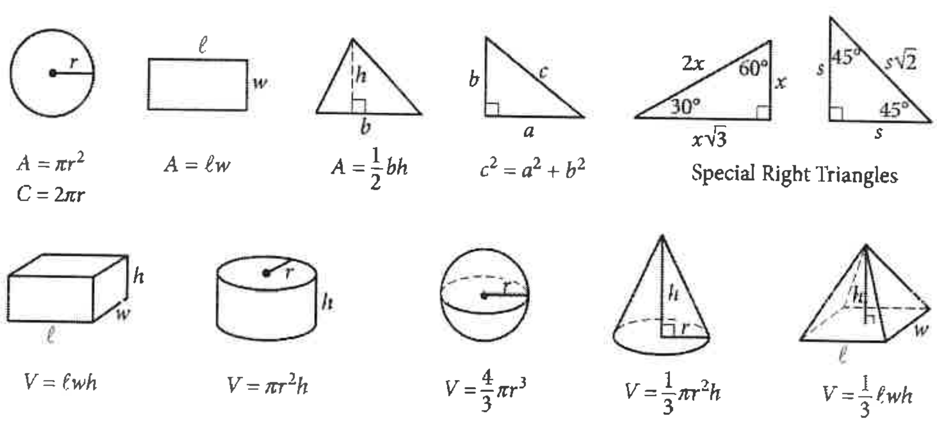
REFERENCE



The number of degrees of arc in a circle is 360.
The number of radians of arc in a circle is 2π.
The sum of the measures in degrees of the angles of a triangle is 180.

2 Module 1 2

For multiple-choice questions, solve each problem, choose the correct answer from the choices provided, and then circle your answer in this book. Circle only one answer for each question. If you change your mind, completely erase the circle. You will not get credit for questions with more than one answer circled, or for questions with no answers circled.

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1

$$\frac{(8+6k-4k)}{2}=7$$

What is the solution to the given equation?

A) 1

B) 3

C) 6

D) 10

2

If $-4 \leq 1-x \leq -2$, what are the possible values of x?

A) $3 \leq x \leq 5$

B) $-5 \leq x \leq -3$

C) $x \leq -5$ or $x \geq -3$

D) $2 \leq x \leq 4$

3

If $f(x)=6x+8$, what is the value of $f(0)$?

A) 14

B) 8

C) 2

D) 9

4

The whale population at the beginning of the study was 2100 whales. The number of whales in the population increased at a rate of approximately 35 per year. Which function f gives the number of whales t years after the beginning of the study?

A) $f(t)=35t+2100$

B) $f(t)=35t-2100$

C) $f(t)=(2100+35)t$

D) $f(t)=2100-35t$

5

$$f(x)=ax^3+x+c$$

The function f is defined above, where a and c are constants. If $f(-1)=3$ and $f(1)-7$, what is the value of a?

A) 5

B) 1

C) 4

D) 6

6

The slope of line l is $-\frac{1}{2}$, and line k is perpendicular to line l in the xy-plane. Which of the following functions could define line k?

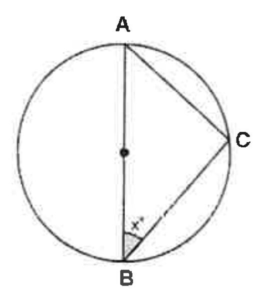
A) $f(x)=2x+5$

B) $f(x)=-2x+3$

C) $f(x)=\frac{1}{2}x+6$

D) $f(x)=-\frac{1}{2}x+5$

7



In the circle above, the diameter AB is equal to 10 and the length of AC is equal to 6. What is the area of triangle ABC?

8

$$f(x) = -3x^2 + 12x + 6$$

For the function above, what is the maximum value?

A) 18

B) 2

C) 15

D) 9

9

Which of the following circles in the xy-plane has a diameter that is not equal to 12?

A) $x^2 + y^2 = 36$

B) $x^2 - 2x + y^2 - 4y = 31$

C) $x^2 - 2x + y^2 - 4y = 36$

D) $x^2 + 4x + y^2 + 6y = 23$

10

Bacteria reproduce by simple asexual binary fission. The initial population of *E.coli* is B and the population doubles every 20 minutes. Which of the following equations represents the total number of *E.coli* bacteria after 3 hours?

A) $y = B(2)^9$

B) $y = B(2)^{180}$

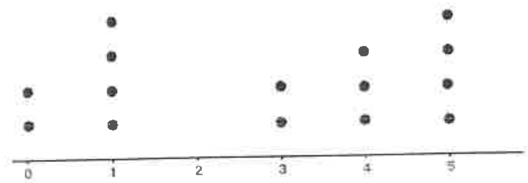
C) $y = B(2)^3$

D) $y = B(2)^{20}$

11

A car traveling at $\frac{1}{2}$ of its usual speed takes an extra 20 minutes to reach its destination. What is the usual time the car takes to cover the same distance at its usual speed?

12



The dot plot represents the distribution of values in a data set. What is the mean value of this data set?

13

A box contains 80 glass beads. 26 beads are blue and 32 beads are red. A bead is picked randomly from the box. What is the probability that the bead is neither blue nor red?

A) $\frac{13}{40}$

B) $\frac{16}{20}$

C) $\frac{11}{40}$

D) $\frac{13}{16}$

14

$$m + 1 = 2m - 616$$

What is the solution to the given equation?

A) 308

B) 206

C) 617

D) 615

2 Module 1 2

15

The function f is defined by $f(x) = (x^2 - 1)(x + 1)$. What is the value of $f(3)$?

A) 32

B) 40

C) 24

D) 12

16

The average of three numbers is 60. If two of the numbers have a sum of 135, what is the third number?

A) 60

B) 45

C) 75

D) 90

17

A side length of a square is 6 inches. What is the area of the square, in square centimeters? (1 inch = 2.54 centimeters)

A) 36

B) 91.44

C) 232.26

D) 165

18

A student went to a bookstore and paid a total of \$150 to purchase books and notebooks. If the student spent 70% of the money on books, how many dollars did he spend on notebooks?

19

Which expression is equivalent to $(6x^3 - 2x^2) - (5x^3 - 2x^2)$?

A) x^3

B) $x^2(x - 4)x$

C) $x^2(11x - 4)$

D) $x^4(x^2 - 1)$

20

Square A has an area of 16 square inches. One side of Square B is 3 times the side of Square A. What is the area of Square B, in square inches?

21

The variable y is the value of x increased by 35% and then decreased by 25%. Which of the following expressions represents the relationship between x and y?

A) $y = x(1 + 35\% - 25\%)$

B) $y = x(1 + 35\%)$

C) $y = x(1 + 35\%)(1 - 25\%)$

D) $y = x \cdot 35\% - x \cdot 25\%$

22

Lucas ran $\frac{4}{5}$ of a kilometer in 6 minutes. How many miles can Lucas run in half an hour? (1 kilometer = 0.62 miles)

A) 4

B) 2.48

C) 15.5

D) 8

No Test Material On This Page

2 Module 2 2

Math

35 MINUTES, 22 QUESTIONS

DIRECTIONS

The questions in this section address a number of important math skills.
Use of a calculator is permitted for all questions.

NOTES

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- All figures lie in a plane.
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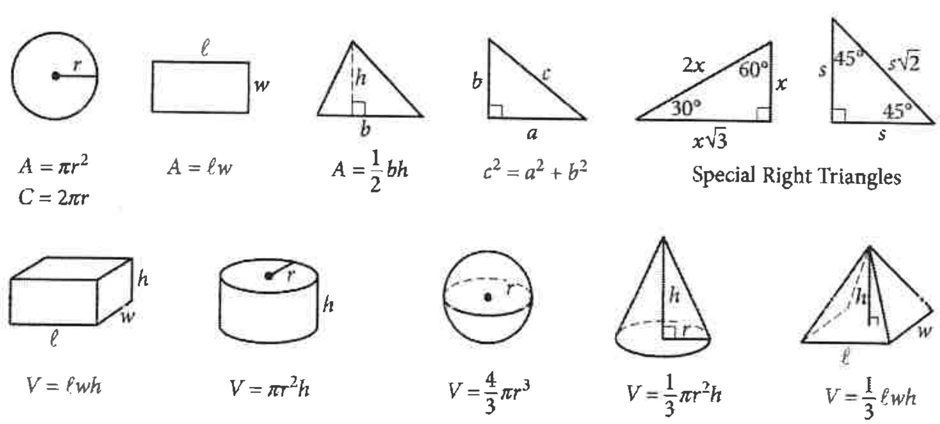
REFERENCE



The number of degrees of arc in a circle is 360.
The number of radians of arc in a circle is 2π.
The sum of the measures in degrees of the angles of a triangle is 180.

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2 Module 2 2

For multiple-choice questions, solve each problem, choose the correct answer from the choices provided, and then circle your answer in this book. Circle only one answer for each question. If you change your mind, completely erase the circle. You will not get credit for questions with more than one answer circled, or for questions with no answers circled.

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- If your answer is a fraction that is too long (over 5 characters for positive, 6 characters for negative), write the decimal equivalent.
- If your answer is a decimal that is too long (over 5 characters for positive, 6 characters for negative), truncate it or round at the fourth digit.
- If your answer is a mixed number (such as $3\frac{1}{2}$), write it as an improper fraction (7/2) or its decimal equivalent (3.5).
- Don't include symbols such as a percent sign, comma, or dollar sign in your circled answer.

2 Module 2 2

1

If $\dfrac{x^2-y^2}{3x+3y}=\dfrac{7}{5}$, what is the value of $5(x-y)$?

2

The function f is defined by $f(x)=kx^2+3x-1$, where k is constant and $f(2)=-19$. What is the value of $f(-2)$?

3

$$y+ax=5$$
$$3y-x=15$$

If the system of equations above has infinite solutions, what is the value of a?

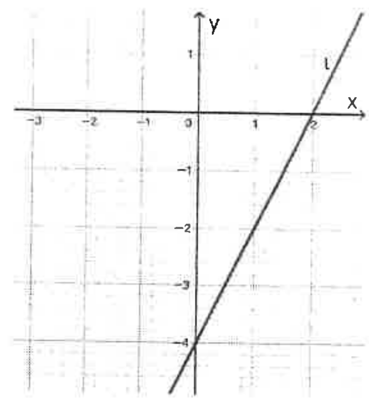
A) $\dfrac{1}{3}$

B) $-\dfrac{1}{3}$

C) 3

D) -3

4



Line l is shown in the xy-plane above. Line p is line l shifted left 2 units. Which equations below represents the line p?

A) $y=2x+4$

B) $y=2x-2$

C) $y=2x$

D) $y=2x-6$

5

The median number of books Sally read each week over the past 9 weeks is 4. Which of the following changes could lead to the same median number of books Sally read per week?

A) Read 2 more books in the first week and 1 less book in the last week

B) Read 1 less book per week

C) Read 2 more books in the week she reads the most

D) Read 2 more books per week

6

Which of the following functions, when graphed in the xy-plane, has exactly one positive x-intercept and one negative x-intercept?

A) $y=x^2+10x+24$

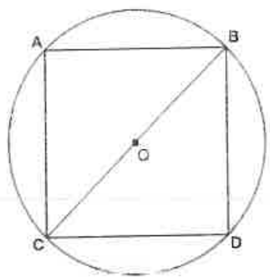
B) $y=x^2-10x+24$

C) $y=x^2-2x-24$

D) $y=x^2-11x+28$

2 Module 2 2

7



The figure above shows a square ABCD inscribed in a circle, where the diameter of the circle is 18. What is the area of the square ABCD?

A) ~~81~~
B) 162
C) 324
D) 250

8

Homemade sugar-salt rehydration solutions should contain a ratio of 1000:20:5 of water-sugar-salt respectively. If the rehydration solution contain 15 grams of sugar, how many grams of salt is needed?

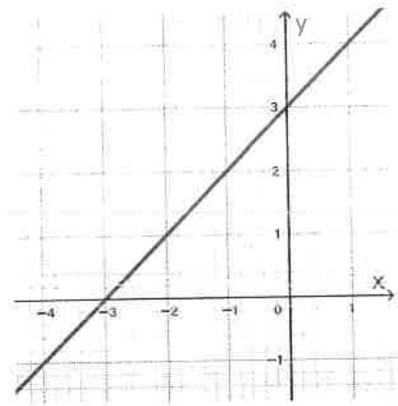
9

$$f(t) = A(1 + 2.5\%)^t$$

An investor deposited some money into his investment account at the end of 2021. The function f models the amount of money in his investment account t years after 2021, where $0 \leq t \leq 5$. Which of the following statements is the best interpretation of the A in the function f?

A) The increase of money every year in the investment account
B) The money the investor deposited at the end of 2021
C) The flat fee for opening the investment account
D) The amount of money in the investment account after 2 years

10



The graph of the linear function f is shown. What is the y-intercept of the graph of f?

A) (3,0)
B) (−3,0)
C) (0,3)
D) (0, − 3)

11

How many seconds is equal to one hour and thirty minutes?

A) 5400
B) 90
C) 3600
D) 6480000

12

Which of the following is equivalent to $2(x^2 - 3) - (x^2 - 5)$?

A) $(x - 1)(x + 1)$
B) $3x^2 + 11$
C) $3x^2 - 1$
D) $x^2 - 11$

2 Module 2 2

13

Georgia and Victoria started from the same coffee shop and walked in opposite directions. Georgia walks 1 mile per hour faster than Victoria. After 2 hours, they are 26 miles apart. What is Georgia's average speed?

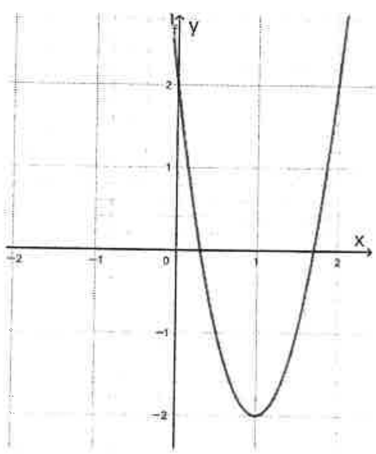
A) 7

B) 6

C) 13

D) 10

14



Which of the following quadratic functions is shown in the graph above?

A) $y = 2(x - 1)^2 - 2$

B) $y = 4(x - 1)^2 - 2$

C) $y = 4(x - 1)^2 + 2$

D) $y = 4(x + 1)^2 - 2$

15

$$y = \frac{1}{2}x - 2$$
$$y = 2x^2 - 4x + 1$$

How many solutions does the given system of equations have?

A) Zero

B) Exactly one

C) Exactly two

D) Infinitely

16

Eight cube blocks each have the same volume of 64 units cubed respectively. When they are stacked together, they form a big cubic block. What is the length of one side of the big cube block?

17

Cocoa Delight sold 80 cups of hot chocolate. Of these, 56 contained marshmallows and 28 contained whipped cream. If 19 cups contained neither marshmallows nor whipped cream, how many cups must have contained both marshmallows and whipped cream?

18

Which expression is equivalent to $(\frac{m}{3} - \frac{n}{2})^2$?

A) $\frac{1}{36}(m^2 - 2mn - n^2)$

B) $\frac{1}{9}m^2 - \frac{1}{4}n^2$

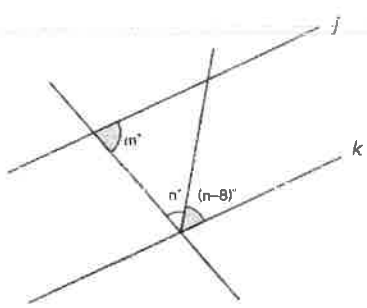
C) $\frac{1}{9}m^2 - \frac{1}{6}mn + \frac{1}{4}n^2$

D) $\frac{1}{9}m^2 - \frac{1}{3}mn + \frac{1}{4}n^2$

19

If $2^m 2^n = 8$, what is the value of $4^m 4^n$, where m and n are positive constants?

20



Lines j and k are parallel in the figure above. Which of the following expresses the value of m in terms of n?

A) $m = 188 - 2n$

B) $m = 172 - 2n$

C) $m = 180 - n$

D) $m = 180 - 2n$

21

There were 3.12 million units of electric vehicle sold globally in 2020. Sales skyrocketed in 2021, increasing by 107% in comparison to 2020. How many vehicles, in million units, were sold globally in 2021? (Round to two decimal places)

A) 3.34

B) 5.30

C) 6.46

D) 8.42

22

The equation $N(t) = 300(1.05)^t$ estimates the wildlife population during 2001-2004, where t represents the number of years since 2001. Which of the following is the best interpretation of (1.05)?

A) The number of wildlife increased by 50% from its previous year during 2001-2004

B) The number of wildlife increased by 5% from its previous year during 2001-2004

C) The number of wildlife increased by 1.05 times its previous year during 2001-2004

D) The number of wildlife decreased by 50% from its previous year during 2001-2004

The SAT

Practice Test 2

- Make time to take the practice test. (It is one of the best ways to get ready for the SAT)
- After you have taken the practice test, score it using answer keys provided at the end of the test.
- This version of the SAT Practice Test is for students using guideline for the real collegeboard digital SAT.
- You will be given 22 questions in each module but two questions won't be scored in the real collegeboard digital SAT (They are for research purposes) but you count all of the 44 questions on both modules to score this practice test to use the score conversion table at the end of the test.
- The use of a calculator is permitted throughout the Math sections

Math

22 QUESTIONS
(TIME: 35 MIN)

DIRECTIONS

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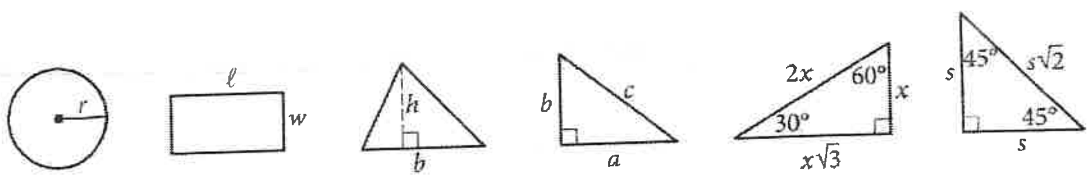


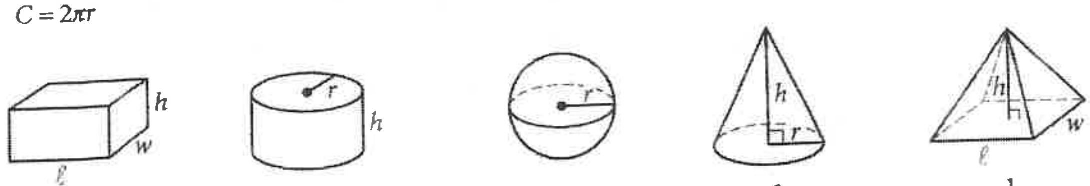
$A = \pi r^2$
$C = 2\pi r$

$A = \ell w$

$A = \frac{1}{2}bh$

$c^2 = a^2 + b^2$

Special Right Triangles



$V = \ell wh$

$V = \pi r^2 h$

$V = \frac{4}{3}\pi r^3$

$V = \frac{1}{3}\pi r^2 h$

$V = \frac{1}{3}\ell wh$

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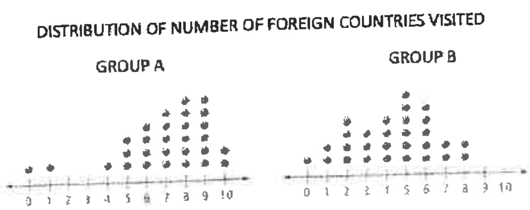
1

$$mx^2 + 10x + 1 = 0$$

In the equation above, c is a constant. If the equation has only one real solution, what is the value of m?

A) -25
B) 25
C) -50
D) 50

2



Which of the following statements must be true?

I. The range of group A is bigger than the range of group B.
II. The mean value describes more accurately in data A than the median value because it has outliers

A) I only
B) II only
C) I and II
D) Neither I or II

3

In a relation between two variables, x and y, they are related such that every increase by 1 in the value of x, the value of y decreases by a factor of 4. If $y = 30$ when $x = 0$, which equation represents the relationship?

A) $y = 30\left(\frac{3}{4}\right)^x$
B) $y = 30(x+1)^4$
C) $y = 30\left(\frac{1}{4}\right)^x$
D) $y = 30(-4)^x$

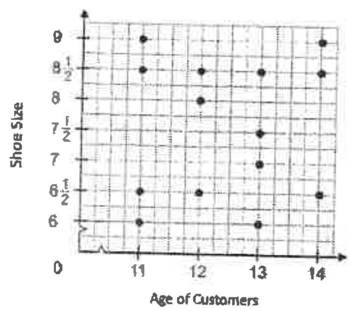
4

Elliott earns k dollars annually and pays r percent of what he earns for income taxes. Which of the following expressions represents the amount of money he earns after tax deduction?

A) $k(100 - r)$ dollars
B) $k(1 - r)$ dollars
C) $k(1 - 0.1r)$ dollars
D) $k(1 - 0.01r)$ dollars

5

Ages and Shoe Size of customers in kid's shoe store



Which of the following statements describes the most appropriate about the scatter plot above?

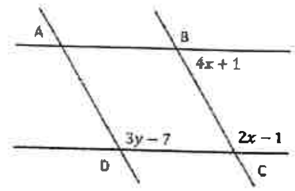
A) The plot shows a positive correlation.
B) The plot shows a negative correlation.
C) The plot shows a constant trend.
D) The plot shows no correlation.

6

In a certain survey, 25% of people were at most 20 years old and 75% were at most 50 years old. If 120 people in the survey were more than 20 years old and at most 50 years old, what was the total number of people surveyed?

A) 120
B) 180
C) 220
D) 240

7



In the diagram above, if $\overline{AB} \parallel \overline{DC}$ and $\overline{AD} \parallel \overline{BC}$, what is the sum of values of x and y?

A) 22
B) 30
C) 52
D) 62

8

Julie's income in 2021 was 40 percent higher than her income in 2020. What is the ratio of her income in 2021 to her income in 2020?

A) 7 to 5
B) 5 to 7
C) 5 to 2
D) 2 to 5

9

$$S(m) = 50{,}000(0.98)^m$$

The function, $S(m)$, above represents the population of bacteria in a certain colony m minutes after exposing them into partial sunlight. Which of the following functions best models the population of bacteria after h hours?

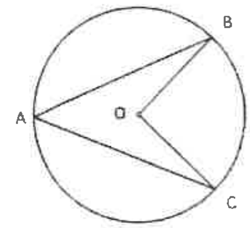
E) $P(h) = 50{,}000\left(\frac{0.98}{60}\right)^h$
F) $P(h) = 50{,}000(0.98)^{\frac{h}{60}}$
G) $P(h) = 50{,}000(0.98)^{60h}$
H) $P(h) = \frac{50{,}000}{60}(0.98)^h$

10

$$f(x) = -2x^2 + 4x + 3$$

In the quadratic function, $f(x)$, shown above, what is the maximum value of the function $f(x)$?

11



The circle O is shown above, if the radius of the circle is 6 and the measure of angle $\angle BAC = 30°$, what is the length of arc BC?

A) 2π
B) 4π
C) 6π
D) 8π

12

The average (arithmetic mean) of three positive integers, $a, b, and\ c$ is 32 where $c < b < a$. If the sum of two smaller numbers is 46, what is the value of a?

13

$$f(x) = (x-1)(x-3)(x+1)$$

The function f is shown above. Which of the following table values represents $y = f(x) + 1$?

A)

x	y
1	2
3	4
-1	0

B)

x	y
1	0
3	0
-1	0

C)

x	y
1	-1
3	-1
-1	-1

D)

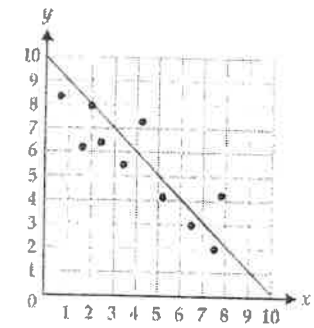
x	y
1	1
3	1
-1	1

14

The number of tiles needed to cover 50 square feet of wall is 100. A bathroom has a total of t square feet on the wall to cover. Which of the following could represent the total number of tiles needed to cover three bathrooms in terms of t?

A) $2t$
B) $3t$
C) $5t$
D) $6t$

15



In the scatterplot shown above, which of the following equations best represents the line of best fit shown?

A) $y = -x + 8$
B) $y = -x + 10$
C) $y = 10 + x$
D) $y = 10 - 2x$

16

$$\frac{(m^{-2}n^{3})(m^{2}n^{-2})^{2}}{(mn^{-1})^{-1}}$$

If the algebraic expression above is simplified to $\frac{m^x}{n^y}$, where x and y are positive integers. What is the value of the sum of x and y?

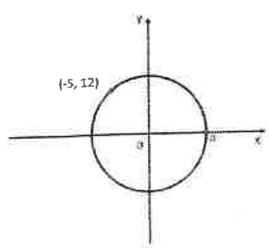
17

$$\frac{1}{r_t}=\frac{1}{r_a}+\frac{1}{r_b}$$

To calculate the total resistance in a parallel circuit, we can use the formula above. Which equation correctly expresses r_t in term of r_a and r_b?

A) $r_t = r_a + r_b$
B) $r_t = \frac{r_a+r_b}{r_a r_b}$
C) $r_t = \frac{r_a r_b}{r_a+r_b}$
D) $r_t = \frac{r_a}{r_a+r_b}$

18



In the circle above, if (-5, 12) is one point on the circle as shown, what is the coordinates of point a?

A) (0,13)
B) (13,0)
C) (5,0)
D) $(\sqrt{159}, 0)$

19

Demographic Characteristics of Education for random 200 participants

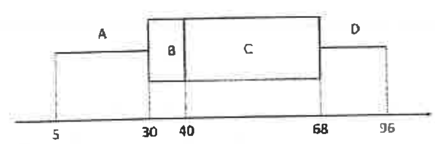
	Male	Female	Total
Less than high school	8	5	
High school degree	30		56
College degree	22	50	72
Bachelor's degree or higher		19	59
Total	100	100	200

In the survey above, some data were deleted by accident. If a female was chosen at random, what is the probability that the person has a high school degree or college degree?

A) 0.38
B) 0.69
C) 0.76
D) 0.81

20

ULTRA POWER COMPANY'S BATTERY LIFE



Battery life in hours

The boxplot represents the distribution of battery life in hours for Ultra Power Company in 2022. Which of the following must be true?

I. The number of batteries in region B is smaller than in region C.
II. Approximately 75% of the batteries have a battery life 30 hours or more.
III. The range of data above is 38.

A) I only
B) II only
C) I and II only
D) I, II, and III

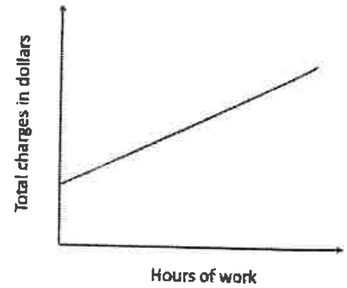
CONTINUE

21

What is the equation of the line which is perpendicular to $y = -\frac{1}{2}x + 3$ and passes through the point (1,4) in the XY-plane?

A) $y = \frac{1}{2}x + \frac{7}{2}$
B) $y = -\frac{1}{2}x + \frac{9}{2}$
C) $y = 2x + 2$
D) $y = -2x + 6$

22



The graph above represents total charges of plumbing work by a certain plumber. The plumber charges the basic fee plus an hourly rate. What is the best interpretation of the y intercept in the graph?

A) The plumber's total charge
B) The plumber's hourly rate
C) One-time basic fee
D) The plumber's minimum hourly rate

STOP

If you finish before time is called, you may check your work on this module only. Do not turn to any other module in the test.

Math

22 QUESTIONS
(TIME: 35 MIN)

DIRECTIONS

The questions in this section address a number of important math skills.
Use of a calculator is permitted for all questions.

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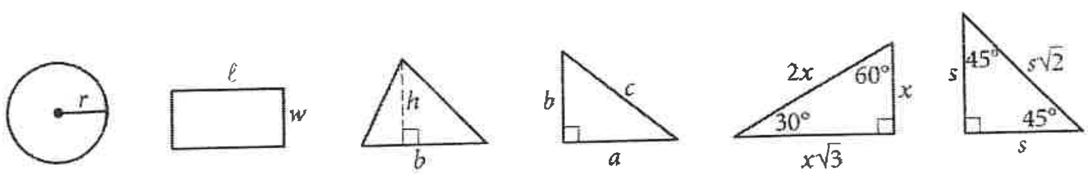
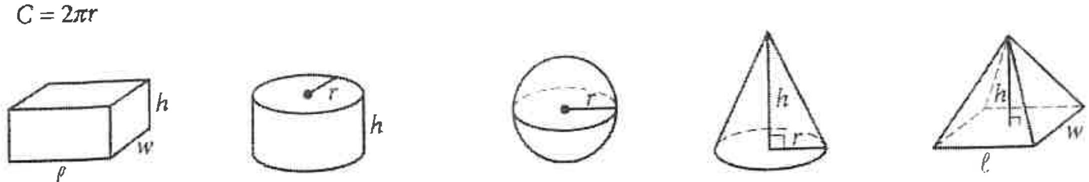
REFERENCE



$A = \pi r^2$ $C = 2\pi r$ | $A = \ell w$ | $A = \frac{1}{2}bh$ | $c^2 = a^2 + b^2$ | Special Right Triangles



$V = \ell wh$ | $V = \pi r^2 h$ | $V = \frac{4}{3}\pi r^3$ | $V = \frac{1}{3}\pi r^2 h$ | $V = \frac{1}{3}\ell wh$

The number of degrees of arc in a circle is 360.
The number of radians of arc in a circle is 2π.
The sum of the measures in degrees of the angles of a triangle is 180.

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1

$$f(t) = 3000(1 + 0.05)^t$$

Margaux opened a CD account for her saving. This account offers a certain annual interest rate. If t is time in year after she deposit money into her account, what is the best interpretation of 0.05 in the function above?

A) Her annual contribution in amount of money.
B) The annual interest rate in decimal for her account
C) The first deposit when she opened her account.
D) The extra one-time bonus from the bank.

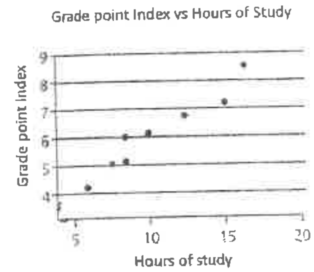
2

x	f(x)
1	22
2	25
3	28

The table shows some values of x and their corresponding f(x) values. If f(x) is a linear function, which of the following function defines f(x)?

A) $f(x) = -3x + 25$
B) $f(x) = 3x + 19$
C) $f(x) = 3x + 22$
D) $f(x) = -3x + 19$

3



The scatterplot above shows the relationship between Grade point index $(0 - 10)$ and Weekly hours of study for a calculus class. Which of the following statements best interprets the graph?

A) It is guaranteed that more hours of study will result in higher grade point index.
B) It is plausible that more hours of study could result in higher grade point index.
C) This scatter plot shows very weak relationship.
D) We can assume that anyone can get at least 5 in grade point index with no hours of study.

4

If a triangle has lengths of sides $3\sqrt{3}, 4\sqrt{3}, and\ \sqrt{75}$ units, what is the area of the triangle, in square units?

A) 18
B) 22.5
C) 30
D) $6\sqrt{3}$

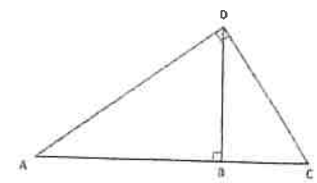
5

$$2(4-x)-5=3-3y$$

From the equation shown above, which of the following equations represents correctly between x *and* y?

A) $\frac{x-4}{y-1}=1$
B) $\frac{x}{y}=\frac{3}{2}$
C) $\frac{y}{x}=\frac{3}{2}$
D) $y=6x$

7



Note: Not drawn to scale.

In the right triangle above, $\overline{AD}$=8 and $\cos A=0.5$. What is the length of $\overline{BD}$?

A) $4\sqrt{3}$
B) $16\sqrt{3}$
C) 6
D) $6\sqrt{3}$

6

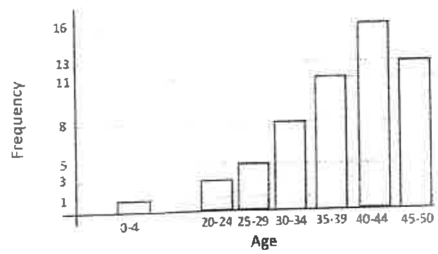
$$2\sqrt[3]{27x^{36}}\cdot\sqrt[5]{-32x^{10}}$$

The expression above could be simplified as kx^m, where k and m are positive numbers. What is the value of $m-k$?

8

The function f in the $xy-plane$ is defined by $f(x)=a\cdot b^x$, where a *and* b are constants.
If the function is translated down 3 units, the y intercept of the function is (0, 5) and passes through (1, 0). What is the value of the product of a *and* b?

9



In the bar graph above, which of the following statements must be true?

I. The mean age of the data is located more left than the median of the data.
II. If the smallest age is removed, the mean value will change more than the median value of the data will change.
III. The range of data is 50.

A) I only
B) II only
C) I and II only
D) I, II, and III

10

$$y < x + 2$$

$$y > -2x - 1$$

If the point $(1, k)$ is a solution to the inequality system above in the $xy-plane$. How many integers of k satisfies the system?

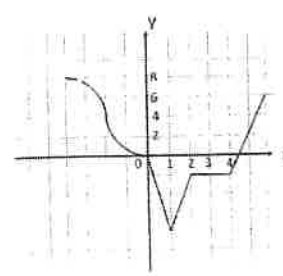
I) 2
J) 4
K) 5
L) 7

11

Adrian studies SAT practice test book for his upcoming test. He finishes one page for 15 minutes in Math and two pages for 10 minutes in English, respectively. He wants to know how long, in minutes, it will take in total to complete the practice book. If the book consists of k pages for Math and w pages for English, which expression best represents the situation?

A) $15k + 10w$
B) $10w + 15k$
C) $15k + 5w$
D) $15k + 20w$

12

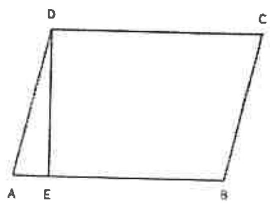


The graph of $y = f(x)$ is shown above in the xy-plane. What is the value of $f(-f(2))$?

A) -2
B) -4
C) 4
D) 1

CONTINUE

13



In the parallelogram ABCD above, $\overline{AD} = 6$ and $\overline{DE} \perp \overline{AB}$. Which of the following expressions represents the height, $\overline{DE}$, of the parallelogram?

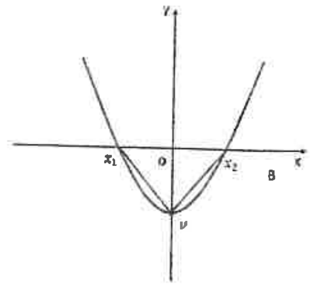
A) $6cos\angle A$
B) $6tan\angle A$
C) $6sin\angle A$
D) $\frac{sin\angle A}{6}$

14

$$x^2 + y^2 = 26$$
$$2xy = 23$$

In the system of non-linear equations above, if x and y are positive numbers, what is the sum of x and y?

15



The graph of $f(x) = 2x^2 - 8$ is shown above in the XY-plane. What is the area of Δx_1x_2v?

16

Each worker is in either department A or department B in XYZ company. If the number of workers in department A is three time the number of workers in department B and the average salary of workers in department A is $50,000 and the average salary for workers in department B is $54,000, what is the average of salary of the entire workers in the company?

A) $51,000
B) $51,500
C) $52,000
D) $52,500

17

Annual percent change in sales for five stores 2006-2008

Store	Percent Change from 2006 to 2007	Percent Change from 2007 to 2008
P	10	−10
Q	−20	9
R	5	12
S	−7	−15
T	17	−8

The table above shows Annual percent change in sales for 5 stores for 2006-2008. Which of the statements must be true?

I. The net percent change for store P from 2006 to 2007 is 0.
II. Total sales amount in dollars in store R for 2006-2008 is larger than total sales amount in store S.

A) I only
B) II only
C) I and II
D) None

18

If $y = 4x$ and $z = -2y$, what is $\frac{x+2y+3z}{5}$ in terms of x?

A) $3x$
B) $-3x$
C) $\frac{3}{5}x$
D) $\frac{9}{5}x$

19

$$2x^2 - 4x + 2y^2 + 8y = -2$$

In the circle equation above, what is the value of the radius of the circle?

20

Thomas Williams	Fran Rodriguez
456	395

The table above shows the result of a survey which was conducted at random for a total of 851 voters for a certain election. If 4,255 people vote for the actual election, by how many votes would Thomas Williams be expected to win the election?

CONTINUE

21

$f(x)$ *equals* 82 *percent of* x

For $x > 0$, the function is defined as above. Which of the following best describes the function?

A) Linearly decreasing
B) Exponentially decreasing
C) Linearly increasing
D) Exponentially increasing

22

Which of the following rational expressions are equivalent to $\frac{1}{x+1} - \frac{x}{x-2} + \frac{x^2+2}{x^2-x-2}$?

A) 0
B) $\frac{2x}{x^2-x-2}$
C) $\frac{-4}{x^2-x-2}$
D) $\frac{x^2-x-1}{x^2+x-3}$

STOP

If you finish before time is called, you may check your work on this module only. Do not turn to any other module in the test.

2 Module 1 2

Math

35 MINUTES, 22 QUESTIONS

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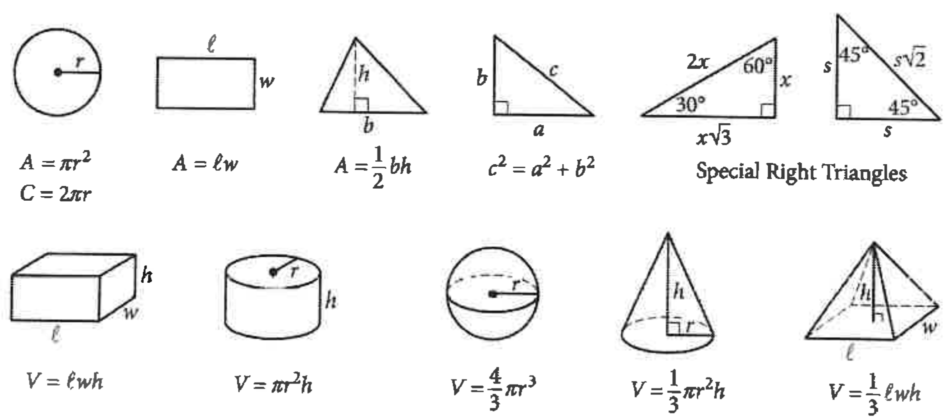

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π.

The sum of the measures in degrees of the angles of a triangle is 180.

2 Module 1 2

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2 Module 1 2

1

If $\frac{5}{3}y + 2 = 7$, what is the value of y?

A) 5

B) $\frac{3}{5}$

C) $\frac{19}{5}$

D) 3

2

A school plans to spend a total of $800 to buy some pens and notebooks. If each pen costs $2 and each notebook costs $5, which inequality represents the possible number of pens, p, and notebooks, n, the school can buy?

A) $2p + 5n < 800$

B) $\frac{1}{2}p + \frac{1}{5}n \leq 800$

C) $2p + 5n \leq 800$

D) $\frac{p}{2} + \frac{n}{5} < 800$

3

If $f(x) = 5x^2 + 3$ and $f(k) = 48$, what is the value of k^2?

A) 3

B) -3

C) 9

D) 5

4

An ice cream vendor sells ice cream at a night market on Sunday night. Each ice cream costs him $4, so he charges $8 for each ice cream at the night market. He also has to pay $20 per hour to rent a booth. If the vendor sells c ice creams and rents the booth for h hours, which of the following is equivalent to the vendor's profit, p, in dollars?

A) $p = 8c - 20h - 4$

B) $p = 4c - 20h$

C) $p = 4h - 8c$

D) $p = 12c - 20h$

5

If $f(x) = x - \sqrt{3x - 1}$, at which of the following values is $f(x)$ undefined?

A) $x \leq \frac{1}{3}$

B) $x < \frac{1}{3}$

C) $x = 1$

D) none

6

In the xy-plane, a circle has center $(0,m)$ and radius 5. The point (4,3) is on the circle. If m is a constant and $m \neq 0$, what is a possible value of m?

A) 6

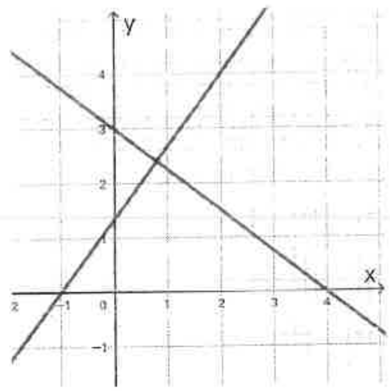
B) 3

C) -6

D) 0

2 Module 1 2

7



The graph shown above represents which of the following systems of equations?

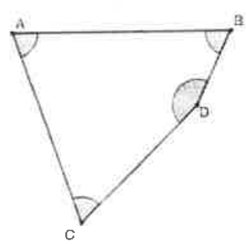
A) $4y - 3x = 12$
$3y + 4x = 4$

B) $4y - 3x = 12$
$3y - 4x = 4$

C) $4y + 3x = 12$
$4y - 3x = 4$

D) $4y + 3x = 12$
$3y - 4x = 4$

8

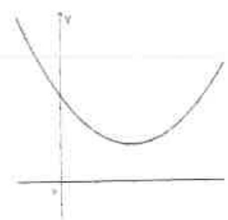


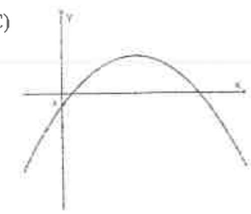
The quadrilateral $ABCD$ is shown, where $\angle A = 70°$, $\angle B = 65°$, and $LC = 63°$. How many degrees is the angle D?

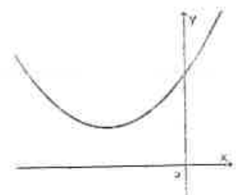
9

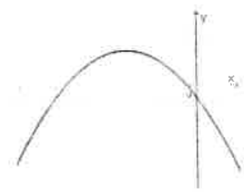
$$f(x) = a(x - m)^2 + n$$

In the function f above, a, m, and n are constants and are greater than zero. Which of the following graphs represents the function f?

A)


C)


B)


D)


10

For a bank savings account, the annual interest rate was 2.5% in 2021 and the annual interest rate was 3.5% in 2022. If Jack deposited \$1,000 into his savings account on January 1, 2021, how many dollars would Jack withdraw on December 31, 2022? (Round to the nearest dollar)

A) 1035

B) 1061

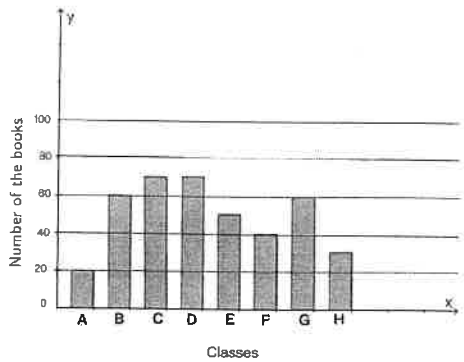
C) 1071

D) 1051

11

A cyclist rides from his house to a park in 5 hours at a constant speed of 12 mph while he rides back home at a constant speed of 15 mph. How many hours did he take to return to his house?

12



The histogram above shows the number of books read by each class during a school reading event. If the students in the 8 classes read a total of 420 books, how many books did the students in class E read?

13

	Vegetarian	Non-vegetarian	Total
Female	160	200	360
Male	50	200	260
Total	210	400	610

A researcher conducted a survey to determine the number of vegetarians in a small community. The results are shown above. If a female is selected at random, what is the probability that the female is vegetarian?

A) $\frac{16}{61}$

B) $\frac{4}{9}$

C) $\frac{36}{61}$

D) $\frac{5}{9}$

14

$$6x + 8 = 20$$

Which equation has the same solutions as the given equation above?

A) $6x = 28$

B) $2x + 4 = 10$

C) $5x + 8 = 18$

D) $3x + 4 = 20$

15

The function is defined by $f(x) = x^3 + 3x^2 - 6$. What is the value of $f(-2)$?

A) -2

B) -10

C) 14

D) -26

16

A person spent a total of \$25,500 to buy two stocks, A and B, and sold both when the values of stock A rose by 15% and the values of stock B fell by 10%, making a total profit of \$1,650. What was the value of stock A when the person first purchased it?

17

The price of potatoes is \$2.25 per pound. How many kilograms of potatoes would cost 20 dollars? (1 pound = 0.45 kilograms)

A) 45

B) 8

C) 4

D) 49.38

18

If 20% of the total books Lucas plans to read this year is 60, how many books does Lucas plan to read ?

A) 24

B) 300

C) 48

D) 150

19

Which expression is equivalent to :

$$(x^5 \cdot y^{-3} \cdot z^2)(x^{-4} \cdot y^2 \cdot z^{-1})?$$

A) $x \cdot y^{-1} \cdot z$

B) $x \cdot y^{-6} \cdot z$

C) $x^{-9} \cdot y^{-5} \cdot z^{-3}$

D) $x \cdot y \cdot z$

20

Circle x has an area of 4π and the perimeter of circle y is 4 times the perimeter of circle x. How many times the radius of circle x is the radius of circle y?

A) 4

B) 2

C) 3

D) 5

21

A company's number of employees increased from 900 at the beginning of the year to 1008 at the end of the year. What is the percent increase in the company's number of employees?

A) 12%

B) 10.7%

C) 11%

D) 13%

22

$$f(x) = -x^2 - 2x + 3$$

The function f is given. Which table of values represents $y = f(x) + 3$?

A)

x	y
-1	7
-2	6
1	4

B)

x	y
-1	7
-2	6
1	5

C)

x	y
-1	7
-2	6
1	3

D)

x	y
-1	7
-2	6
1	2

No Test Material On This Page

2 Module 2 2

Math

35 MINUTES, 22 QUESTIONS

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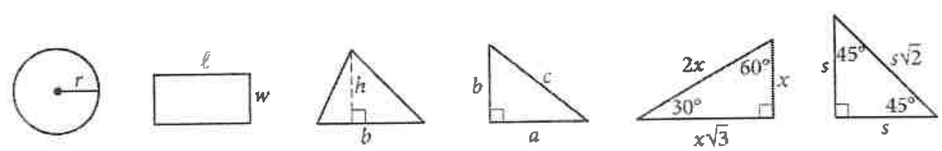
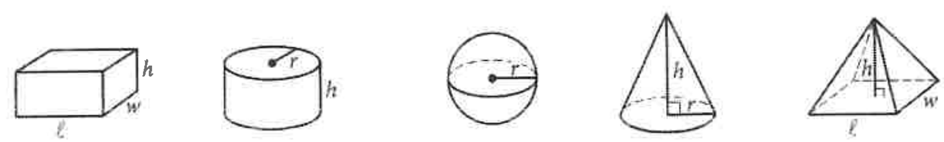


$A = \pi r^2$, $C = 2\pi r$ $A = \ell w$ $A = \frac{1}{2}bh$ $c^2 = a^2 + b^2$



$V = \ell wh$ $V = \pi r^2 h$ $V = \frac{4}{3}\pi r^3$ $V = \frac{1}{3}\pi r^2 h$ $V = \frac{1}{3}\ell wh$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π.

The sum of the measures in degrees of the angles of a triangle is 180.

2 Module 2 2

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2 Module 2 2

1

If $6x - 3y = 4$, what is the value of $\frac{3y+4}{3x}$?

2

$f(x) = x^2 + 8x + 17$, if $y = f(x)$ in the xy-plane, what is the value of $f(x)$?

A) 1

B) 65

C) -31

D) 4

3

$$y - 2x = 2$$
$$2y + 4x = 5$$

Based on the system of equations above, what is the value of $2(y^2 - 4x^2)$?

A) 9

B) 11

C) 5

D) 10

4

In a graph of the xy-plane, line l is perpendicular to line m. Which of the following systems of equations represents the two lines?

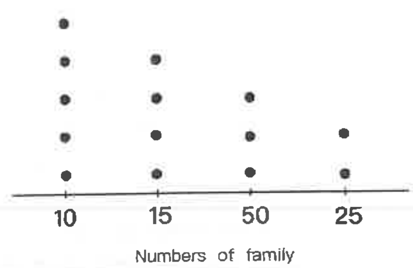
A) $5y - 4x = 20$
$4y + 5x = 15$

B) $4y - 2x = 5$
$4y + 2x = 6$

C) $3y - x = 5$
$2y + x = 3$

D) $y - 3x = 2$
$y + 3x = 3$

5



100 families in a certain area were asked how many children they had, and the results are shown in the dot plot above. What is the median number of children per family for the 100 families?

6

$$x^2 + 6x + c = 0$$

In the given equation, c is a constant. If the equation has no real solution, what is the value of c?

A) 4

B) 9

C) 10

D) 8

7

In the right triangle MQP, the angle Q has a measure of 90° and the two legs of the triangle have lengths of 4 and 8, respectively. What is the measure of the largest acute angle?

A) 45°

B) 60°

C) 30°

D) 20°

8

What number is 120% of 80?

2 Module 2 2

9

$$P(x) = 150(F)^x$$

The function $P(x)$ models the specific population of wildlife t years after the start of the study, where F is a constant. If the population of wildlife decreases by 12% each year, what is the value of F?

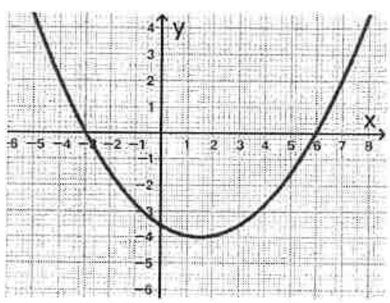
A) 1.12

B) 0.88

C) 0.12

D) 0.10

10



What is the y-intercept of the graph shown?

A) $(-3.5, 0)$

B) $(-3, 0)$

C) $(0, -3.5)$

D) $(6, 0)$

11

Alex walked at a constant speed of 8 kilometres per hour for thirty minutes and slowly walked for another thirty minutes at an average rate of 5 miles per hour. What is the total distance Alex traveled, in miles? (1 kilometer = 0.62 miles)

A) 4.98

B) 6.5

C) 8.03

D) 7

12

In the xy-plane, if $y = f(x) = x^2 + 6x - 3$, and $y = g(x) = x^2 - 11x + 14$, $f(x)$ and $g(x)$ intersect at point(m, n), what is the value of n?

13

From Los Angeles to Vancouver, the driving distance is 2055 kilometers, which takes approximately 24.5 hours. The flight distance is 1740 kilometers and the flight time is approximately 3 hours. Approximately how many times the average driving speed is the average flying speed? (To the nearest whole number)

A) 8

B) 7

C) 10

D) 15

14

$$f(x) = ax^2 - 16x + 24$$

The given equation $y = f(x)$ intersect x-axis at $(2, 0)$ and $(m, 0)$ in the xy-plane, where a and m are constants. What is the value of m?

15

$$x - 6 = -10$$
$$y = x^2 - 3x - 9$$

Which ordered pair (x, y) is a solution to the given system of equations?

A) $(-4, -5)$

B) $(-4, 19)$

C) $(4, 5)$

D) $(4, 19)$

2 Module 2 2

16

An isosceles triangle has a height of 4 inches and a base of 6 inches. What is the perimeter of this triangle, in inches?

A) 12

B) 16

C) 13

D) 32

17

Mike and Alex typed at constant speeds and together they typed a total of 1440 words in 4 minutes. If Mike typed 14 less words per minute than Alex, how many words did Mike type per minute?

18

Which expression is equivalent to $(5x^3+9)-(5x^3+x^2)$?

A) $10x^3-x^2+9$

B) $(3+x)(3-x)$

C) $10x^3+x^2+9$

D) x^2-9

19

Which expression is equivalent to $\sqrt[3]{p^2}\cdot\sqrt[15]{p^5}$, where $p>0$?

A) $\sqrt[18]{p^7}$

B) $\sqrt[12]{p^3}$

C) $\sqrt[45]{p}$

D) p

20

Line l is tangent to the circle $x^2+y^2=2$ at the point $(1,-1)$. Which of the following equations represents the line l?

A) $x-y=4$

B) $x-y=2$

C) $x+y=2$

D) $x+y=4$

21

	English Test			French Test		
Marks	A	B	C	A	B	C
Number of Students	18	15	5	15	17	6

Grade 11 has 38 students. The table above shows the results of the recent English and French tests. If one of the students is chosen at random, what is the probability that the person received a B on the French test?

A) $\frac{17}{38}$

B) $\frac{17}{76}$

C) $\frac{17}{32}$

D) $\frac{32}{38}$

22

Victor reads books at an average rate of 3 days per book. Which function, y, models the number of days it will take Victor to read x books at this rate?

A) $y=3x$

B) $y=\frac{1}{3}x$

C) $y=x-\frac{1}{3}$

D) $y=x+\frac{1}{3}$

Math

22 QUESTIONS
(TIME: 35 MIN)

DIRECTIONS

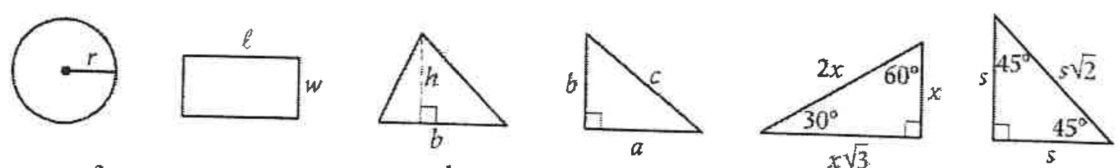
The questions in this section address a number of important math skills.
Use of a calculator is permitted for all questions.

NOTES

Unless otherwise indicated:

- All variables and expressions represent real numbers.
- Figures provided are drawn to scale.
- All figures lie in a plane.
- The domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

REFERENCE



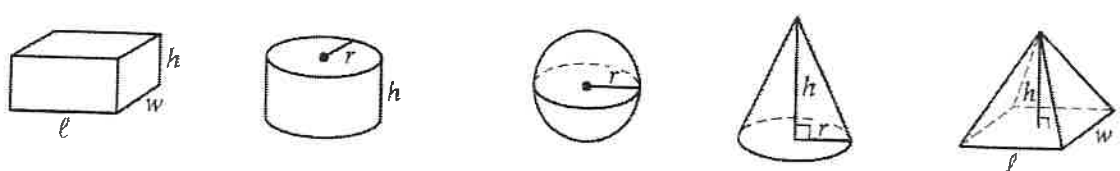
$A = \pi r^2$
$C = 2\pi r$

$A = \ell w$

$A = \frac{1}{2}bh$

$c^2 = a^2 + b^2$

Special Right Triangles



$V = \ell wh$

$V = \pi r^2 h$

$V = \frac{4}{3}\pi r^3$

$V = \frac{1}{3}\pi r^2 h$

$V = \frac{1}{3}\ell wh$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π.

The sum of the measures in degrees of the angles of a triangle is 180.

1

$$\frac{3x-9}{x^2-x-20} \cdot \frac{x^2-25}{x^2+2x-15}$$

If $x \neq 5, -5, and -4$, Which of the following expressions is equivalent to the above?

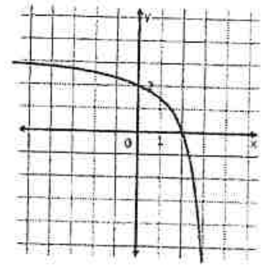
A) $\frac{3}{-x-4}$
B) $\frac{3(x-3)}{(x+4)(x-5)}$
C) $\frac{3}{x+4}$
D) $\frac{3}{x-5}$

2

Two dice are rolled at the same time. If both dice show the same numbers, then the player wins $70. Otherwise, the player receives $4. The cost of game is $10. What is the expected value per game considering the cost?

A) $50
B) $15
C) $5
D) $3.33

3



The partial graph of rational function $f(x) = \frac{k}{x+m} + 3$ is shown above, where k *and* m are constants. What is the value of $k+m$?

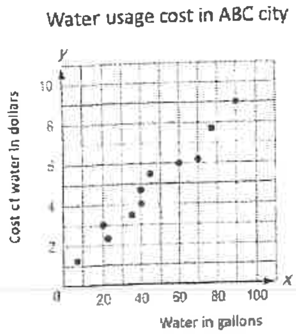
A) 3
B) -3
C) -2
D) 0

4

The maximum value of m is 8 less than two times the quantity of $n-4$. Which inequality represents the values of m?

A) $m \geq 2(n-4)-8$
B) $m \leq 2(n-4)-8$
C) $m \leq 8-2(n-4)$
D) $m \geq 8-2(n-4)$

5



The scatterplot above shows the relationship between the amount of water used in gallons and the cost of water usage in ABC city. If you predict the cost using the line of best fit, what is the closest amount, in dollars, to the cost of a 120 gallons water usage?

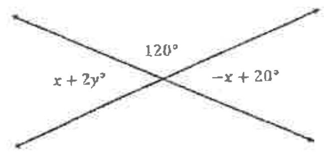
A) $11
B) $15
C) $20
D) $22

6

For an exponential function f, if $f(0)$ is a, where a is a constant. Which of the following equivalent forms of the function f shows the value of a as the coefficient of the function?

A) $f(x) = 23(1.2)^{x-1}$
B) $f(x) = -2(0.9)^{x}$
C) $f(x) = 3(1.1)^{x+1}$
D) $f(x) = 1.2(0.23)^{x-1}$

7



In the diagram above, what is the sum of values of x and y?

E) 10
F) 20
G) 90
H) 120

8

$$x^2 + 4x + k = 0$$

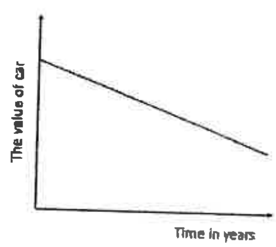
In the quadratic equation above, k is a constant. The equation has no real solution if $k > m$. What is the greatest possible value of m?

9

PSA test is a blood test that is commonly used to detect possible prostate cancer but it is known to produce false positive result which induces a patient undergoes unnecessary biopsy to confirm they don't have prostate cancer. If PSA test generates one false positive in 20 tests and a doctor conducts the test to three patients, which of the following shows the probability that none of patients will generate false positive?

A) $\left(\frac{1}{20}\right)\left(\frac{1}{20}\right)\left(\frac{1}{20}\right)$
B) $\left(\frac{19}{20}\right)\left(\frac{19}{20}\right)\left(\frac{19}{20}\right)$
C) $\left(\frac{3}{20}\right)$
D) $\left(\frac{1}{8000}\right)$

10



The graph above represents the values of a certain car over the years. If the value of the car follows the graph, which of the following statements best interprets the slope of the line?

A) The price when it was purchased
B) One time charge from the dealership
C) The annual decrease rate of the value of the car over years
D) The value of the car over years

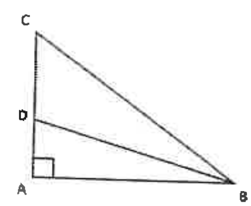
11

$$x = \frac{-b \pm \sqrt{D}}{2a}$$

In the quadratic equation of $ax^2 + bx + c = 0$, we can derive the quadratic formula as shown above. Which of the following equations shows D in terms of $a, b, and\ x$?

A) $D = \left(x + \frac{b}{2a}\right)^2$
B) $D = (b + 2ax)^2$
C) $D = \left(x - \frac{b}{2a}\right)^2$
D) $D = \frac{(x+b)^2}{2a}$

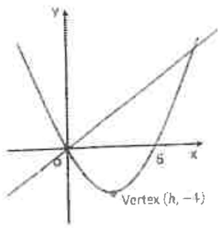
12



Note: Not drawn to scale.

In the figure above, $\overline{BD}$ *bisects* $\angle ABC$. If the length of $\overline{BD}$ is twice the length of $\overline{AD}$ and $\overline{AB} = 4$, What is the length of $\overline{BC}$?

13



The graphs of linear and quadratic functions in the XY-plane are shown above. If the vertex of parabola graph is $(h, -4)$ and the equation of line is $y = 4x$, what is the coordinates of the point of intersection?

A) $(3,12)$
B) $(10,20)$
C) $(12,40)$
D) $(15,60)$

14

$$\frac{m^{\frac{2}{3}}(m^{-2}m^{4})^{3}}{m^{4}}$$

Which of the following expressions is equivalent to the expression above, where $m > 0$?

A) $\sqrt[3]{m^8}$
B) $\sqrt[8]{m^3}$
C) $\sqrt[4]{m^3}$
D) $\sqrt{m^{-3}}$

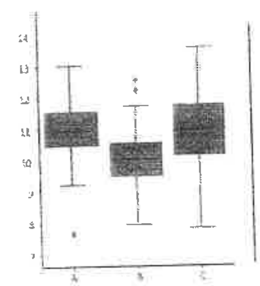
15

$$7, 4, 5, 4, 5, 6, 6, 7, 3, 10$$

Ten people were selected at random to participate in a survey. The numbers represent the scale of satisfaction of life from 1 to 10. The list above shows the result of the survey. What is the fraction of the number of people who gave a number of 5 or higher to the number of people who gave a number of lower than 5?

A) $\frac{3}{7}$
B) $\frac{7}{3}$
C) $\frac{3}{2}$
D) $\frac{2}{3}$

16



The box plots summarize the number of vacation days for the employees for three companies. Which of following statements represents correctly for the plots above?

I. The range of numbers of vacation days of company C is the largest.
II. The mode of number of vacation days is 11 for two companies.
III. The median value of vacation days for company B is the smallest.

A) I only
B) I and II only
C) I and III only
D) I, II, and III

CONTINUE

17

x	f(x)
0	4
2	36
4	324

For the exponential function f(x), if $f(x) = a \cdot b^x$, where a and b are constants. Which of the following functions represents the data in the table above?

E) $f(x) = 3 \cdot 4^x$
F) $f(x) = 4 \cdot 3^x$
G) $f(x) = 3 \cdot \left(\frac{1}{4}\right)^x$
H) $f(x) = 4 \cdot \left(\frac{1}{3}\right)^x$

18

The average price per pound of lamb meat in a certain butcher shop started at $7.20. However, it was increased at a constant rate each week for several weeks until it reached $8.45 because of high inflation. The equation, $7.20 + 0.25x = 8.45$, represents this situation, where x stands for the number of weeks after the average price per pound started to increase. Which of the following statements best represents 0.25 in this context?

A) The average price per pound of lamb meat
B) The weekly rate of change in the average price per pound of lamb meat
C) The percent change in average price per pound of lamb meat
D) The total increase in average price per pound while it reached $8.45.

19

The monthly payment for a certain gym in 2021 was 1.45 times the monthly payment in 2020. By how much percent did the monthly payment increase from 2020 to 2021?

A) 0.45%
B) 1.45%
C) 145%
D) 45%

20

$$-2ax + by - 3 = 0$$

In the linear equation above, where a and b are non-zero constants. If $2a + b = 0$, which of the following statements correctly describes the graph in the XY-plane?

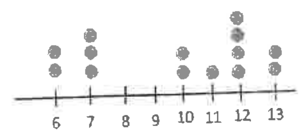
A) The y intercept of the graph is positive.
B) The slope of the graph is positive.
C) The slope of the graph is zero.
D) The slope of the graph is -1.

21

The television broadcasting company conducts a survey for a tax raise issue. A telephone number is provided and respondents are asked to call and register their opinions by pressing 1 if they support it and 2 if they oppose it. Which of the following statements best represents this survey?

A) This survey is non-biased because the survey is open to anyone.
B) This survey could be biased because only those who feel strongly either for or against will call and register their opinions voluntarily.
C) This survey is well designed because of simple random samples.
D) This survey is not trustworthy because some radio broadcasting company can do the same survey.

22



In the dot-plots shown above, what is the value of median of data?

STOP

If you finish before time is called, you may check your work on this module only. Do not turn to any other module in the test.

CONTINUE

Math

22 QUESTIONS
(TIME: 35 MIN)

DIRECTIONS

The questions in this section address a number of important math skills.
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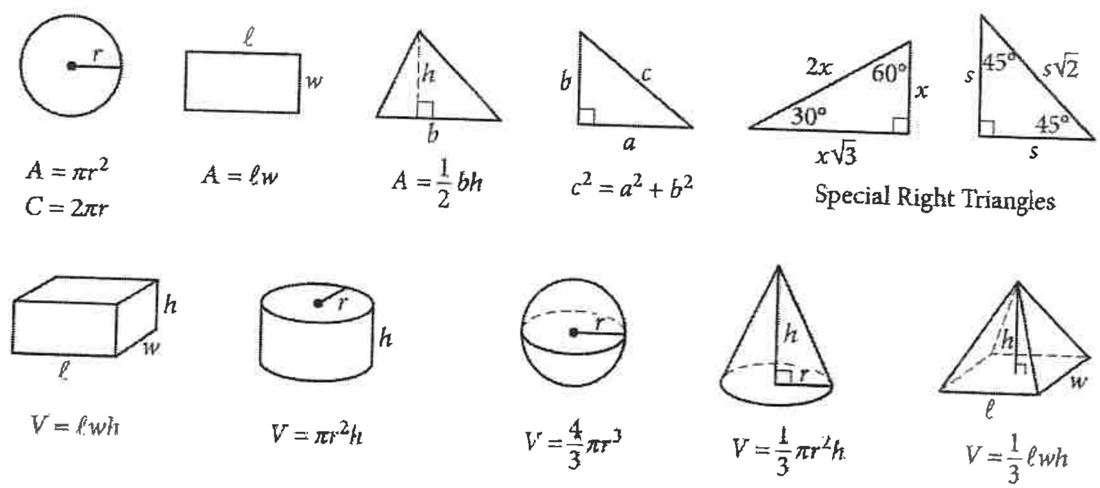
REFERENCE



The number of degrees of arc in a circle is 360.
The number of radians of arc in a circle is 2π.
The sum of the measures in degrees of the angles of a triangle is 180.

Module 2

For multiple-choice questions, solve each problem, choose the correct answer from the choices provided, and then circle your answer in this book. Circle only one answer for each question. If you change your mind, completely erase the circle. You will not get credit for questions with more than one answer circled, or for questions with no answers circled.

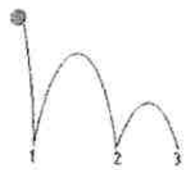
For student-produced response questions, solve each problem and write your answer next to or under the question in the test book as described below.

- Once you've written your answer, circle it clearly. You will not receive credit for anything written outside the circle, or for any questions with more than one circled answer.
- If you find **more than one correct answer**, write and circle only one answer.
- Your answer can be up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer, but no more.
- If your answer is a **fraction** that is too long (over 5 characters for positive, 6 characters for negative), write the decimal equivalent.
- If your answer is a **decimal** that is too long (over 5 characters for positive, 6 characters for negative), truncate it or round at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), write it as an improper fraction (7/2) or its decimal equivalent (3.5).
- Don't include **symbols** such as a percent sign, comma, or dollar sign in your circled answer.

CONTINUE

1

Ball Free-fall diagram



A tennis ball bounces on the ground as shown. The ball is released from the initial height, h_0 and reaches out approximately two third the previous height after each bounce. Which of the following equations best represents the height of the ball after n bounce, where n is the number of bounce after it is released?

A) $h(n) = h_0\left(\frac{1}{3}\right)^n$
B) $h(n) = h_0\left(\frac{2}{3}\right)^n$
C) $h(n) = h_0 - \frac{h_0}{3^n}$
D) $h(n) = h_0\left(1 - \left(\frac{2}{3}\right)^n\right)$

2

The chance of an accident on the freeway 23 on a dry day is 0.10%. but the chance of an accident will go up to 0.50% on a rainy day. If there is a 30% chance of the weather being rainy today, what is the probability that there will be an accident on the freeway 23 today?

A) 0.22
B) 0.0022
C) 0.38
D) 0.0038

3

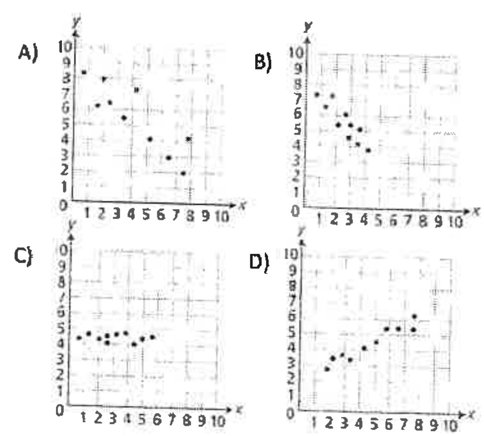
$$4r^2q^4 + r^3 - 4(q^2 + rq^2)^2$$

Which of the followings is equivalent to the expression above?

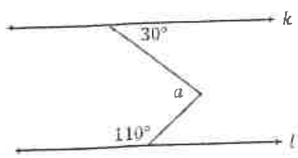
A) $r^3 - 4q^4 + 8rq^4$
B) $r^3 - 4q^4 - 8r^2q^4$
C) $r^3 - 4q^4 - 8rq^4$
D) $r^3 - 4q^4 - 4rq^4$

4

Which of the following scatter plots could be described the line of best fit $y = kx$, where k is a constant?



5



In the diagram above, lines k and l are parallel. What is the measure of angle a in degrees?

A) 80°
B) 90°
C) 100°
D) 110°

7

$$|x-2| = 8$$
$$-|y+3| = -4$$

In the system of absolute equations above, what is the least possible value of $x+y$?

A) -5
B) -7
C) -13
D) -17

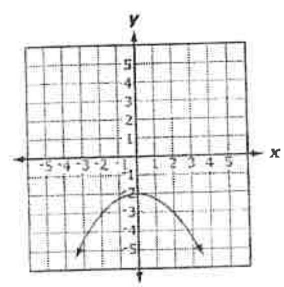
6

$$-x^2 + 5x + 3 = 0$$

In the quadratic equation above, what is the solution to the equation?

A) $\frac{-5\pm\sqrt{37}}{2}$
B) $\frac{5\pm\sqrt{37}}{-2}$
C) $\frac{5\pm\sqrt{37}}{2}$
D) $\frac{\sqrt{37}\pm5}{2}$

8



Which of the following equations represents the graph above?

A) $y = -\frac{1}{4}x^2 - 2$
B) $y = -4x^2 - 2$
C) $y = -\frac{1}{2}x^2 - 2$
D) $y = \frac{1}{4}x^2 - 2$

CONTINUE

9

Which of the following equations is correct for a circle with center $(2, 0)$ and diameter 6?

A) $(x+2)^2 + y^2 = 9$
B) $(x-2)^2 + y^2 = 36$
C) $(x+2)^2 + y^2 = 36$
D) $(x-2)^2 + y^2 = 9$

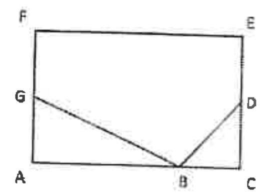
10

$$f(x) = -(2)^x + 1$$

What are the x intercept and y intercept, respectively, for the exponential function above?

A) $(0, 0)$ on both
B) $(1, 0), (0, -1)$
C) $(-1, 0), (1, 0)$
D) $(1, 0), (0, 0)$

11



In the rectangle shown above, D and G are the midpoints of $\overline{CE}$ and $\overline{AF}$, respectively and the length of $\overline{BC}$ is the same as the length of $\overline{DC}$. If the length of $\overline{BG}$ is twice the length of $\overline{AG}$, what is the measure of $\angle DBG$, in degrees?

12

$$80, 39, 210, 152, 244, 239, 232, 112$$

The internet store manager kept on track of the number of items sold over an eight-week period. The results are shown above. What is the positive difference between mean and median of these lists?

A) 11.5
B) 17.5
C) 46.5
D) 48.5

13

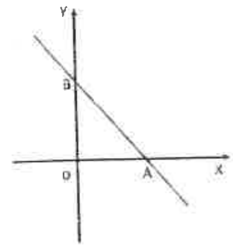
If $2xy = 3$, what is the value of $\left(\frac{y}{3x}\right) \cdot (2x)^2$?

14

If a function f, is defined as $y = f(x)$. Which of the following statements is correctly described the transformation for $y = -f(x+1)$?

A) Translate the graph horizontally 1 unit to the right and then reflect it over the x axis.
B) Reflect the graph over the x axis and then translate it horizontally 1 unit to the left.
C) Translate the graph horizontally 1 unit to the right and then reflect it over the x axis.
D) Reflect the graph over the y axis and then translate it vertically 1 unit up.

15

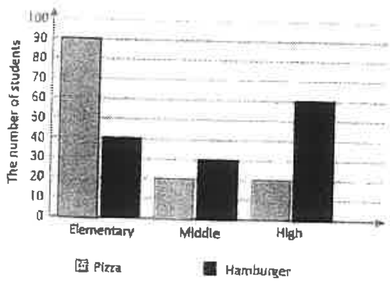


If a line passes through two points $A(n, 0)$ *and* $B(0, m)$ on the axes in the XY-plane as shown above. Which one is correct expression for $\cos\angle OAB$ in terms of m *and* n?

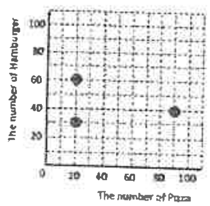
A) $\frac{n}{\sqrt{m^2+n^2}}$
B) $\frac{m}{\sqrt{m^2+n^2}}$
C) $\frac{m}{n}$
D) $\frac{\sqrt{m^2+n^2}}{m}$

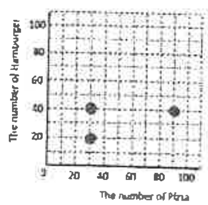
CONTINUE

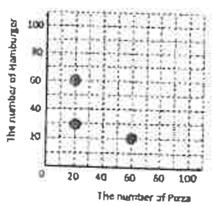
16

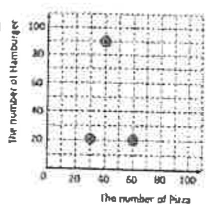


The bar graph above shows the distribution of lunch choices for 130 orders of pizza and 130 orders of hamburger in ABC school district. Which of the scatter plots correctly represents the bar graph?

A) 

B) 

C) 

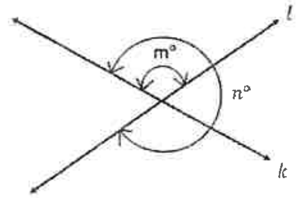
D) 

17

Julie bought a new phone with promotion. She paid a promotion discount price that was 30% less than the list price (\$1,200). She also paid a sales tax equal to 9% of the list price. What is the total amount of cost in dollars for her purchase?

A) \$915.60
B) \$948.00
C) \$764.40
D) \$756.00

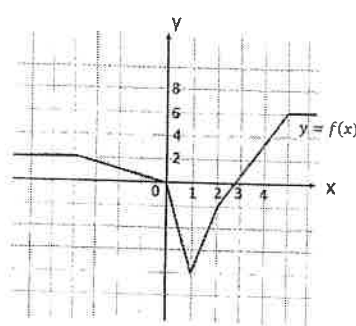
18



Note: Not drawn to scale.

In the figure above, lines l and k intersect at one point and form angles. If $n = 3m$, what is the measure of angle m, in degrees?

19



The graph of f in the XY-plane is shown above. What is the value of $-f(2) + 2 \cdot f(-5)$?

E) 3
F) 4
G) 5
H) 6

CONTINUE

20

If $sin(2a - 15)° = cos(a + 30)°$, what is the value of a?

A) 25
B) 35
C) 45
D) 55

21

$$C(x) = \frac{450x}{270 - x}$$

The cost C, in thousands of dollars, to pave $x\%$ of the entire dirt road in a certain rural area with brand-new asphalt is modeled by the function above. Approximately what percent of the entire dirt road could be paved with 120 thousand dollars budget?

A) 50
B) 57
C) 62
D) 67

22

$$f(k) = 2.88(0.94)^k$$

The function above represents the number of ICU (Internal Combustion Units), in billions, in a certain country, where k is the number of years since 2020. Which of the following best represents the number 0.94 in this context?

A) The estimated percent decrease of ICU in the entire country in 2020.
B) The number shows that the number of ICU in the country is decreasing 6% from the number of ICU in the previous year.
C) The number shows that the number of ICU in the country is increasing 94% annually for the number of ICU in the previous year.
D) The estimated number of ICU in billions in k years after 2020.

STOP

If you finish before time is called, you may check your work on this module only. Do not turn to any other module in the test.

2 Module 1 2

Math

35 MINUTES, 22 QUESTIONS

DIRECTIONS

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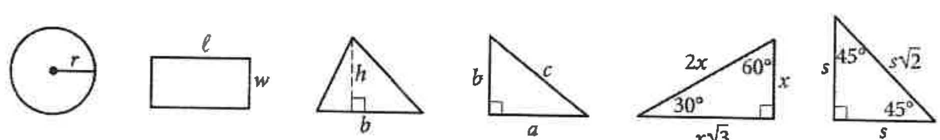
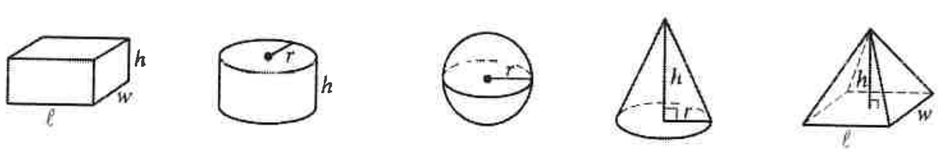
REFERENCE



$A = \pi r^2$, $C = 2\pi r$ | $A = \ell w$ | $A = \frac{1}{2}bh$ | $c^2 = a^2 + b^2$ | Special Right Triangles



$V = \ell wh$ | $V = \pi r^2 h$ | $V = \frac{4}{3}\pi r^3$ | $V = \frac{1}{3}\pi r^2 h$ | $V = \frac{1}{3}\ell wh$

The number of degrees of arc in a circle is 360.
The number of radians of arc in a circle is 2π.
The sum of the measures in degrees of the angles of a triangle is 180.

2 Module 1 2

For multiple-choice questions, solve each problem, choose the correct answer from the choices provided, and then circle your answer in this book. Circle only one answer for each question. If you change your mind, completely erase the circle. You will not get credit for questions with more than one answer circled, or for questions with no answers circled.

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2 Module 1 2

1

If $\frac{1}{2}x + \frac{2}{3}x = 7$, what is the value of $\frac{5}{6}x$?

A) 5

B) 1

C) 6

D) $\frac{7}{6}$

2

$$|m - 3| < 6$$

How many integers m satisfy the inequality above?

A) 10

B) 12

C) 11

D) 13

3

$f(x) = \frac{(x^2 - 2x + 1)}{x - 1}$, where $x \neq 1$. Which of the following statements is true?

A) $f(138) = 139$

B) $f(138) = 137$

C) $f(138) = 2567$

D) $f(138) = \frac{5}{137}$

4

Anne reads books every day. She reads x pages each weekday. Every Saturday and Sunday, she reads 4 times the number of pages she reads on each weekday. If Anne read for 30 days, of which 8 days were Saturdays and Sundays, how many pages, y, did Anne read?

A) $y = 38x$

B) $y = 30x$

C) $y = 22x$

D) $y = 54x$

5

If $g(x) = 3x^3 - 8x + 1$ and $f(x) = g(3x)$, what is $f(1)$?

A) -12

B) 58

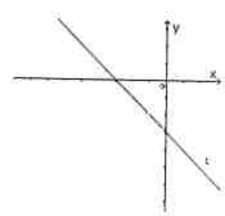
C) 106

D) 56

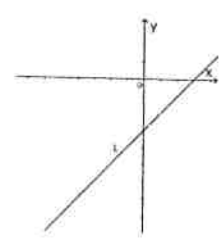
6

Which of the following graphs represent $y = kx - 2$, where $k < 0$?

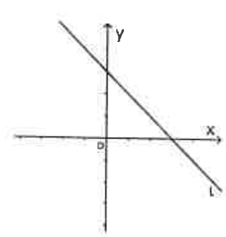
A)



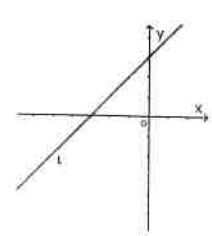
C)



B)



D)



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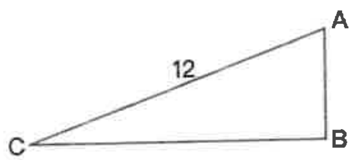


In the right triangle ABC above, AC=12. Which of the following expressions represents the length of AB ?

A) $\frac{\cos(A)}{12}$

B) $12 \cdot \tan(A)$

C) $12 \cdot \cos(A)$

D) $12 \cdot \sin(A)$

8

x	$f(x)$
2	2
6	6
8	14

Some values of $f(x)$ and their corresponding values of x are provided in the table shown above, If $f(x)$ is a quadratic function $f(x) = ax^2 + bx + c$, which of the following equations defines $f(x)$?

A) $f(x) = \frac{1}{2}x^2 + 3x + 6$

B) $f(x) = \frac{1}{2}x^2 - 3x + 6$

C) $f(x) = \frac{1}{2}x^2 - 3x - 6$

D) $f(x) = -\frac{1}{2}x^2 + 3x + 6$

9

Line l intersects circle $x^2 + y^2 = r^2$ and (3, 4) is a point on the circle. What is the diameter of the circle?

10

Item	Price (each)
A bottle of wine	30
A bar of chocolate	5
A papaya	15
A basket	5

A grocery store sells holiday baskets with various products. The table above shows the regular price of each of the items. Each basket contains 2 bottles of wine, 4 bars of chocolate, and 5 papayas. If the price of the basket was 15% off the sum of each of the items' prices, what is the sale price for each basket?

11

It takes Clark 30 minutes to walk from his house to school. His speed for the first half of his walk is 100 meters per minute, and his speed for the second half his walk is 150 meters per minute. What is the total distance, in meters, between Clark's house and the school?

A) 3600

B) 1800

C) 900

D) 2400

2 Module 1 2

12

	Bachelor's	Master's	Total
Java	300	61	361
Python	280	57	337
Github	84	18	102
Total	664	136	800

A tech company posted information regarding their employees' software preferences and the post-secondary degrees they have. If one employee is chosen at random, what is the probability that the employee prefers using Java and has a bachelor's degree?

A) $\frac{300}{361}$

B) $\frac{75}{156}$

C) $\frac{3}{8}$

D) $\frac{83}{100}$

13

Andrew wants to purchase a precisely 8.5 feet long ladder. When he arrives at the store, he realizes that the marked lengths of all the ladders are in inches. How many inches long should the ladder he purchases be?

14

$$3y + x = 0$$

If (x, y) is a solution to the equation above and $y \neq 0$, what is the ratio of $\frac{x}{y}$?

A) 3

B) $\frac{1}{3}$

C) -3

D) $-\frac{1}{3}$

15

$$3x^2 - 5x - 12 = 0$$

How many solutions does the given equation have?

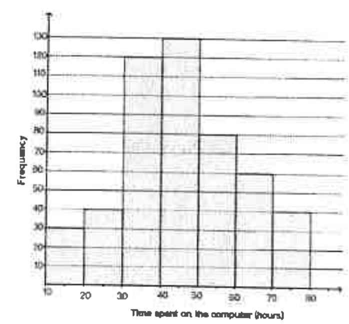
A) One

B) Two

C) Three

D) None

16



In a recent survey, 500 college students were asked how many hours they spent on their computers each week. The results are shown in the histogram above. How many of the 500 students spend more than 40 hours on their computer each week?

A) 130

B) 310

C) 250

D) 120

17

There are two types of guest rooms in a hotel. The triple room charges \$25 per person per night and double rooms charge \$35 per person per night. A tourist group of 50 people stayed in the hotel and paid a total of \$1510 for one night. How many double rooms were booked? (assume each room they booked was full)

A) 8

B) 21

C) 13

D) 15

2 Module 1 2

18

A company gave holiday gifts to its clients. 40% of the gifts, 20 gifts, were given to VIP clients. How many gifts were given in total?

19

$$\frac{2x(2x-3)+3(2x-3)}{(4x^2-9)}$$

If $x^2 \neq \frac{9}{4}$, which expression is equivalent to the equation above?

A) $\frac{2x-3}{2x+3}$

B) 1

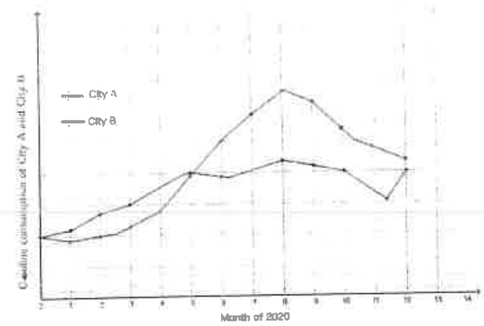
C) 2

D) $\frac{4x^2+12x-9}{4x^2-9}$

20

Triangles ABC and EFG are similar, where side AB of triangle ABC corresponds to side EF of triangle EFG and $\frac{AB}{EF} = 3$. The area of triangle ABC is how many times the area of triangle EFG?

21



The graph above shows gasoline consumption of City A and City B from Jan, 2020 to Dec, 2020. In which month did the two city have the greatest difference in gasoline consumption?

A) Sept., 2020

B) Mar., 2020

C) Aug., 2020

D) Nov., 2020

22

$$y = 4900 - 70x$$

An assistant was asked to type up a handwritten report with a total of 4900 words. If the assistant types the same number of words every hour, there would be y words left to type after x hours, where $0 \leq x \leq 70$. Which of the following statements best describes the meaning of 70 in the context?

A) The number of words left after x hours

B) The average number of words the assistant typed each hour

C) The decreasing number of words the assistant typed

D) The total number of words the assistant typed in x hours

No Test Material On This Page

2 Module 2 2

Math

35 MINUTES, 22 QUESTIONS

DIRECTIONS

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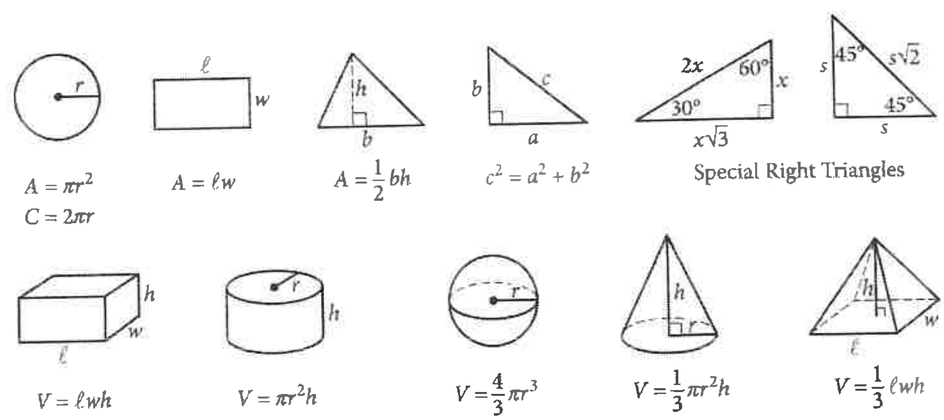

The number of degrees of arc in a circle is 360.
The number of radians of arc in a circle is 2π.
The sum of the measures in degrees of the angles of a triangle is 180.

---

2 Module 2 2

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2 Module 2 2

1

If $2m^2 - 4m + 2 = 50$, what is the negative value of $2(m - 1)$?

2

In the xy-plane, quadratic equation $y = ax^2 + bx + 11$ passes through the points $(1, 5)$ and $(3, 5)$. What is the value of $a - b$?

A) 10

B) -6

C) -10

D) 6

3

Which of the following systems of equations has no solution?

A) $y - 2x = 5$
$2y - 2x = 2$

B) $y - 2x = 5$
$2y - 4x = 2$

C) $y - 2x = 5$
$y - x = 6$

D) $y - 2x = 5$
$2x - 3y = 1$

4

In the xy-plane, points m, $(14, a)$, and n, $(2, 1)$, are on the line l. If points m and n are 13 units apart, which of the following equations represent line l?

A) $y = -\frac{5}{12}x + \frac{11}{6}$

B) $y = \frac{5}{12}x + \frac{1}{6}$

C) $y = \frac{1}{6}x + 3$

D) $y = -\frac{5}{12}x - \frac{1}{6}$

5

Distance (km)	Number of days
22	8
18	4
10	5
5	3

A long-distance runner runs varying distances each day. The number of days she ran each distance is shown in the table above. What is the difference between the mode and the range of the data set?

A) 5

B) 22

C) 17

D) 18

6

The equation of $(m - 1)x^2 + 2x - 5 = 0$ has exactly one solution, where m is a constant. What is the value of m?

A) 5

B) -5

C) $\frac{4}{5}$

D) $-\frac{4}{5}$

2 Module 2 2

7

In a right triangle MOP, one of the acute angles M has a measure of 30° and the other acute angle is $\angle P$. What is the value of $\sin(P)$?

A) $\frac{1}{2}$

B) $\frac{\sqrt{2}}{2}$

C) 1

D) $\frac{\sqrt{3}}{2}$

8

The price of a stock increased by 25% in one year and decreased by 12% the next year. By what percent did the price increase over the two-year period?

A) 13%

B) 25%

C) 10%

D) 12%

9

When initially purchased, a car valued \$56,200. However, it depreciated at a rate of approximately 15% per year within 5 years. What is the value of the car 5 years after purchase? (Neglect the decimals)

10

The function f is defined by $f(x) = x - 6$. What is the x-intercept of the graph of $y = f(x)$ in the xy-plane?

A) (0,6)

B) (0, − 6)

C) (6,0)

D) (−6,0)

11

A water tank has dimensions of width 14 feet, length 20 feet, and height 8 feet. How much water, in cubic meters, is needed to fill the tank? (1 foot = 0.3 meters)

A) 60.48

B) 672

C) 201.6

D) 2240

12

$$3 - \frac{2}{3}x = 8$$

Which equation has the same solution as the given equation?

A) $3 - 2x = 24$

B) $6 - 2x = 16$

C) $9 - 2x = 24$

D) $2x - 9 = 24$

13

An electric vehicle (EV) traveled at an average speed of 66 miles per hour for 3 hours and had an average energy consumption of 0.25 kilowatt-hour (kwh) per mile. Approximately how many kwh did the EV consume over the entire 3 hour drive?

A) 18.5

B) 49.5

C) 5.5

D) 0.75

14

Which of the following expressions below is equivalent to $5x^5 - 26x + 5$?

I. $(5x - 5)(x - 1)$
II. $(x - 5)(5x - 1)$
III. $(x - 1)(5x + 1)$

A) I only

B) II only

C) III only

D) None

15

$$y + 2x < 3$$
$$x + 2y > 2$$

Which of the following pairs of (x_0, y_0) satisfies the system of inequalities above in the xy-plane?

A) $(-1, -2)$

B) $(-1, 2)$

C) $(2, 3)$

D) $(-3, -1)$

16

If a circle has a radius of 6 units, what is the length of an arc with a central angle of 120°?

A) $\frac{2}{3}\pi$

B) 2π

C) 4π

D) 3π

17

A department store sells a toaster at a regular price and makes a profit of \$5 per toaster. If the profit of selling 4 toasters at 10% off of the regular price is equal to the profit of selling 5 toasters at \$3 less than the regular price for each toaster, what is the regular price of a toaster?

18

Which expression is equivalent to $\frac{x(y-6)}{xy^2 - 6xy} - \frac{3(y-6)}{y-6}$, where $y \neq 0$?

A) $\frac{1}{3}y + 3$

B) $\frac{3xy - 12}{xy^2 - 6xy}$

C) $\frac{1}{y} - 3$

D) $\frac{4x^2}{2x - 1}$

19

Which expression is equivalent to $\frac{m^{-4} \cdot n^8 \cdot p^{12}}{m^{-2} \cdot n^2 \cdot p^{-2}}$, where m, n, and p are positive?

A) $m^{-6} \cdot n^6 \cdot p^{14}$

B) $m^{-6} \cdot n^6 \cdot p^{10}$

C) $m^{-2} \cdot n^6 \cdot p^{14}$

D) $m^2 \cdot n^4 \cdot p^{-6}$

20

What is the diameter of the circle $x^2 - 10x + y^2 - 6y + 22 = 0$ in the xy-plane?

A) $2\sqrt{3}$

B) 12

C) $4\sqrt{3}$

D) 6

21

Mode of transportation	Number of students
Drive	
Bike	33
Walk	22
Other	19
Total	100

A university conducted a survey in which 100 students were selected randomly and asked about the mode of transportation they take to get to campus. The results are shown in the incomplete table above. If the university has a total of 8000 students, how many of them drive to the campus?

22

The function f is defined by $f(x) = 8x^2 + 3$. In the xy-plane, the graph of $y = p(x)$ is the result of shifting the graph of $y = f(x)$ down 8 units. Which equation defines the function p?

A) $p(x) = 8x^2 + 11$

B) $p(x) = 8x^2 - 5$

C) $p(x) = x^2 + 3$

D) $p(x) = x^2 - 3$

35:00

Section 2, Module 1: Math

Annotate

1

Mark for Review

Which of the following points represent the intersections of the function $f(x) = 3x^2 + 36x + 96$ with the x axis?

(A) $(4, 0)$

(B) $(0, -4)$

(C) $(-8, 0)$

(D) $(0, 0)$

TESTQUBE Question 1 of 22 >

Section 2, Module 1: Math

Annotate

2

Mark for Review

Jimmy is booking a hotel which costs $\$80$ per night and a $\$100$ one-time fee. Considering Jimmy's budget of $\$1200$, what is the maximum number of days Jimmy can stay at the hotel?

(A) 12

(B) 13

(C) 14

(D) 15

TESTQUBE Question 2 of 22 >

Section 2, Module 1: Math

Annotate

3

Mark for Review

James created a unique shape by combining without overlapping three identical equilateral triangles. If each triangle has a side length of 10 centimeters, what is the **closest** total area of this special shape?

(A) 140cm^2

(B) 43.30cm^2

(C) 129.90cm^2

(D) 229.90cm^2

TESTQUBE Question 3 of 22 >

Section 2, Module 1: Math

Annotate

4

Mark for Review

A bag contains 18 marbles, 13 of which are red and 5 of which are blue. Three marbles are drawn at random, without replacement. What is the probability that all three marbles are red?

(A) $1/18$

(B) $143/408$

(C) $13/126$

(D) $1/4$

TESTQUBE Question 4 of 22 > Back Next

Section 2, Module 1: Math

Annotate

5

Mark for Review

In a random selection of 230 voters, they were asked about their satisfaction with a policy. Out of these voters, 80 expressed dissatisfaction. If a total of $17,250$ people were to vote for the policy, what is the best estimate of votes that indicate satisfaction with the policy?

(A) 11250

(B) 6000

(C) 16470

(D) 8000

Section 2, Module 1: Math

Annotate

6

Mark for Review

In triangle ABC, the following information is given: Angle B measures 90 degrees, and Angle C measures 30 degrees. If the length of side AB is 20, what is the length of side AC?

(A) $20\sqrt{3}$

(B) 40

(C) 60

(D) 15

Section 2, Module 1: Math

Annotate

7

Mark for Review

What is the product of the solutions for x in the given equation?

$$x(x+8) = 4x^2 + 45x + 20$$

(A) $37/3$

(B) 37

(C) $20/3$

(D) 20

Section 2, Module 1: Math

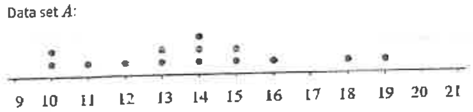
Annotate

8

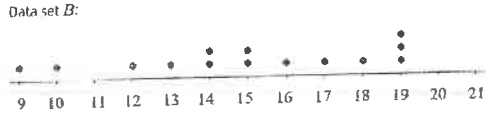
Mark for Review

Based on the dot plot provided, which of the following descriptions best characterizes the distribution of data sets A and B?

Data set A:



Data set B:



(A) Data set A has greater mean and standard deviation than Data set B.

(B) Data set A has greater mean but lower standard deviation than Data set B.

(C) Data set A has smaller mean but greater standard deviation than Data set B.

(D) Data set A has smaller mean and standard deviation than Data set B.

TESTBQUBE Question 8 of 22 >

Section 2, Module 1: Math

Annotate

9

Mark for Review

The given function $f(x) = (x+4)(x+3)$ is translated up by 3 units. At which y-coordinate does this graph intersect $x = 3$?

(A) 45

(B) 48

(C) 50

(D) 52

TESTBQUBE Question 9 of 22 >

Section 2, Module 1: Math

Annotate

10

Mark for Review

If $2x + 4 = 12$, what is the value of $x + 9$?

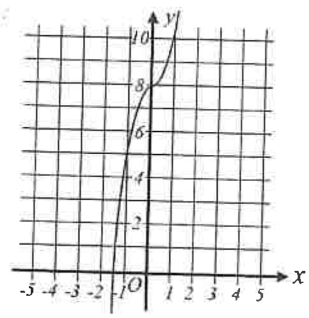
TESTBQUBE Question 10 of 22 > Back Next

Section 2, Module 1: Math

Annotate

11 Mark for Review

What is the y-intercept for the given graph below?



Ⓐ 8

Ⓑ 7

Ⓒ 1

Ⓓ −1

Section 2, Module 1: Math

Annotate

12 Mark for Review

Chris rides his bicycle at a constant speed of 30 feet per minute. How long, in seconds, does it approximately take him to reach a point that is 70 feet away?

Ⓐ 125

Ⓑ 130

Ⓒ 140

Ⓓ 160

Section 2, Module 1: Math

Annotate

13 Mark for Review

The length of the garden is 20 meters, and it is 20% longer than its width. What is the approximate width of the garden in meters? (Round your answer to the nearest tenth.)

Section 2, Module 1: Math

Annotate

14

Mark for Review

James boards a train leaving a station at $10:00$ AM and observes that the train is traveling at a speed of 60 miles per hour. Meanwhile, Eve boards another train from the same station, departing one hour later, and her train travels in the same direction at a speed of 70 miles per hour. What time will Eve's train catch up to James' train?

(A) $1:00$ PM

(B) $2:30$ PM

(C) $4:00$ PM

(D) $5:00$ PM

TESTQUBE Question 14 of 22 >

Section 2, Module 1: Math

Annotate

15

Mark for Review

The store is selling a shirt for $\$20$. The store is offering a 20% discount on the shirt. If there is a sales tax of 5% on the discounted price, what will be the final price of the shirt?

(A) $\$15.00$

(B) $\$15.20$

(C) $\$16.00$

(D) $\$16.80$

TESTQUBE Question 15 of 22 >

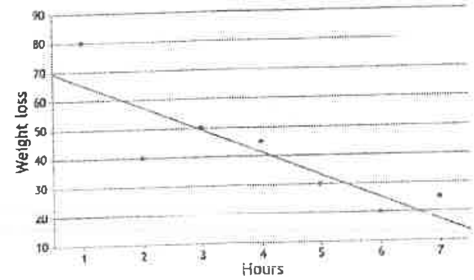
Section 2, Module 1: Math

Annotate

16

Mark for Review

The scatter plot provided illustrates the relationship between hours of exercise and the corresponding weight loss. Additionally, the graph includes a line of best fit as well. Which of the following best approximates the slope of the line of best fit?



(A) 8

(B) -8

(C) 10

(D) -10

TESTQUBE Question 16 of 22 > Back Next

Section 2, Module 1: Math

Annotate

17

Mark for Review

For the quadratic function $f(x)$,
$f(1) = 4$, $f(2) = 7$
Which equation could define $f(x)$?

(A) $f(x) = x^2 + 3$

(B) $f(x) = 2x^2 - 1$

(C) $f(x) = x^2 + x + 3$

(D) $f(x) = 3x^2 - 2$

TESTQUBE Question 17 of 22 >

Section 2, Module 1: Math

Annotate

18

Mark for Review

A hotel has a rectangular swimming pool with dimensions 20 meters in length and 15 meters in width. The manager fills the pool with water to a depth of 2 meters. What is the volume of water in the pool, measured in cubic meters?

(A) 450 cubic meters

(B) 400 cubic meters

(C) 300 cubic meters

(D) 600 cubic meters

TESTQUBE Question 18 of 22 >

Section 2, Module 1: Math

Annotate

19

Mark for Review

With the given system of equations
$2x + 3y = 25$
$x = 2y$

What is the value of x?

TESTQUBE Question 19 of 22 > Back Next

I

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VII

20

Mark for Review

The table provided displays the preference distribution of students from three classes. If a student is chosen randomly from the total student population, what is the probability that the selected student prefers Math class?

	Math	English	Science	Total
Class A	40	20	15	75
Class B	80	50	30	160
Class C	30	10	20	60
Total	150	80	65	295

(A) 160/295

(B) 60/295

(C) 80/295

(D) 150/295

21

Mark for Review

Considering the given function $f(x) = 1.5x + 30$, if there exists a linear function g that is parallel to function f and intersects at the point $(-4, 2)$, what is the y-intercept of function g?

22

Mark for Review

Alice is trying to determine the optimal allocation of hours between her two jobs. She aims to work a total of 30 hours per week while earning \$300 in total. The first job offers a pay rate of \$5 per hour, and the second job pays \$15 per hour. How many hours should Alice work for the first job in order to achieve her desired earnings and total hours?

(A) 20 hours

(B) 15 hours

(C) 10 hours

(D) 30 hours

Move on to the Next Section

35:00

Section 2, Module 2: Math

Annotate

1

Mark for Review

What value of x is the solution for the following equation below?

$$6x - 3x - 15 = 75$$

Section 2, Module 2: Math

Annotate

2

Mark for Review

The solution to the given system of equations is (m, n). What is the value of $m - n$?

$$3m + 4n = 7$$
$$4m + 3n = 10$$

(A) -3

(B) -1

(C) 1

(D) 3

Section 2, Module 2: Math

Annotate

3

Mark for Review

Alex takes a tram and bicycle from home to his workplace. The tram travels at a constant speed of 15 miles per hour and the bicycle at 10 miles per hour. The distance between his home and workplace is 10 miles and it takes 45 minutes for him to travel one-way. How long did Alex travel on the tram?

(A) 12 minutes

(B) 15 minutes

(C) 24 minutes

(D) 30 minutes

Section 2, Module 2: Math

Annotate

4

Mark for Review

Which of the systems of equations have infinitely many solutions?

(A) $2x + 4y = 3$ and $6x + 12y = 9$

(B) $-x - y = 5$ and $x - y = 5$

(C) $4x - y = -2$ and $4x - y = 2$

(D) $3x + 3y = 1$ and $3x - 3y = -1$

I

Section 2, Module 2: Math

Annotate

5

Mark for Review

The function g is defined by $g(x) = 3x^2 - 6x + 4$. For what value of x does $g(x) = 28$?

(A) -4

(B) -2

(C) 3

(D) 5

TESTQUBE Question 5 of 22 >

Section 2, Module 2: Math

Annotate

7

Mark for Review

The given equation relates the numbers a, b, and c. Which of the following correctly expresses b in terms of a and c?

$c = 3a^2 - 2b$

(A) $b = \frac{3a^2+c}{2}$

(B) $b = \frac{3}{2a^2-c}$

(C) $b = \frac{3a^2-c}{2}$

(D) $b = 3a^2 - c$

TESTQUBE Question 7 of 22 >

II

III

IV

Section 2, Module 2: Math

Annotate

6

Mark for Review

What value of b would result in no real solution for the given equation below?

$3x^2 - bx + 2 = 0$

(A) 4

(B) 5

(C) 6

(D) 7

TESTQUBE Question 6 of 22 >

Section 2, Module 2: Math

Annotate

8

Mark for Review

Jake is shopping at a store and plans to purchase both cookies and chips. The price of each cookie is $\$2.5$, while each chip costs $\$1.5$. Jake intends to buy 6 chips and wishes to spend a minimum of $\$26$ but no more than $\$27$ in total. How many cookies should Jake include in the purchase?

TESTQUBE Question 8 of 22 > Back Next

V

VI

VII

Section 2, Module 2: Math

Annotate

9

Mark for Review

Which of the following expressions is equivalent to $\frac{(a^4b^3c^2)(a^{-1}b^2c^5)}{(ab^2c^{-1})}$, where a, b, and c are positive?

(A) $a^{-4}b^{12}c^{-10}$

(B) $a^{-4}b^{3}c^{-10}$

(C) $a^{4}b^{7}c^{6}$

(D) $a^{2}b^{3}c^{8}$

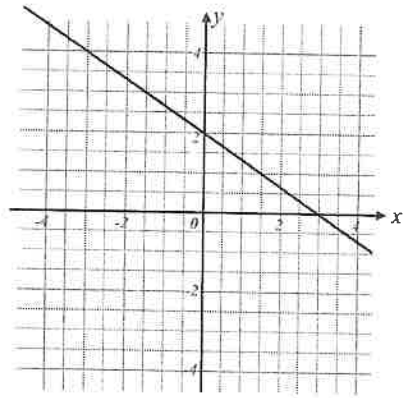
Section 2, Module 2: Math

Annotate

10

Mark for Review

What is the slope for the given linear function below?



(A) $-3/2$

(B) $-2/3$

(C) $2/3$

(D) $3/2$

I

II

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Section 2, Module 2: Math

Annotate

11

Mark for Review

The function f is defined by $f(x) = -2x + 6$ and function g is defined by $g(x) = -f(x)$. What is the value of $g(-1)$?

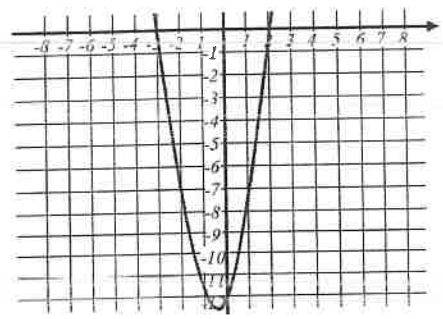
Section 2, Module 2: Math

Annotate

12

Mark for Review

Which of the following equations defines the function f as shown in the graph below?



Ⓐ $f(x) = -2x^2 + 2x + 12$

Ⓑ $f(x) = -2x^2 - 2x + 12$

Ⓒ $f(x) = 2x^2 - 2x - 12$

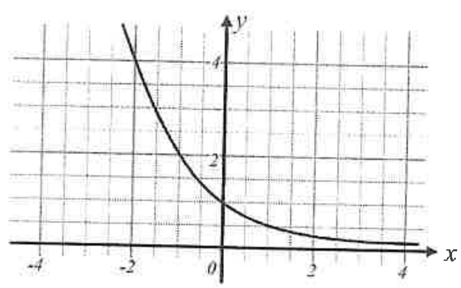
Ⓓ $f(x) = 2x^2 + 2x - 12$

Section 2, Module 2: Math

Annotate

13 Mark for Review

The function $f(x)$ is illustrated below. What is the value of $f(-2)$?



(A) -4

(B) -2

(C) 2

(D) 4

TEST QUBE Question 13 of 22 >

Section 2, Module 2: Math

Annotate

14 Mark for Review

To produce 10 grams of dough, a mixture is prepared by combining 8 grams of flour, 1.5 grams of water, and 0.5 grams of sugar. If there is a total of 65 grams of dough, how many grams of water is included in the mixture? (Ignore the gram sign.)

TEST QUBE Question 14 of 22 >

Section 2, Module 2: Math

Annotate

15 Mark for Review

Jeremy ordered a steak and wine at a restaurant. The steak costs 28 dollars and wine 12 dollars. Jeremy must pay 10 percent tax to the total amount of food he paid and an additional tip of 6 dollars above that. How much money does Jeremy have to pay at the restaurant? (ignore the dollar sign)

TEST QUBE Question 15 of 22 > Back Next

Section 2, Module 2: Math

Annotate

16

Mark for Review

Class A comprises 50 students, where 22 students can speak Spanish, 16 students can speak French, and 4 students can speak both languages. What is the probability that a randomly selected student from Class A speak neither Spanish nor French?

(A) 16%

(B) 24%

(C) 32%

(D) 36%

Section 2, Module 2: Math

Annotate

17

Mark for Review

Anna weighs 125 pounds. Determine Anna's weight in tons. (1 pound = 0.45 kilograms and 1 kilogram = 0.001 ton)

(A) 0.0056 tons

(B) 0.056 tons

(C) 0.56 tons

(D) 5.6 tons

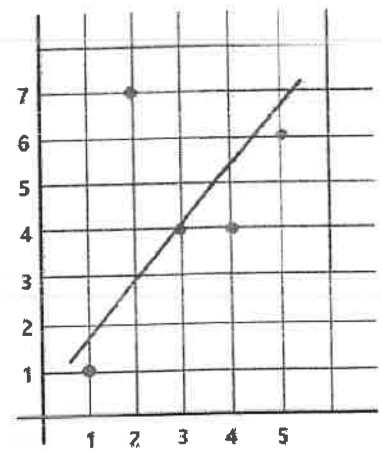
Section 2, Module 2: Math

Annotate

18

Mark for Review

Which of the following equations best represents the line of best fit for the scatterplot below?



(A) $y = 1.2x + 0.5$

(B) $y = -1.2x + 0.5$

(C) $y = 1.2x - 0.5$

(D) $y = -1.2x - 0.5$

Section 2, Module 2: Math

Annotate

19

Mark for Review

What is the volume of a pyramid with a length of 6, a width of 4 and a height of 5?

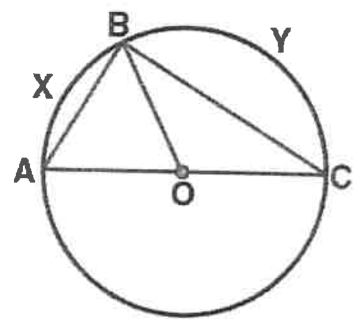
Section 2, Module 2: Math

Annotate

20

Mark for Review

In the diagram below, triangle ABC is circumscribed by circle with a center O. If $\angle BAC$ is 40°, what is $\angle BCA$?



(A) 30°

(B) 40°

(C) 50°

(D) 60°

I

Section 2, Module 2: Math

Annotate

21

Mark for Review

II

A ladder, initially 15 feet long, is leaning exactly at the top of the building with the base of the ladder positioned 9 feet from the building. When the ladder is pulled 3 feet farther from the building, the top of the ladder drops to a new height. Let's denote the angle between the newly adjusted ladder and the ground as A. Find the value of $\sin(A)$.

III

IV

TEST QUBE Question 21 of 22 >

Section 2, Module 2: Math

Annotate

V

22

Mark for Review

Consider an equilateral triangle A with a side length of $4cm$. If triangle B is similar to triangle A and has side lengths that are 50 percent longer than that of triangle A, what is the area of triangle B?

VI

A) $4\sqrt{3}$

B) $6\sqrt{3}$

C) $8\sqrt{3}$

D) $9\sqrt{3}$

VII

TEST QUBE Question 22 of 22 >

Section 2, Module 1: Math

Annotate

1

Mark for Review

How do the mean and standard deviation of Class A compare to those of Class B based on the scores of their students given in the list below?

Class A	60	70	65	78	62
Class B	78	35	45	50	40

(A) Class A has a greater mean but lower standard deviation than Class B.

(B) Class A has a greater mean and standard deviation than Class B.

(C) Class A has a lower mean but greater standard deviation than Class B.

(D) Class A has a lower mean and lower standard deviation than Class B.

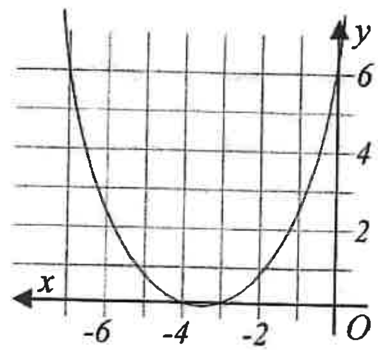
Section 2, Module 1: Math

Annotate

2

Mark for Review

What is the most appropriate function to describe the graph depicted below?



(A) $(x+4)(x+3)$

(B) $(x+4)(x+3)+6$

(C) $0.5(x+4)(x+3)$

(D) $2(x+4)(x+3)$

I

Section 2, Module 1: Math

Annotate

3

Mark for Review

Out of 300 residents in a town, a sample was randomly selected and asked if they were satisfied with the air quality. 30% of those surveyed responded positively. Based on this result, what is the most accurate estimate of the total number of residents in the town who are satisfied with the air quality?

(A) 60

(B) 30

(C) 90

(D) 120

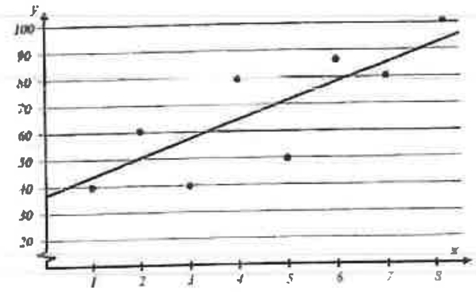
Section 2, Module 1: Math

Annotate

4

Mark for Review

Which equation provides the best estimate for the slope of the line of best fit for the given graph below?



(A) −8

(B) 20

(C) −20

(D) 8

II

III

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Section 2, Module 1: Math

Annotate

5

Mark for Review

Which expression is equivalent to $(a^4b^3c^{-1})(b^2c^{-3})$, where a,b,and c are positive?

(A) $a^8b^{-9}c^3$

(B) $a^4b^5c^{-4}$

(C) $a^4b^6c^3$

(D) $a^8b^3c^{-2}$

TEST QUBE Question 5 of 22 >

Section 2, Module 1: Math

Annotate

6

Mark for Review

The function g represents the distance, in miles, from home to school after driving m miles. Based on this model, what is the initial distance in miles from home to school?

$g(m) = -0.5m + 20$

(A) 15

(B) 10

(C) 20

(D) None of the above

TEST QUBE Question 6 of 22 >

Section 2, Module 1: Math

Annotate

7

Mark for Review

Given that a table provides the distribution of favorite classes and grade levels of 100 students, what is the likelihood of randomly selecting a student who is a sophomore and has Math as their favorite class?

Grade	Favorite Class			
	Math	English	Science	Total
Freshman	9	8	5	22
Sophomore	8	9	12	29
Junior	4	10	8	22
Senior	9	13	5	27
Total	30	40	30	100

(A) 2/25

(B) 3/10

(C) 29/100

(D) 1/2

TEST QUBE Question 7 of 22 > Back Next

I

Section 2, Module 1: Math

Annotate

8

Mark for Review

A right triangle ABC is similar to right triangle DEF, where angle B corresponds to angle E and angle A corresponds to angle D. If, the length of AB is 3, BC is 4, and DE is 6, what is the length of DF? (Angles B and E are right angles.)

Ⓐ 12

Ⓑ 5

Ⓒ 10

Ⓓ 8

TESTQUBE Question 8 of 22 >

II

III

IV

Section 2, Module 1: Math

Annotate

9

Mark for Review

Which of the following would be the **approximate** value of James' house after a year if it is currently worth $\$200,000$ and its value increases by 25 percent every three months?

Ⓐ $\$250,000$

Ⓑ $\$390,625$

Ⓒ $\$312,500$

Ⓓ $\$488,281$

TESTQUBE Question 9 of 22 >

V

VI

VII

Section 2, Module 1: Math

Annotate

10

Mark for Review

What is the y-coordinate of the y-intercept of function g if it is perpendicular to $h(x) = 0.5x + 48$ and passes through the x-intercept at $(3, 0)$?

Ⓐ 6

Ⓑ -2

Ⓒ 48

Ⓓ -6

TESTQUBE Question 10 of 22 >

Section 2, Module 1: Math

Annotate

11

Mark for Review

On the election ballot, there was a proposal for a new grading policy. The results showed that there were twice as many students who voted in favor of the proposal as those who voted against it. If there were $2,000$ students who voted against the proposal, how many students voted in favor of it?

Ⓐ 1000

Ⓑ 2000

Ⓒ 4000

Ⓓ 6000

TESTQUBE Question 11 of 22 > Back Next

Section 2, Module 1: Math

Annotate

12

Mark for Review

What is the volume, in cubic centimeters, of a rectangular prism with a length of $10cm$, a width of $6cm$, and a height of $8cm$?

TESTQUBE Question 12 of 22 >

Section 2, Module 1: Math

Annotate

13

Mark for Review

How many real solutions does the equation below have?

$8x^2 + 17x + 3 = 0$

(A) Exactly one

(B) Zero

(C) Infinitely many

(D) Exactly two

TESTQUBE Question 13 of 22 >

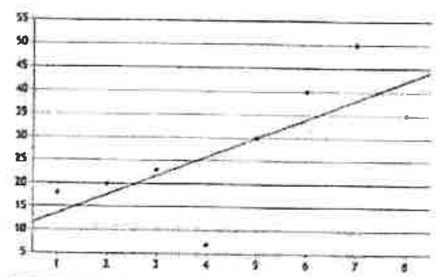
Section 2, Module 1: Math

Annotate

14

Mark for Review

Which equation among the options below best represents the line of best fit in the scatterplot displaying the relationship between variables x and y? (Note: The graph below does not illustrate $x = 0$.)



(A) $1.25x + 20$

(B) $-1.25x + 20$

(C) $4x + 10$

(D) $-4x - 10$

TESTQUBE Question 14 of 22 > Back Next

I

Section 2, Module 1: Math

Annotate

15

Mark for Review

II

Bob bought a phone that was on sale at a store. The phone was on 70% discount but included a 20% tax which ended up costing Bob 900 dollars. What was the original price of the phone?

Ⓐ $1000

Ⓑ $1500

III

Ⓒ $2500

Ⓓ $3000

IV

TESTQUBE

Question 15 of 22

Section 2, Module 1: Math

Annotate

16

Mark for Review

V

What is the area, in square meters, of a rectangular garden that has a perimeter of 100 meters and where the length is 10 meters more than the width?

VI

VII

TESTQUBE

Question 16 of 22

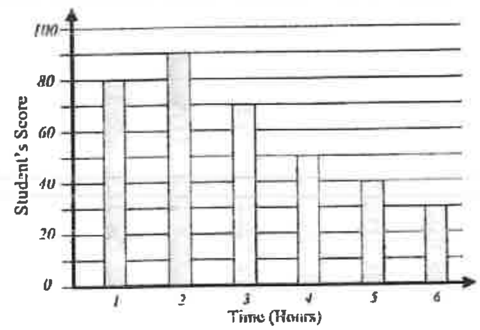
Section 2, Module 1: Math

Annotate

17

Mark for Review

The bar graph below shows the scores for each student in Mr.Jackson's math class where each student spent different amounts of time preparing for the exam. What is the average score of the students based on the bar graph?



TESTQUBE

Question 17 of 22

Back

Next

Section 2, Module 1: Math

Annotate

18

Mark for Review

In a game of rock paper scissors, two players, Bob and Alice, choose one of the three choices at random. What is the probability that both players choose the same shape?

(A) 1/3

(B) 1/9

(C) 2/3

(D) 1

TESTQUBE Question 18 of 22 >

Section 2, Module 1: Math

Annotate

19

Mark for Review

A bag contains 10 red marbles, 5 blue marbles, and 4 green marbles. If two marbles are drawn at random, without replacement, what is the probability that both marbles are red?

(A) 10/19

(B) 5/19

(C) 3/11

(D) 3/18

TESTQUBE Question 19 of 22 >

Section 2, Module 1: Math

Annotate

20

Mark for Review

James owns a rectangular garden that has a length of 10 meters and a width of 8 meters. He wants to put a circular pond in the center of the garden, and the pond has a radius of 2 meters. Which of the following best approximates the area of the garden that is not covered by the pond?

(A) 40.32 square meters

(B) 64.47 square meters

(C) 80.43 square meters

(D) 67.43 square meters

TESTQUBE Question 20 of 22 >

Section 2, Module 1: Math

Annotate

21

Mark for Review

A circle passes through the points (-3, 3) and (9, 3). The two points, when connected together by a line, passes through the center of the circle. Find the radius of the circle.

TESTQUBE Question 21 of 22 > Back Next

I

Section 2, Module 1: Math

Annotate

22

Mark for Review

II

An elevator can carry a maximum of $1,000$ pounds. The elevator operator weighs 200 pounds and each passenger weighs 150 pounds. What is the maximum number of passengers that the elevator can carry if the elevator is already carrying a load of 300 pounds?

Ⓐ 4 passengers

III

Ⓑ 3 passengers

Ⓒ 2 passengers

Ⓓ 5 passengers

IV

V

VI

VII

TESTQUBE Question 22 of 22 >

Move on to the Next Section

35:00

Section 2, Module 2: Math

Annotate

1

Mark for Review

What is the value of x that satisfies the two systems of equations given below? $(x \geq 0)$

$$x^2 - y = 18$$
$$x = 2y + 8$$

(A) -4

(B) -2

(C) 2

(D) 4

TESTQUBE Question 1 of 22 >

Section 2, Module 2: Math

Annotate

2

Mark for Review

Elizabeth is participating in a quiz show. For every question she gets correct, she earns 2 points. For every question she gets incorrect, she loses 1 point. If there are a total of 20 questions and she earned 19 points in total, how many questions did she answer incorrectly?

(A) 6

(B) 7

(C) 8

(D) 9

TESTQUBE Question 2 of 22 >

Section 2, Module 2: Math

Annotate

3

Mark for Review

If $6x - 3 = 15$, what is the value of $15x - 35$?

TESTQUBE Question 3 of 22 >

Section 2, Module 2: Math

Annotate

4

Mark for Review

Rectangle A has a width $3cm$ shorter than the length. If the perimeter of rectangle A is 26, what is the width of rectangle A?

(A) $5cm$

(B) $6cm$

(C) $7cm$

(D) $8cm$

TESTQUBE Question 4 of 22 > Back Next

I

Section 2, Module 2: Math

Annotate

5

Mark for Review

A travel agency is selling two types of tickets, Ticket A and Ticket B, for the observatory deck. Ticket A costs $\$15$ each, and Ticket B costs $\$25$ each. In one day, the travel agency sold a total of 57 tickets and earned a total revenue of $\$1,065$. How many Ticket B's were sold that day?

Ⓐ 7

Ⓑ 14

Ⓒ 21

Ⓓ 28

TESTQUBE Question 5 of 22 >

II

Section 2, Module 2: Math

Annotate

7

Mark for Review

Which of the following is not a factor of $2x^3 + 7x^2 - 19x - 60$?

Ⓐ -4

Ⓑ 3

Ⓒ $-5/2$

Ⓓ $2/3$

TESTQUBE Question 7 of 22 >

III

IV

Section 2, Module 2: Math

Annotate

6

Mark for Review

In the given equation below, b is a constant. The equation has one real solution. What is the value of b when $b > 0$?

$3x^2 - bx + 3 = 0$

V

VI

Section 2, Module 2: Math

Annotate

8

Mark for Review

Which of the following expressions is equivalent to $a^6 \div a^4$?

Ⓐ a^{6+4}

Ⓑ a^{6-4}

Ⓒ $a^{6\times4}$

Ⓓ $a^{6/4}$

VII

TESTQUBE Question 6 of 22 >

TESTQUBE Question 8 of 22 > Back Next

Section 2, Module 2: Math

Annotate

9

Mark for Review

In the xy-plane, what is the area of a polygon that satisfies the condition of the three inequalities shown below?

$y \leq \frac{4}{3}x + 4$
$y \leq -x + 4$
$y \geq 0$

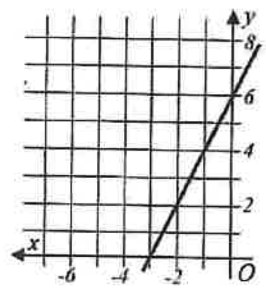
Section 2, Module 2: Math

Annotate

10

Mark for Review

What is the x-intercept of the graph below?



Ⓐ $(-3, 0)$

Ⓑ $(3, 0)$

Ⓒ $(0, 6)$

Ⓓ $(6, 0)$

I
II
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VII

Section 2, Module 2: Math

Annotate

11

Mark for Review

The function f is defined by $f(x) = 2x - 5$. In the xy-plane, the graph $f(x)$ is shifted 1 unit to the left and 4 units up. What is the x-intercept of the new function?

(A) -1

(B) -0.5

(C) 0.5

(D) 1

TESTQUBE Question 11 of 22

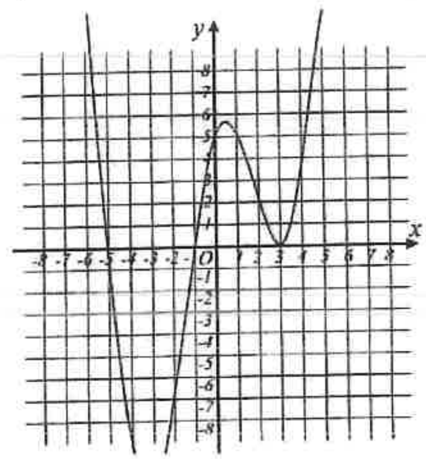
Section 2, Module 2: Math

Annotate

12

Mark for Review

The graph of $y = g(x)$ is shown below. For how many values of x does $g(x) = 0$?



(A) 0

(B) 1

(C) 2

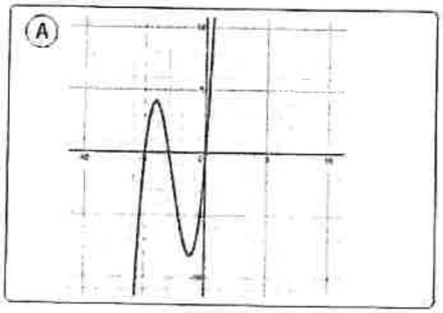
(D) 3

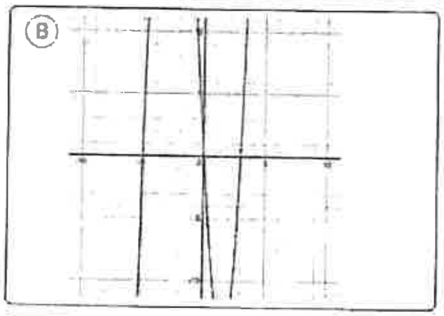
TESTQUBE Question 12 of 22 Back Next

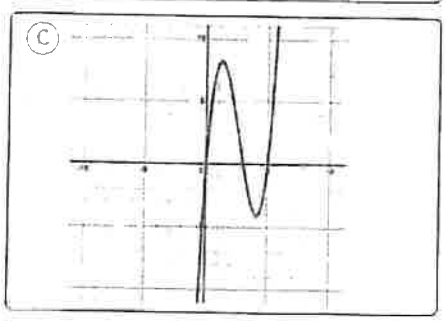
Section 2, Module 2: Math
Annotate

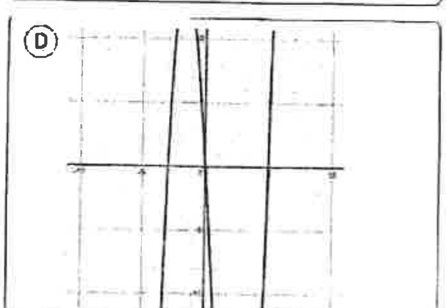
13 Mark for Review

Which of the following graphs correctly represents the function $f(x) = x(x-3)(x+5)$?

(A)

A

(B)

B

(C)

C

(D)

D

TESTQUBE Question 13 of 22 >

Section 2, Module 2: Math
Annotate

14 Mark for Review

The equation below defines the function g. What is the maximum value of $g(x)$?

$g(x) = \frac{-5}{3}x^2 - 10x + 9$

TESTQUBE Question 14 of 22 >

Section 2, Module 2: Math
Annotate

15 Mark for Review

For a particular factory that manufactures pens, 6 out of every 100 pens are defective. If this machine produces 500 pens a day, how many defects in total are expected to be found in a week? (The machine produces all seven days a week from Monday to Sunday.)

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I

Section 2, Module 2: Math

Annotate

16

Mark for Review

II

Alex is depositing his money at a bank. Alex estimates that, starting from present, the value of money will increase by 0.5 percent every 10 years. If the present amount of money deposited is \$7,500, which of the following represents the estimate of the amount of money, in dollars, x years from now?

III

(A) $7,500(1.05)^{x/10}$

(B) $7,500(1.005)^{x/10}$

(C) $7,500(1.05)^{10/x}$

(D) $7.500(1.005)^{10/x}$

IV

V

VI

VII

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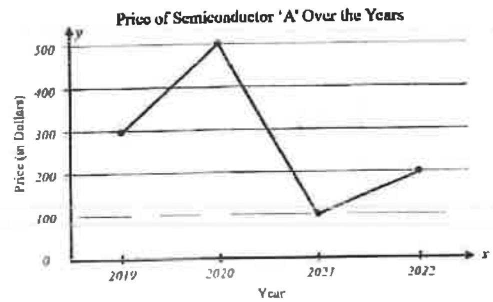
Annotate

17

Mark for Review

The line graph below shows the price of semiconductor A over the years from 2019 to 2022. Which time interval, spanning from 2019 to 2022, exhibits the largest difference in the price of semiconductor A?



(A) 2019 to 2020

(B) 2020 to 2021

(C) 2021 to 2022

(D) None of the above

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18

Mark for Review

There is a 12-sided die which is labeled with a number from 1 to 12, with a different number on each side. If the die is rolled once, what is the probability the number is either an odd or even number?

(A) 0

(B) 1/12

(C) 1/2

(D) 1

TESTQUBE

Question 18 of 22

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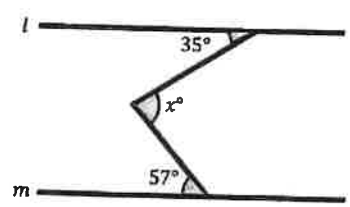
Annotate

19

Mark for Review

In the diagram below, the lines l and m run parallel to each other. What is the measure of angle x in degrees?



TESTQUBE

Question 19 of 22

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20

Mark for Review

A circle has an equation of $x^2 + 6x + y^2 - 10y + 18 = 0$. What is the radius of this circle?

(A) 2

(B) 4

(C) 8

(D) 16

TESTQUBE

Question 20 of 22

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21

Mark for Review

What is the circumference of a circle with an area of 16π?

(A) 2π

(B) 4π

(C) 8π

(D) 16π

TESTQUBE

Question 21 of 22

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Section 2, Module 2: Math

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22

Mark for Review

Find the area of a regular hexagon with each side length of 4. Round your answer to the nearest tenth.